MATH50003 Numerical Analysis

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Chapter I

Calculus on a Computer

In this first chapter we explore the basics of mathematical computing and numerical analysis. In particular we investigate the following mathematical problems which can not in general be solved exactly:

1. Integration. General integrals have no closed form expressions. Can we instead use a computer to approximate the values of definite integrals? Numerical integration underpins much of modern scientific computing and simulations of physical systems modelled by partial differential equations.

2. Differentiation. Differentiating a formula as in calculus is usually algorithmic, however, it is often needed to compute derivatives without access to an underlying formula, eg, a function defined only in code. Can we use a computer to approximate derivatives? A very important application is in Machine Learning, where there is a need to compute gradients in training neural networks.

3. Root finding. There is no general formula for finding roots (zeros) of arbitrary functions, or even polynomials that are of degree 5 (quintics) or higher. Can we compute roots of general functions using a computer?

Each chapter is divided into sections that roughly correspond to individual lectures. In this chapter we investigate solving the above computational problems:

1. I.1 Rectangular rule: we review the rectangular rule for integration and deduce the *convergence rate* of the approximation. In the lab/problem sheet we investigate its implementation as well as extensions to the Trapezium rule.

2. I.2 Divided differences: we investigate approximating derivatives by a divided difference and again deduce the convergence rates. In the lab/problem sheet we extend the approach to the central differences formula and computing second derivatives. We also observe a mystery: the approximations may have significant errors in practice, and there is a limit to the accuracy.

3. I.3 Dual numbers: we introduce the algebraic notion of a *dual number* which allows the implementation of *forward-mode automatic differentiation*, a high accuracy alternative to divided differences for computing derivatives.

4. I.4 Newton's method: Newton's method is a basic approach for computing roots/zeros of a function. We use dual numbers to implement this algorithm.

Each week there are labs and problem sheets that further explore the mathematical material introduced in each section. The labs generally explore practical implementation and the impact of implementing methods in computer arithmetic. The problem sheets dig deeper into analysis of other methods and phenomena observed in the labs. The material introduced in the labs and problem sheets is also examinable so it's important to study these as well.

I.1 Rectangular rule

One possible definition for an integral is the limit of a Riemann sum, for example:

$$\int_a^b f(x)\mathrm{d}x = \lim_{n\to\infty} h \sum_{j=1}^{n} f(x_j)$$

where $x_j = a + jh$ are evenly spaced points dividing up the interval $[a, b]$, that is with the *step size* $h = (b - a)/n$. This suggests an algorithm known as the *(right-sided) rectangular rule* for approximating an integral: choose n large so that

$$\int_a^b f(x)\mathrm{d}x \approx h \sum_{j=1}^{n} f(x_j).$$

We will show that the error in approximation is bounded by C/n for some constant C. This can be expressed using "Big-O" notation:

$$\int_a^b f(x)\mathrm{d}x = h \sum_{j=1}^{n} f(x_j) + O(1/n).$$

In these notes we consider the "Analysis" part of "Numerical Analysis": we want to *prove* the convergence rate of the approximation, including finding an explicit expression for the constant C.

To tackle this question we consider the error incurred on a single panel (x_{j-1}, x_j), then sum up the errors on rectangles.

Now for a secret. There are only so many tools available in analysis (especially at this stage of your career), and one can make a safe bet that the right tool in any analysis proof is either (1) integration-by-parts, (2) geometric series or (3) Taylor series. In this case we use (1):

Lemma 1 ((Right-sided) Rectangular Rule error on one panel)**.** *Assuming f is differentiable on $[a, b]$ and its derivative is integrable we have*

$$\int_a^b f(x)\mathrm{d}x = (b - a)f(b) + \delta$$

where $|\delta| \leq M(b - a)^2$ for $M = \sup_{a\leq x\leq b} |f'(x)|$.

Proof We write

$$\begin{aligned}\int_a^b f(x)\mathrm{d}x &= \int_a^b (x-a)' f(x)\mathrm{d}x = [(x-a)f(x)]_a^b - \int_a^b (x-a)f'(x)\mathrm{d}x \\ &= (b-a)f(b) + \underbrace{\left(-\int_a^b (x-a)f'(x)\mathrm{d}x\right)}_{\delta}.\end{aligned}$$

Recall that we can bound the absolute value of an integral by the supremum of the integrand times the width of the integration interval:

$$\left|\int_a^b g(x)\mathrm{d}x\right| \leq (b-a)\sup_{a\leq x\leq b}|g(x)|.$$

The lemma thus follows since

$$\begin{aligned}\left|\int_a^b (x-a)f'(x)\mathrm{d}x\right| &\leq (b-a)\sup_{a\leq x\leq b}|(x-a)f'(x)| \\ &\leq (b-a)\sup_{a\leq x\leq b}|x-a|\sup_{a\leq x\leq b}|f'(x)| \\ &\leq M(b-a)^2.\end{aligned}$$

■

Now summing up the errors in each panel gives us the error of using the Rectangular rule:

Theorem 1 (Rectangular Rule error)**.** *Assuming f is differentiable on $[a,b]$ and its derivative is integrable we have*

$$\int_a^b f(x)\mathrm{d}x = h\sum_{j=1}^n f(x_j) + \delta$$

where $|\delta| \leq M(b-a)h$ for $M = \sup_{a\leq x\leq b}|f'(x)|$, $h = (b-a)/n$ and $x_j = a + jh$.

Proof We split the integral into a sum of smaller integrals:

$$\int_a^b f(x)\mathrm{d}x = \sum_{j=1}^n \int_{x_{j-1}}^{x_j} f(x)\mathrm{d}x = \sum_{j=1}^n [(x_j - x_{j-1})f(x_j) + \delta_j] = h\sum_{j=1}^n f(x_j) + \underbrace{\sum_{j=1}^n \delta_j}_{\delta}$$

where δ_j, the error on each panel as in the preceding lemma, satisfies

$$|\delta_j| \leq (x_j - x_{j-1})^2 \sup_{x_{j-1}\leq x\leq x_j}|f'(x)| \leq Mh^2.$$

Thus using the triangular inequality we have

$$|\delta| = \left|\sum_{j=1}^n \delta_j\right| \leq \sum_{j=1}^n |\delta_j| \leq Mnh^2 = M(b-a)h.$$

■

Note a consequence of this lemma is that the approximation converges as $n \to \infty$ (i.e. $h \to 0$). In the labs and problem sheets we will consider the left-sided rule:

$$\int_a^b f(x)\mathrm{d}x \approx h\sum_{j=0}^{n-1} f(x_j).$$

We also consider the *Trapezium rule*. Here we approximate an integral by an affine function:

$$\int_a^b f(x)\mathrm{d}x \approx \int_a^b \frac{(b-x)f(a) + (x-a)f(b)}{b-a}\mathrm{d}x = \frac{b-a}{2}\left[f(a) + f(b)\right].$$

Subdividing an interval $a = x_0 < x_1 < \ldots < x_n = b$ and applying this approximation separately on each subinterval $[x_{j-1}, x_j]$, where $h = (b-a)/n$ and $x_j = a + jh$, leads to the approximation

$$\int_a^b f(x)\mathrm{d}x \approx \frac{h}{2}f(a) + h\sum_{j=1}^{n-1} f(x_j) + \frac{h}{2}f(b)$$

We shall see both experimentally and provably that this approximation converges faster than the rectangular rule.

I.1.1 Lab and problem sheet

In the lab, we explore the practical implementation of the right-sided rectangular rule and extensions to other rules like the left-sided rectangular rule and trapezium rule. We also see how linear convergence ($O(h) = O(1/n)$) can be deduced *experimentally*: by comparing an implementation of the rule to specific integrals with known formulæ we can compute the error, and determine its rate of decay visually by plotting it. In particular, we deduce that the Trapezium rule converges much faster to the true value of the integral than the other rules. In the problem sheet we explore the *analysis* of these other rules, proving that the Trapezium rule converges to the true integral at a faster quadratic ($O(h^2)$) error rate. This is a guarantee that the integral can be computed much more accurately for the same amount of work by taking into account the analysis, highlighting the important contribution of analysis in the construction of algorithms.

I.2 Divided Differences

Given a function, how can we approximate its derivative at a point? We consider an intuitive approach to this problem using *(Right-sided) Divided Differences*:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Note by the definition of the derivative we know that this approximation will converge to the true derivative as $h \to 0$. But in numerical approximations we also need to consider the rate of convergence.

Now in the previous section I mentioned there are three basic tools in analysis: (1) integration-by-parts, (2) geometric series or (3) Taylor series. In this case we use (3):

Proposition 1 (divided differences error)**.** *Suppose that f is twice-differentiable on the interval $[x, x+h]$. The error in approximating the derivative using divided differences is*

$$f'(x) = \frac{f(x+h) - f(x)}{h} + \delta$$

where $|\delta| \leq Mh/2$ for $M = \sup_{x \leq t \leq x+h} |f''(t)|$.

Proof Follows immediately from Taylor's theorem: recall that

$$f(x+h) = f(x) + f'(x)h + \frac{f''(t)}{2}h^2$$

for some $t \in [x, x+h]$. Rearranging we get

$$f'(x) = \frac{f(x+h) - f(x)}{2} + \underbrace{\left(-\frac{f''(t)}{2h^2}\right)}_{\delta}.$$

We then bound:

$$|\delta| \leq \left|\frac{f''(t)}{2}h\right| \leq \frac{Mh}{2}.$$

■

Unlike the rectangular rule, the computational cost of computing the divided difference is independent of h! We only need to evaluate a function f twice and do a single division. Here we are assuming that the computational cost of evaluating f is independent of the point of evaluation. Later we will investigate the details of how computers work with numbers via floating point, and confirm that this is a sensible assumption.

In the lab we investigate the convergence rate of these approximations (in particular, that central differences is more accurate than standard divided differences) and observe that they too suffer from unexplained (for now) loss of accuracy as $h \to 0$. In the problem sheet we prove the theoretical convergence rate, which is never realised because of these errors.

I.2.1 Lab and problem sheet

In the labs and problem sheets we explore alternative versions of divided differences. Left-side divided differences evaluates to the left of the point where we wish to know the derivative:

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

and central differences evaluates both left and right:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

We can further arrive at an approximation to the second derivative by composing a left- and right-sided finite difference:

$$f''(x) \approx \frac{f'(x+h) - f'(x)}{h} \approx \frac{\frac{f(x+h)-f(x)}{h} - \frac{f(x)-f(x-h)}{h}}{h} = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$$

The lab explores these approximations *experimentally*, and we will observe that central differences converges much faster to the true value of the derivative as h becomes moderately small.

An important distinction between rectangular rules and divided difference is that the computational cost of divided differences is independent of h: we can choose h arbitrarily and the approximation will take the same amount of time. This raises a question: why not just set h ridiculously small so that the approximation is extremely accurate? Unfortunately, we will observe in the lab a serious issue: if h becomes too small, the error mysteriously starts to grow, and hence these rules do not actually converge to the true value of the derivatives! Thus there is a limitation to how accurate one can approximate a derivative using divided differences, an issue we will overcome in the next section by re-thinking derivatives in an algebraic way.

The problem sheet explores the *analysis* of divided difference rules, proving the precise theoretical convergence rates observed for moderately small h. This presents a bit of a conundrum: why does the theory say the method converges but in practice it diverges, and spectacularly so! This is a mystery that we will return to later, by understanding how computer arithmetic with real numbers works.

I.3 Dual Numbers

In this section we introduce a mathematically beautiful alternative to divided differences for computing derivatives: *dual numbers*. These are a commutative ring that *exactly* compute

derivatives, which when implemented on a computer gives very high-accuracy approximations to derivatives. They underpin forward-mode automatic differentation. Automatic differentiation is a basic tool in Machine Learning for computing gradients necessary for training neural networks.

Definition 1 (Dual numbers)**.** Dual numbers $\mathbb{D}$ are a commutative ring (over $\mathbb{R}$) generated by 1 and ϵ such that $\epsilon^2 = 0$, that is,

$$\mathbb{D} := \{a + b\epsilon \quad : \quad a, b \in \mathbb{R}, \quad \epsilon^2 = 0\}.$$

This is very much analoguous to complex numbers, which are a field generated by 1 and i such that $\mathrm{i}^2 = -1$, that is,

$$\mathbb{C} := \{a + b\mathrm{i} \quad : \quad a, b \in \mathbb{R}, \quad \mathrm{i}^2 = -1\}.$$

Compare multiplication of each number type which falls out of the rules of the generators:

$$\begin{aligned}(a + b\mathrm{i})(c + d\mathrm{i}) &= ac + (bc + ad)\mathrm{i} + bd\mathrm{i}^2 = ac - bd + (bc + ad)\mathrm{i},\\ (a + b\epsilon)(c + d\epsilon) &= ac + (bc + ad)\epsilon + bd\epsilon^2 = ac + (bc + ad)\epsilon.\end{aligned}$$

And just as we view $\mathbb{R} \subset \mathbb{C}$ by equating $a \in \mathbb{R}$ with $a + 0\mathrm{i} \in \mathbb{C}$, we can view $\mathbb{R} \subset \mathbb{D}$ by equating $a \in \mathbb{R}$ with $a + 0\epsilon \in \mathbb{D}$.

Conceptually, dual numbers can be thought of as introducing an infinitesimally small ϵ, where ϵ^2 is so small it is treated as zero. This is the intuitive reason they allow for differentiation of functions. But we do not need to appeal to this calculus-like interpretation, instead, their construction and relationship to differentiation can be accomplished using purely algebraic reasoning.

I.3.1 Differentiating polynomials

Polynomials evaluated on dual numbers are well-defined as they depend only on the operations $+$ and $*$. From the formula for multiplication of dual numbers we deduce that evaluating a polynomial at a dual number $a + b\epsilon$ tells us the derivative of the polynomial at a:

Theorem 2 (polynomials on dual numbers)**.** *Suppose p is a polynomial. Then*

$$p(a + b\epsilon) = p(a) + bp'(a)\epsilon$$

Proof

First consider $p(x) = x^n$ for $n \geq 0$. The cases $n = 0$ and $n = 1$ are immediate. For $n > 1$ we have by induction:

$$(a + b\epsilon)^n = (a + b\epsilon)(a + b\epsilon)^{n-1} = (a + b\epsilon)(a^{n-1} + (n - 1)ba^{n-2}\epsilon) = a^n + bna^{n-1}\epsilon.$$

For a more general polynomial

$$p(x) = \sum_{k=0}^{n} c_k x^k$$

the result follows from linearity:

$$p(a+b\varepsilon) = \sum_{k=0}^{n} c_k(a+b\epsilon)^k = c_0+\sum_{k=1}^{n} c_k(a^k+kba^{k-1}\epsilon) = \sum_{k=0}^{n} c_k a^k+b\sum_{k=1}^{n} c_k ka^{k-1}\epsilon = p(a)+bp'(a)\epsilon.$$

■

Example 1 (differentiating polynomial)**.** Consider computing $p'(2)$ where

$$p(x) = (x-1)(x-2) + x^2.$$

We can use dual numbers to differentiate, avoiding expanding in monomials or applying rules of differentiating:

$$p(2+\epsilon) = (1+\epsilon)\epsilon + (2+\epsilon)^2 = \epsilon + 4 + 4\epsilon = 4 + \underbrace{5}_{p'(2)}\epsilon.$$

I.3.2 Differentiating other functions

We can extend real-valued differentiable functions to dual numbers in a similar manner. First, consider a standard function with a Taylor series (e.g. cos, sin, exp, etc.)

$$f(x) = \sum_{k=0}^{\infty} f_k x^k$$

so that a is inside the radius of convergence. This leads naturally to a definition on dual numbers:

$$\begin{aligned} f(a+b\epsilon) &= \sum_{k=0}^{\infty} f_k (a+b\epsilon)^k = f_0 + \sum_{k=1}^{\infty} f_k (a^k + ka^{k-1}b\epsilon) = \sum_{k=0}^{\infty} f_k a^k + \sum_{k=1}^{\infty} f_k k a^{k-1} b\epsilon \\ &= f(a) + bf'(a)\epsilon. \end{aligned}$$

More generally, given a differentiable function (which may not have a Taylor series) we can extend it to dual numbers:

Definition 2 (dual extension)**.** Suppose a real-valued function $f : \Omega \to \mathbb{R}$ is differentiable in $\Omega \subset \mathbb{R}$. We can construct the *dual extension* $\underline{f} : \Omega + \epsilon\mathbb{R} \to \mathbb{D}$ by defining

$$\underline{f}(a+b\epsilon) := f(a) + bf'(a)\epsilon.$$

By viewing $\mathbb{R} \subset \mathbb{D}$, it is natural to reuse the notation f for the dual extension, hence when there's no chance of confusion we will identify $f(a+b\epsilon) \equiv \underline{f}(a+b\epsilon)$.

Thus, for basic functions we have natural extensions:

$$\begin{aligned} \exp(a+b\epsilon) &:= \exp(a) + b\exp(a)\epsilon && (a,b \in \mathbb{R}) \\ \sin(a+b\epsilon) &:= \sin(a) + b\cos(a)\epsilon && (a,b \in \mathbb{R}) \\ \cos(a+b\epsilon) &:= \cos(a) - b\sin(a)\epsilon && (a,b \in \mathbb{R}) \\ \log(a+b\epsilon) &:= \log(a) + \frac{b}{a}\epsilon && (a \in (0,\infty), b \in \mathbb{R}) \\ \sqrt{a+b\epsilon} &:= \sqrt{a} + \frac{b}{2\sqrt{a}}\epsilon && (a \in (0,\infty), b \in \mathbb{R}) \\ |a+b\epsilon| &:= |a| + b\,\mathrm{sign} a\,\epsilon && (a \in \mathbb{R}\backslash\{0\}, b \in \mathbb{R}) \end{aligned}$$

provided the function is differentiable at a. Note the last example does not have a convergent Taylor series (at 0) but we can still extend it where it is differentiable.

Going further, we can add, multiply, and compose such dual-extensions. And the beauty is these automatically satisfy the right properties to be dual-extensions themselves, thus

allowing for differentiation of complicated functions built from basic differentiable building blocks.

The following lemma shows that addition and multiplication in some sense "commute" with the dual-extension, hence we can recover the product rule from dual number multiplication:

Lemma 2 (addition/multiplication)**.** *Suppose $f, g : \Omega \to \mathbb{R}$ are differentiable for $\Omega \subset \mathbb{R}$ and $c \in \mathbb{R}$. Then for $a \in \Omega$ and $b \in \mathbb{R}$ we have*

$$\begin{aligned} \underline{f+g}(a+b\epsilon) &= \underline{f}(a+b\epsilon) + \underline{g}(a+b\epsilon) \\ \underline{cf}(a+b\epsilon) &= c\underline{f}(a+b\epsilon) \\ \underline{fg}(a+b\epsilon) &= \underline{f}(a+b\epsilon)\underline{g}(a+b\epsilon) \end{aligned}$$

Proof The first two are immediate due to linearity:

$$\begin{aligned} \underline{(f+g)}(a+b\epsilon) &= (f+g)(a) + b(f+g)'(a)\epsilon \\ &= (f(a) + bf'(a)\epsilon) + (g(a) + bg'(a)\epsilon) = \underline{f}(a+b\epsilon) + \underline{g}(a+b\epsilon), \\ \underline{cf}(a+b\epsilon) &= (cf)(a) + b(cf)'(a)\epsilon = c(f(a) + bf'(a)\epsilon) = c\underline{f}(a+b\epsilon). \end{aligned}$$

The last property essentially captures the product rule of differentiation:

$$\begin{aligned} \underline{fg}(a+b\epsilon) &= f(a)g(a) + b(f(a)g'(a) + f'(a)g'(a))\epsilon \\ &= (f(a) + bf'(a)\epsilon)(g(a) + bg'(a)\epsilon) = \underline{f}(a+b\epsilon)\underline{g}(a+b\epsilon). \end{aligned}$$

■

Furthermore composition recovers the chain rule:

Lemma 3 (composition)**.** *Suppose $f : \Gamma \to \mathbb{R}$ and $g : \Omega \to \Gamma$ are differentiable in $\Omega, \Gamma \subset \mathbb{R}$. Then*

$$\underline{(f \circ g)}(a+b\epsilon) = \underline{f}(\underline{g}(a+b\epsilon))$$

Proof Again it falls out of the properties of dual numbers:

$$\underline{(f \circ g)}(a+b\epsilon) = f(g(a)) + bg'(a)f'(g(a))\epsilon = \underline{f}(g(a) + bg'(a)\epsilon) = \underline{f}(\underline{g}(a+b\epsilon))$$

■

A simple corollary is that any function defined in terms of addition, multiplication, composition, etc. of basic functions with dual-extensions will be differentiable via dual numbers. In this following example we see a practical realisation of this, where we differentiate a function by just evaluating it on dual numbers, implicitly, using the dual-extension for the basic build blocks:

Example 2 (differentiating non-polynomial)**.** Consider differentiating $f(x) = \exp(x^2 + \cos x)$ at the point $a = 1$, where we automatically use the dual-extension of exp and cos. We can differentiate f by simply evaluating on the duals:

$$f(1+\epsilon) = \exp(1 + 2\epsilon + \cos 1 - \sin 1\epsilon) = \exp(1 + \cos 1) + \exp(1 + \cos 1)(2 - \sin 1)\epsilon.$$

Therefore we deduce that

$$f'(1) = \exp(1 + \cos 1)(2 - \sin 1).$$

I.3.3 Lab and problem sheet

In the lab we explore how one can turn this mathematical idea into a practical implementation on a computer, giving a basic version of *forward-mode automatic differentiation*. This is a concept that underpins machine learning, which uses *reverse-mode automatic differentiation* to compute gradients when performing stochastic gradient descent. In order to implement dual numbers, we will introduce the concept of a *type*: a data structure with fields. For example, we will implement a type `Rat` for representing rationals p/q, where the type has two fields (`p` and `q`). Basic arithmetic operations like `+` and `*` can be implemented to correctly do rational arithmetic. We will then create a new type that can represent a dual number $a + b\epsilon$, where the the type has two fields (`a` and `b`). By implementing basic arithmetic operations as well as more complicated functions like `exp` we can efficiently, and extremely accurately, compute derivatives of quite general functions.

In the problem sheet, we explore how dual numbers can also be used for pen-and-paper calculations of derivatives. This gives an alternative to traditional differentiation rules like chain and product rule, that while it is mathematically equivalent feels very different in practice. (I prefer it because it is much more algorithmic!) Make sure when doing the problem sheet to only use dual numbers and not fall back to the more traditional rules. We also see that one can extend the concept to a 2D-analogue of dual numbers, which allows for computation of gradients.

I.4 Newton's method

In school you may recall learning Newton's method: a way of approximating zeros/roots to a function by using a local approximation by an affine function. That is, approximate a function $f(x)$ locally around an initial guess x_0 by its first order Taylor series:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

and then find the root of the right-hand side which is

$$f(x_0) + f'(x_0)(x - x_0) = 0 \Leftrightarrow x = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

We can then repeat using this root as the new initial guess. In other words we have a sequence of *hopefully* more accurate approximations:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

Thus *if* we can compute derivatives, we can (sometimes) compute roots.

In terms of analysis, we can guarantee convergence provided our initial guess is accurate enough. The first step is the bound the error of an iteration in terms of the previous error:

Theorem 3 (Newton error)**.** *Suppose f is twice-differentiable in a neighbourhood B of r such that $f(r) = 0$, and f' does not vanish in B. Denote the error of the k-th Newton iteration as $\varepsilon_k := r - x_k$. If $x_k \in B$ then*

$$|\varepsilon_{k+1}| \leq M|\varepsilon_k|^2$$

where

$$M := \frac{1}{2} \sup_{x \in B} |f''(x)| \sup_{x \in B} \left| \frac{1}{f'(x)} \right|.$$

Proof Using Taylor's theorem we find that

$$0 = f(r) = f(x_k + \varepsilon_k) = f(x_k) + f'(x_k)\varepsilon_k + \frac{f''(t)}{2}\varepsilon_k^2.$$

for some $t \in B$ between r and x_k. Rearranging this we get an expression for $f(x_k)$ that tells us that

$$\varepsilon_{k+1} = r - \underbrace{x_{k+1}}_{x_k - f(x_k)/f'(x_k)} = \varepsilon_k + \frac{f(x_k)}{f'(x_k)} = -\frac{f''(t)}{2f'(x_k)}\varepsilon_k^2.$$

Taking absolute values of each side gives the result.

■

This result says that the error decays *quadratically*, which in this case means that the number of digits roughly doubles each iteration. That is, if the error and one step is about 10^{-3} then the error at the next step is about 10^{-6} and the step after about 10^{-12}: this is a drastic improvement! Hidden in this result is a guarantee of convergence provided x_0 is sufficiently close to r.

Corollary 1 (Newton convergence)**.** *If $x_0 \in B$ is sufficiently close to r then $x_k \to r$.*

Proof

Suppose $x_k \in B$ satisfies $|\varepsilon_k| = |r - x_k| \leq M^{-1}$. Then

$$|\varepsilon_{k+1}| \leq M|\varepsilon_k|^2 \leq |\varepsilon_k|,$$

hence $x_{k+1} \in B$. Thus from induction if x_0 satisfies the condition $|\varepsilon_0| < M^{-1}$ condition then $x_k \in B$ for all k and satisfies $|\varepsilon_k| \leq M^{-1}$. Thus we find (for large enough k)

$$|\varepsilon_k| \leq M|\varepsilon_{k-1}|^2 \leq M^3|\varepsilon_{k-2}|^4 \leq M^7|\varepsilon_{k-3}|^8 \leq \ldots \leq M^{2^k-1}|\varepsilon_0|^{2^k} = \frac{1}{M}(M|\varepsilon_0|)^{2^k}.$$

Provided x_0 satisfies the strict inequality $|\varepsilon_0| < M^{-1}$ this will go to zero as $k \to \infty$.

■

I.4.1 Lab and problem sheet

In the lab we explore using Newton's method for some simple root finding problems. We also see that automatic differentiation via dual numbers can be used effectively to compute the derivatives. This is in some sense a baby version of how Machine Learning algorithms train neural networks; but whilst Newton uses derivatives (or in higher-dimensions, gradients) to find roots of functions Machine Learning uses gradients to (very roughly) minimise functions that represent the error between a neural network and training data. Minimisation problems are very closely related to root finding problems (essentially the minima are associated with roots of the gradient) and there are specialised training algorithms in ML built on a randomised version of Newton's method.

In the problem sheet we see how the error bound for Newton iteration can be extended to the degenerate case where the second derivative also vanishes, but now we no longer achieve quadratic convergence, but it still decays exponentially with the number of iterations (which is called *linear convergence*).

Chapter II

Representing Numbers

In this chapter we aim to answer the question: when can we rely on computations done on a computer? Why are some computations (differentiation via divided differences), extremely inaccurate whilst others (integration via rectangular rule) accurate up to about 16 digits? In order to address these questions we need to dig deeper and understand at a basic level what a computer is actually doing when manipulating numbers.

Before we begin it is important to have a basic model of how a computer works. Our simplified model of a computer will consist of a Central Processing Unit (CPU)—the brains of the computer—and Memory—where data is stored. Inside the CPU there are registers, where data is temporarily stored after being loaded from memory, manipulated by the CPU, then stored back to memory. Memory is a sequence of bits: `1`s and `0`s, essentially "on/off" switches, and memory is *finite*. Finally, if one has a p-bit CPU (eg a 32-bit or 64-bit CPU), each register consists of exactly p-bits. Most likely $p = 64$ on your machine.

Thus representing numbers on a computer must overcome three fundamental limitations:

1. CPUs can only manipulate data p-bits at a time.
2. Memory is finite (in particular at most 2^p bytes).
3. There is no such thing as an "error": if anything goes wrong in the computation we must use some of the p-bits to indicate this.

This is clearly problematic: there are an infinite number of integers and an uncountable number of reals! Each of which we need to store in precisely p-bits. Moreover, some operations are simply undefined, like division by 0. This chapter discusses the solution used to this problem, alongside the mathematical analysis that is needed to understand the implications, in particular, that computations have *error*.

In particular we discuss:

1. II.1 Reals: real numbers are approximated by floating point numbers, which are a computers version of scientific notation.
2. II.2 Floating Point Arithmetic: arithmetic with floating point numbers is exact up-to-rounding, which introduces small-but-understandable errors in the computations. We explain how these errors can be analysed mathematically to get rigorous bounds.

3. II.3 Interval Arithmetic: rounding can be controlled in order to implement *interval arithmetic*, a way to compute rigorous bounds for computations. In the lab, we use this to compute up to 15 digits of $\mathrm{e} \equiv \exp 1$ rigorously with precise bounds on the error.

II.1 Reals

In this chapter, we introduce the IEEE Standard for Floating-Point Arithmetic. There are multiplies ways of representing real numbers on a computer, as well as the precise behaviour of operations such as addition, multiplication, etc. One can use

1. Fixed-point arithmetic: essentially representing a real number as an integer where a decimal point is inserted at a fixed position. This turns out to be impractical in most applications, e.g., due to loss of relative accuracy for small numbers.
2. Floating-point arithmetic: essentially scientific notation where an exponent is stored alongside a fixed number of digits. This is what is used in practice.
3. Level-index arithmetic: stores numbers as iterated exponents. This is the most beautiful mathematically but unfortunately is not as useful for most applications and is not implemented in hardware.

Before the 1980s each processor had potentially a different representation for floating-point numbers, as well as different behaviour for operations. IEEE introduced in 1985 standardised this across processors so that algorithms would produce consistent and reliable results.

This chapter may seem very low level for a mathematics course but there are two important reasons to understand the behaviour of floating-point numbers in details:

1. Floating-point arithmetic is precisely defined, and can even be used in rigorous computations as we shall see in the labs. But it is not exact and its important to understand how errors in computations can accumulate.
2. Failure to understand floating-point arithmetic can cause catastrophic issues in practice, with the extreme example being the explosion of the Ariane 5 rocket.

II.1.1 Real numbers in binary

We begin by describing how both integers and real numbers can be written in binary, that is, base-2. In this case the digits are either 0 or 1, which matches how a computer stores data.

Integers can be written in binary as follows:

Definition 3 (binary format)**.** For $B_0, \ldots, B_p \in \{0, 1\}$ denote an integer in *binary format* by:

$$\pm(B_p \ldots B_1 B_0)_2 := \pm \sum_{k=0}^{p} B_k 2^k$$

Reals can also be presented in binary format, that is, a sequence of `0`s and `1`s alongside a decimal point:

Definition 4 (real binary format)**.** For $b_1, b_2, \ldots \in \{0, 1\}$, Denote a non-negative real number in *binary format* by:

$$(B_p \ldots B_0.b_1b_2b_3\ldots)_2 := (B_p \ldots B_0)_2 + \sum_{k=1}^{\infty} \frac{b_k}{2^k}.$$

Example 3 (rational in binary)**.** Consider the number `1/3`. In decimal recall that:

$$1/3 = 0.3333\ldots = \sum_{k=1}^{\infty} \frac{3}{10^k}$$

We will see that in binary

$$1/3 = (0.010101\ldots)_2 = \sum_{k=1}^{\infty} \frac{1}{2^{2k}}$$

Both results can be proven using the geometric series:

$$\sum_{k=0}^{\infty} z^k = \frac{1}{1-z}$$

provided $|z| < 1$. That is, with $z = \frac{1}{4}$ we verify the binary expansion:

$$\sum_{k=1}^{\infty} \frac{1}{4^k} = \frac{1}{1-1/4} - 1 = \frac{1}{3}$$

A similar argument with $z = 1/10$ shows the decimal case.

II.1.2 Floating-point numbers

Floating-point numbers are a subset of real numbers that are representable using a fixed number of bits.

Definition 5 (floating-point numbers)**.** Given integers σ (the *exponent shift*), Q (the number of *exponent bits*) and S (the *precision*), define the set of *Floating-point numbers* by as the union of *normal*, *sub-normal*, and *special* floating point numbers:

$$F_{\sigma,Q,S} := F^{\text{normal}}_{\sigma,Q,S} \cup F^{\text{sub}}_{\sigma,Q,S} \cup F^{\text{special}}.$$

The *normal numbers* $F^{\text{normal}}_{\sigma,Q,S} \subset \mathbb{R}$ are

$$F^{\text{normal}}_{\sigma,Q,S} := \{\pm 2^{q-\sigma} \times (1.b_1b_2b_3\ldots b_S)_2 : 1 \leq q < 2^Q - 1\}.$$

The *sub-normal numbers* $F^{\text{sub}}_{\sigma,Q,S} \subset \mathbb{R}$ are

$$F^{\text{sub}}_{\sigma,Q,S} := \{\pm 2^{1-\sigma} \times (0.b_1b_2b_3\ldots b_S)_2\}.$$

The *special numbers* $F^{\text{special}} \not\subset \mathbb{R}$ are

$$F^{\text{special}} := \{\infty, -\infty, \text{NaN}\}$$

where NaN is a special symbol representing "not a number", essentially an error flag.

Note this set of real numbers has no nice *algebraic structure*: it is not closed under addition, subtraction, etc. On the other hand, we can control errors effectively hence it is extremely useful for analysis.

Floating-point numbers are stored in $1 + Q + S$ total number of bits, in the format

$$s \; q_{Q-1} \dots q_0 \; b_1 \dots b_S$$

The first bit (s) is the *sign bit*: 0 means positive and 1 means negative. The bits $q_{Q-1} \dots q_0$ are the *exponent bits*: they are the binary digits of the unsigned integer q:

$$q = (q_{Q-1} \dots q_0)_2.$$

Finally, the bits $b_1 \dots b_S$ are the *significand bits*. If $1 \leq q < 2^Q - 1$ then the bits represent the normal number

$$x = \pm 2^{q-\sigma} \times (1.b_1 b_2 b_3 \dots b_S)_2.$$

If $q = 0$ (i.e. all bits are 0) then the bits represent the sub-normal number

$$x = \pm 2^{1-\sigma} \times (0.b_1 b_2 b_3 \dots b_S)_2.$$

If $q = 2^Q - 1$ (i.e. all bits are 1) then the bits represent a special number. If all sigificand bits are 0 then it represents $\pm\infty$. Otherwise if any significand bit is 1 then it represents `NaN`.

Remark A common point of confusion is the difference between a *number* whose digits are 0 and 1 but may have a sign ($\pm$) and a decimal point and a *sequence of bits*, which is how a number is stored in memory in a computer.

II.1.3 IEEE floating-point numbers

Definition 6 (IEEE floating-point numbers)**.** IEEE has 3 standard floating-point formats: 16-bit (half precision), 32-bit (single precision) and 64-bit (double precision) defined by (you *do not* need to memorise these):

$$\begin{aligned} F_{16} &:= F_{15,5,10} \\ F_{32} &:= F_{127,8,23} \\ F_{64} &:= F_{1023,11,52} \end{aligned}$$

We now see a simple example of relating the bits of a floating point number with the number it represents:

Example 4 (interpreting 16-bits as a float)**.** Consider the number with bits

$$0 \; 10000 \; 1010000000$$

assuming it is a half-precision float (F_{16}). Since the sign bit is `0` it is positive. The exponent bits encode

$$q = (10000)_2 = 2^4$$

hence the exponent is

$$q - \sigma = 2^4 - 15 = 1$$

and the number is:

$$2^1 (1.1010000000)_2 = 2(1 + 1/2 + 1/8) = 3 + 1/4 = 3.25.$$

Example 5 (rational to 16-bits)**.** How is the number 1/3 stored in F_{16}? Recall that

$$1/3 = (0.010101\ldots)_2 = 2^{-2}(1.0101\ldots)_2 = 2^{13-15}(1.0101\ldots)_2$$

and since $13 = (1101)_2$ the exponent bits are `01101`. For the significand we round the last bit to the nearest element of F_{16}, (the exact rule for rounding is explained in detail later), so we have

$$1.0101010101010101010101\ldots \approx 1.0101010101 \in F_{16}$$

and the significand bits are `0101010101`. Thus the stored bits for 1/3 are:

0 01101 0101010101

II.1.4 Sub-normal and special numbers

For sub-normal numbers, the simplest example is zero, which has $q = 0$ and all significand bits zero: `0 00000 0000000000`. Unlike integers, we also have a negative zero, which has bits: `1 00000 0000000000`. This is treated as identical to positive `0` (except for degenerate operations as explained in the lab).

Example 6 (subnormal in 16-bits)**.** Consider the number with bits

1 00000 1100000000

assuming it is a half-precision float (F_{16}). Since all exponent bits are zero it is sub-normal. Since the sign bit is `1` it is negative. Hence this number is:

$$-2^{1-\sigma}(0.1100000000)_2 = -2^{-14}(2^{-1} + 2^{-2}) = -3 \times 2^{-16}$$

The special numbers extend the real line by adding $\pm\infty$ but also a notion of "not-a-number" NaN. Whenever the bits of q of a floating-point number are all 1 then they represent an element of F^{special}. If all $b_k = 0$, then the number represents either $\pm\infty$. All other special floating-point numbers represent NaN.

Example 7 (special in 16-bits)**.** The number with bits

1 11111 0000000000

has all exponent bits equal to 1, and significand bits 0 and sign bit 1, hence represents $-\infty$. On the other hand, the number with bits

1 11111 0000000001

has all exponent bits equal to 1 but does not have all significand bits equal to 0, hence is one of many representations for NaN.

II.1.5 Lab and problem sheet

In the lab we explore how integers and floating point numbers are stored in a computer via different *types*. We begin with a description both signed and unsigned integers, whose mathematical behaviour are detailed in the (non-examinable) appendix. We see how different

sequences of bits can be reinterpreted as different numbers, and explore using string manipulation to construct different types of numbers. We also see how integers can sometimes be output in hexadecimal (base-16) format, which aligns better with the underlying binary storage. We then explore floating point numbers including construction by specifying the bits directly. We finally investigate some of the degenerate behaviour such as arithmetic with special numbers. In the problem sheet we investigate some simple examples of representing numbers in floating point format.

II.2 Floating Point Arithmetic

We now turn our attention to how arithmetic operations (`+`, `*`, `-`, `/`, etc.) work with floating point arithmetic, in particular, how they cope with the fact that some calculations involving floating point numbers result in numbers that are not floating point (like `1/3`). The answer is that arithmetic operations on floating-point numbers are rounded, and are guaranteed to be *exact up to rounding*. There are three basic rounding strategies: round up/down/nearest. Mathematically we introduce a function to capture the notion of rounding:

Definition 7 (rounding)**.** The function $\mathrm{fl}^{\mathrm{up}}_{\sigma,Q,S} : \mathbb{R} \to F_{\sigma,Q,S}$ rounds a real number up to the nearest floating-point number that is greater than or equal:

$$\mathrm{fl}^{\mathrm{up}}_{\sigma,Q,S}(x) := \min\{y \in F_{\sigma,Q,S} : y \geq x\}.$$

The function $\mathrm{fl}^{\mathrm{down}}_{\sigma,Q,S} : \mathbb{R} \to F_{\sigma,Q,S}$ rounds a real number down to the nearest floating-point number that is less than or equal:

$$\mathrm{fl}^{\mathrm{down}}_{\sigma,Q,S}(x) := \max\{y \in F_{\sigma,Q,S} : y \leq x\}.$$

The function $\mathrm{fl}^{\mathrm{nearest}}_{\sigma,Q,S} : \mathbb{R} \to F_{\sigma,Q,S}$ denotes the function that rounds a real number to the nearest floating-point number. In case of a tie, it returns the floating-point number whose least significant bit is equal to zero. We use the notation fl when σ, Q, S and the rounding mode are implied by context, with $\mathrm{fl}^{\mathrm{nearest}}$ being the default rounding mode.

In more detail on the behaviour of nearest mode, if a positive number x is between two normal floats $x_- \leq x \leq x_+$ we can write its expansion as

$$x = 2^{q-\sigma}(1.b_1 b_2 \ldots b_S b_{S+1} \ldots)_2$$

where

$$\begin{aligned} x_- &:= \mathrm{fl}^{\mathrm{down}}(x) = 2^{q-\sigma}(1.b_1 b_2 \ldots b_S)_2 \\ x_+ &:= \mathrm{fl}^{\mathrm{up}}(x) = x_- + 2^{q-\sigma-S} \end{aligned}$$

Write the half-way point as:

$$x_{\mathrm{h}} := \frac{x_+ + x_-}{2} = x_- + 2^{q-\sigma-S-1} = 2^{q-\sigma}(1.b_1 b_2 \ldots b_S 1)_2$$

If $x_- \leq x < x_{\mathrm{h}}$ then $\mathrm{fl}(x) = x_-$ and if $x_{\mathrm{h}} < x \leq x_+$ then $\mathrm{fl}(x) = x_+$. If $x = x_{\mathrm{h}}$ then it is exactly half-way between x_- and x_+. The rule is if $b_S = 0$ then $\mathrm{fl}(x) = x_-$ and otherwise $\mathrm{fl}(x) = x_+$.

In IEEE arithmetic, the arithmetic operations `+`, `-`, `*`, `/` are defined by the property that they are exact up to rounding. Mathematically we denote these operations as $\oplus, \ominus, \otimes, \oslash : F_{\sigma,Q,S} \times F_{\sigma,Q,S} \to F_{\sigma,Q,S}$ as follows:

$$\begin{aligned} x \oplus y &:= \mathrm{fl}(x + y) \\ x \ominus y &:= \mathrm{fl}(x - y) \\ x \otimes y &:= \mathrm{fl}(x * y) \\ x \oslash y &:= \mathrm{fl}(x / y) \end{aligned}$$

Note also that `^` and `sqrt` are similarly exact up to rounding. Also, note that when we convert a Julia command with constants specified by decimal expansions we first round the constants to floats, e.g., `1.1 + 0.1` is actually reduced to

$$\mathrm{fl}(1.1) \oplus \mathrm{fl}(0.1)$$

This includes the case where the constants are integers (which are normally exactly floats but may be rounded if extremely large).

Example 8 (decimal is not exact)**.** On a computer `1.1+0.1` is close to but not exactly the same thing as `1.2`. This is because $\mathrm{fl}(1.1) \neq 1 + 1/10$ and $\mathrm{fl}(0.1) \neq 1/10$ since their expansion in *binary* is not finite. For F_{16} we have:

$$\begin{aligned} &\mathrm{fl}(1.1) = \mathrm{fl}((1.0001100110011\ldots)_2) = (1.0001100110)_2 \\ &\mathrm{fl}(0.1) = \mathrm{fl}(2^{-4}(1.1001100110011\ldots)_2) = 2^{-4} * (1.1001100110)_2 = (0.00011001100110)_2 \end{aligned}$$

Thus when we add them we get

$$\mathrm{fl}(1.1) + \mathrm{fl}(0.1) = (1.0011001100011)_2$$

where the red digits indicate those beyond the 10 significant digits representable in F_{16}. In this case we round down and get

$$\mathrm{fl}(1.1) \oplus \mathrm{fl}(0.1) = (1.0011001100)_2$$

On the other hand,

$$\mathrm{fl}(1.2) = \mathrm{fl}((1.0011001100110011001100\ldots)_2) = (1.0011001101)_2$$

which differs by 1 bit.

WARNING (non-associative) These operations are not associative! E.g. $(x \oplus y) \oplus z$ is not necessarily equal to $x \oplus (y \oplus z)$. Commutativity is preserved, at least.

II.2.1 Bounding errors in floating point arithmetic

We will now see that the error introduced by rounding can be bounded, giving a means to guarantee that some algorithms yield accurate results despite the errors introduced. When dealing with normal numbers there are some important constants that we will use to bound errors.

Definition 8 (machine epsilon/smallest positive normal number/largest normal number)**.** *Machine epsilon* is denoted

$$\epsilon_{\mathrm{m},S} := 2^{-S}.$$

When S is implied by context we use the notation ϵ_{m}. The *smallest positive normal number* is $q = 1$ and b_k all zero:

$$\min |F^{\mathrm{normal}}_{\sigma,Q,S}| = 2^{1-\sigma}$$

where $|A| := \{|x| : x \in A\}$. The *largest (positive) normal number* is

$$\max F^{\mathrm{normal}}_{\sigma,Q,S} = 2^{2^Q-2-\sigma}(1.11\ldots)_2 = 2^{2^Q-2-\sigma}(2-\epsilon_{\mathrm{m}})$$

We can bound the error of basic arithmetic operations in terms of machine epsilon, provided a real number is close to a normal number:

Definition 9 (normalised range)**.** The *normalised range* $\mathcal{N}_{\sigma,Q,S} \subset \mathbb{R}$ is the subset of real numbers that lies between the smallest and largest normal floating-point number:

$$\mathcal{N}_{\sigma,Q,S} := \{x : \min |F^{\mathrm{normal}}_{\sigma,Q,S}| \leq |x| \leq \max F^{\mathrm{normal}}_{\sigma,Q,S}\}$$

When σ, Q, S are implied by context we use the notation $\mathcal{N}$.

We can use machine epsilon to determine bounds on rounding:

Proposition 2 (round bound)**.** *If $x \in \mathcal{N}$ then*

$$\mathrm{fl}^{\mathrm{mode}}(x) = x(1+\delta_x^{\mathrm{mode}})$$

where the relative error *is bounded by:*

$$|\delta_x^{\mathrm{nearest}}| \leq \frac{\epsilon_{\mathrm{m}}}{2}$$
$$|\delta_x^{\mathrm{up/down}}| < \epsilon_{\mathrm{m}}.$$

Proof

We will show this result for the nearest rounding mode. Note first that

$$\mathrm{fl}(-x) = -\mathrm{fl}(x)$$

and hence it suffices to prove the result for positive x. Write

$$x = 2^{q-\sigma}(1.b_1b_2\ldots b_S b_{S+1}\ldots)_2.$$

Define

$$\begin{aligned} x_- &:= \mathrm{fl}^{\mathrm{down}}(x) = 2^{q-\sigma}(1.b_1b_2\ldots b_S)_2 \\ x_+ &:= \mathrm{fl}^{\mathrm{up}}(x) = x_- + 2^{q-\sigma-S} \\ x_{\mathrm{h}} &:= \frac{x_+ + x_-}{2} = x_- + 2^{q-\sigma-S-1} = 2^{q-\sigma}(1.b_1b_2\ldots b_S 1)_2 \end{aligned}$$

so that $x_- \leq x \leq x_+$. We consider two cases separately.

(**Round Down**) First consider the case where x is such that we round down: $\mathrm{fl}(x) = x_-$. Since $2^{q-\sigma} \leq x_- \leq x \leq x_{\mathrm{h}}$ we have

$$|\delta_x| = \frac{x - x_-}{x} \leq \frac{x_{\mathrm{h}} - x_-}{x_-} \leq \frac{2^{q-\sigma-S-1}}{2^{q-\sigma}} = 2^{-S-1} = \frac{\epsilon_{\mathrm{m}}}{2}.$$

(**Round Up**) If $\mathrm{fl}(x) = x_+$ then $2^{q-\sigma} \leq x_- < x_{\mathrm{h}} \leq x \leq x_+$ and hence

$$|\delta_x| = \frac{x_+ - x}{x} \leq \frac{x_+ - x_{\mathrm{h}}}{x_-} \leq \frac{2^{q-\sigma-S-1}}{2^{q-\sigma}} = 2^{-S-1} = \frac{\epsilon_{\mathrm{m}}}{2}.$$

■

This immediately implies relative error bounds on all IEEE arithmetic operations, e.g., if $x + y \in \mathcal{N}$ then we have

$$x \oplus y = (x + y)(1 + \delta_1)$$

where (assuming the default nearest rounding) $|\delta_1| \leq \frac{\epsilon_{\mathrm{m}}}{2}$.

II.2.2 Idealised floating point

With a complicated formula it is mathematically inelegant to work with normalised ranges: one cannot guarantee apriori that a computation always results in a normal float. Extending the bounds to subnormal numbers is tedious, rarely relevant, and beyond the scope of this module. Thus to avoid this issue we will work with an alternative mathematical model:

Definition 10 (idealised floating point)**.** An idealised mathematical model of floating point numbers for which the only subnormal number is zero can be defined as:

$$F_{\infty,S} := \{\pm 2^q \times (1.b_1 b_2 b_3 \ldots b_S)_2 : q \in \mathbb{Z}\} \cup \{0\}$$

Note that $F^{\mathrm{normal}}_{\sigma,Q,S} \subset F_{\infty,S}$ for all $\sigma, Q \in \mathbb{N}$. The definition of rounding $\mathrm{fl}^{mode}_{\infty,S} : \mathbb{R} \to F_{\infty,S}$ naturally extend to $F_{\infty,S}$ and hence we can consider bounds for floating point operations such as $\oplus$, $\ominus$, etc. And in this model the round bound is valid for all real numbers (including $x = 0$).

Example 9 (bounding a simple computation)**.** We show how to bound the error in computing $(1.1 + 1.2) * 1.3 = 2.99$ and we may assume idealised floating-point arithmetic $F_{\infty,S}$. First note that `1.1` on a computer is in fact $\mathrm{fl}(1.1)$, and we will always assume nearest rounding unless otherwise stated. Thus this computation becomes

$$(\mathrm{fl}(1.1) \oplus \mathrm{fl}(1.2)) \otimes \mathrm{fl}(1.3)$$

We will show the *absolute error* is given by

$$(\mathrm{fl}(1.1) \oplus \mathrm{fl}(1.2)) \otimes \mathrm{fl}(1.3) = 2.99 + \delta$$

where $|\delta| \leq 23\epsilon_{\mathrm{m}}$. First we find

$$\begin{aligned} \mathrm{fl}(1.1) \oplus \mathrm{fl}(1.2) &= (1.1(1 + \delta_1) + 1.2(1 + \delta_2))(1 + \delta_3) \\ &= 2.3 + \underbrace{1.1\delta_1 + 1.2\delta_2 + 2.3\delta_3 + 1.1\delta_1\delta_3 + 1.2\delta_2\delta_3}_{\varepsilon_1}. \end{aligned}$$

While $\delta_1\delta_3$ and $\delta_2\delta_3$ are absolutely tiny in practice we will bound them rather naïvely by eg.

$$|\delta_1\delta_3| \leq \epsilon_{\mathrm{m}}^2/4 \leq \epsilon_{\mathrm{m}}/4.$$

Further we round up constants to integers in the bounds for simplicity (we won't be concerned here with deriving the sharpest error bounds). We thus have the bound

$$|\varepsilon_1| \leq (2+2+3+1+1)\frac{\epsilon_{\mathrm{m}}}{2} \leq 5\epsilon_{\mathrm{m}}.$$

Writing $\mathrm{fl}(1.3) = 1.3(1+\delta_4)$ and also incorporating an error from the rounding in $\otimes$ we arrive at

$$\begin{aligned}(\mathrm{fl}(1.1) \oplus \mathrm{fl}(1.2)) \otimes \mathrm{fl}(1.3) &= (2.3+\varepsilon_1)1.3(1+\delta_4)(1+\delta_5) \\ &= 2.99 + \underbrace{1.3(\varepsilon_1 + 2.3\delta_4 + 2.3\delta_5 + \varepsilon_1\delta_4 + \varepsilon_1\delta_5 + 2.3\delta_4\delta_5 + \varepsilon_1\delta_4\delta_5)}_{\delta}\end{aligned}$$

We use the bounds

$$\begin{aligned}|\varepsilon_1\delta_4|, |\varepsilon_1\delta_5| &\leq 5\epsilon_{\mathrm{m}}^2/2 \leq 5\epsilon_{\mathrm{m}}/2, \\ |\delta_4\delta_5| &\leq \epsilon_{\mathrm{m}}^2/4 \leq \epsilon_{\mathrm{m}}/4, \\ |\varepsilon_1\delta_4\delta_5| &\leq 5\epsilon_{\mathrm{m}}^3/4 \leq 5\epsilon_{\mathrm{m}}/4.\end{aligned}$$

Thus the *absolute error* is bounded (bounding 1.3 by 3/2) by

$$|\delta| \leq (3/2)(5+3/2+3/2+5/2+5/2+3/4+5/4)\epsilon_{\mathrm{m}} \leq 23\epsilon_{\mathrm{m}}.$$

II.2.3 Divided differences floating point error bound

We saw experimentally that divided differences resulted in a large error which was not consistent with the error bound derived assuming exact real arithmetic. Here we see how we can derive a bound incorporating the behaviour of floating point arithmetic, which reflects the large growth in error when h became small.

We assume that the function we are attempting to differentiate is computed using floating point arithmetic in a way that has a small absolute error.

Theorem 4 (divided difference error bound)**.** *Assume we are working in idealised floating-point arithmetic $F_{\infty,S}$. Let f be twice-differentiable in a neighbourhood of $x \in F_{\infty,S}$ and assume that*

$$f(x) = f^{\mathrm{FP}}(x) + \delta_x^f$$

where $f^{\mathrm{FP}} : F_{S,\infty} \to F_{S,\infty}$ has uniform absolute accuracy in that neighbourhood, that is:

$$|\delta_x^f| \leq c\epsilon_{\mathrm{m}}$$

for a fixed constant $c \geq 0$. The divided difference approximation partially implemented with floating point satisfies

$$\frac{f^{\mathrm{FP}}(x+h) \ominus f^{\mathrm{FP}}(x)}{h} = f'(x) + \delta_{x,h}^{\mathrm{FD}}$$

where

$$|\delta_{x,h}^{\mathrm{FD}}| \leq \frac{|f'(x)|}{2}\epsilon_{\mathrm{m}} + Mh + \frac{4c\epsilon_{\mathrm{m}}}{h}$$

for $M = \sup_{x \leq t \leq x+h} |f''(t)|$.

Proof

We have

$$\begin{aligned}(f^{\mathrm{FP}}(x+h) \ominus f^{\mathrm{FP}}(x))/h &= \frac{f(x+h) - \delta^f_{x+h} - f(x) + \delta^f_x}{h}(1+\delta_1) \\ &= \frac{f(x+h) - f(x)}{h}(1+\delta_1) + \frac{\delta^f_x - \delta^f_{x+h}}{h}(1+\delta_1)\end{aligned}$$

where $|\delta_1| \leq \epsilon_{\mathrm{m}}/2$. Applying Taylor's theorem we get

$$(f^{\mathrm{FP}}(x+h) \ominus f^{\mathrm{FP}}(x))/h = f'(x) + \underbrace{f'(x)\delta_1 + \frac{f''(t)}{2}h(1+\delta_1) + \frac{\delta^f_x - \delta^f_{x+h}}{h}(1+\delta_1)}_{\delta^{\mathrm{FD}}_{x,h}}$$

The bound then follows, using the very pessimistic bound $|1+\delta_1| \leq 2$.

■

The previous theorem neglected some errors due to rounding, which was done for simplicity. This is justified under fairly general restrictions:

Corollary 2 (divided differences in practice)**.** *We have*

$$(f^{\mathrm{FP}}(x \oplus h) \ominus f^{\mathrm{FP}}(x)) \oslash h = \frac{f^{\mathrm{FP}}(x+h) \ominus f^{\mathrm{FP}}(x)}{h}$$

whenever $h = 2^{j-n}$ *for* $0 \leq n \leq S$ *and the last binary place of* $x \in F_{\infty,S}$ *is zero, that is* $x = \pm 2^j(1.b_1 \ldots b_{S-1}0)_2$.

Proof

We first confirm $x \oplus h = x + h$. If $b_S = 0$ the worst possible case is that we increase the exponent by one as we are just adding 1 to one of the digits $b_1, \ldots, b_S$. This would cause us to lose the last digit. But if that is zero no error is incurred when we round.

Now write $y := (f^{\mathrm{FP}}(x \oplus h) \ominus f^{\mathrm{FP}}(x)) = \pm 2^\nu(1.c_1 \ldots c_S)_2 \in F_{\infty,S}$. We have

$$y/h = \pm 2^{\nu+n-j}(1.c_1 \ldots c_S)_2 \in F_{\infty,S} \Rightarrow y/h = y \oslash h.$$

■

The three-terms of this bound tell us a story: the first term is a fixed (small) error, the second term tends to zero as $h \to 0$, while the last term grows like ϵ_{m}/h as $h \to 0$. Thus we observe convergence while the second term dominates, until the last term takes over. Of course, a bad upper bound is not the same as a proof that something grows, but it is a good indication of what happens *in general* and suffices to choose h so that these errors are balanced (and thus minimised). Since in general we do not have access to the constants c and M we employ the following heuristic to balance the two sources of errors:

Heuristic (divided difference with floating-point step) Choose h proportional to $\sqrt{\epsilon_{\mathrm{m}}}$ in divided differences so that Mh and $\frac{4c\epsilon_{\mathrm{m}}}{h}$ are (roughly) the same magnitude.

In the case of double precision $\sqrt{\epsilon_{\mathrm{m}}} \approx 1.5 \times 10^{-8}$, which is close to when the observed error begins to increase in the examples we saw before.

Remark While divided differences is of debatable utility for computing derivatives, it is extremely effective in building methods for solving differential equations, as we shall see

later. It is also very useful as a "sanity check" if one wants something to compare with other numerical methods for differentiation.

Remark It is also possible to deduce an error bound for the rectangular rule showing that the error caused by round-off is on the order of $n\epsilon_{\mathrm{m}}$, that is it does in fact grow but the error without round-off which was bounded by M/n will be substantially greater for all reasonable values of n.

II.2.4 Lab and problem sheet

In the lab we see how we can set the rounding mode of floating point calculations. This will set the ground-work for the next section where we implement interval arithmetic, which automatically computes rigorous error bounds. We also see that the `BigFloat` type allows for high-precision computations. In the problem sheet we bound the errors in some simple floating point expressions by-hand, and deduce an error bound for central differences capturing the impact of rounding. Finally, we see how addition and multiplication of many floating point numbers can also be bounded, which will lay the ground-work for understanding errors in linear algebra with floating point numbers.

II.3 Interval Arithmetic

We will now see how the details of floating point arithmetic give us a means of performing *computer-assisted proofs*, essentially turning the computer into a rigorous mathematician who can do millions (or even billions) of inequalities in seconds. To do this we will use the ability to set the rounding mode of floating point arithmetic to establish bounds. As an example we consider computing the digits of e with rigorous error bounds.

We first review set arithmetic. For sets $X, Y \subseteq \mathbb{R}$, the set arithmetic operations are defined as

$$\begin{aligned} X + Y &:= \{x + y : x \in X, y \in Y\}, \\ XY &:= \{xy : x \in X, y \in Y\}, \\ X/Y &:= \{x/y : x \in X, y \in Y\}. \end{aligned}$$

We will use floating point arithmetic to construct approximate set operations $\oplus$, $\otimes$ so that

$$\begin{aligned} X + Y &\subseteq X \oplus Y, \\ XY &\subseteq X \otimes Y, \\ X/Y &\subseteq X \oslash Y. \end{aligned}$$

Thereby a complicated algorithm can be run on sets and the true result is guaranteed to be a subset of the output.

When our sets are intervals we can deduce simple formulæ for basic arithmetic operations. For simplicity we only consider the case where all values are positive, leaving the generalisation to the problem sheet.

Proposition 3 (interval bounds)**.** *For intervals $X = [a, b]$ and $Y = [c, d]$ satisfying $0 < a \leq b$*

and $0 < c \leq d$*, and* $n > 0$*, we have:*

$$\begin{aligned} X + Y &= [a + c, b + d], \\ X/n &= [a/n, b/n], \\ XY &= [ac, bd]. \end{aligned}$$

Proof We first show $X + Y \subseteq [a + c, b + d]$. If $z \in X + Y$ then $z = x + y$ such that $a \leq x \leq b$ and $c \leq y \leq d$ and therefore $a + c \leq z \leq c + d$ and $z \in [a + c, b + d]$. Equality follows from convexity. First note that $a + c, b + d \in X + Y$. Any point $z \in [a + b, c + d]$ can be written as a convex combination of the two endpoints: there exists $0 \leq t \leq 1$ such that

$$z = (1 - t)(a + c) + t(b + d) = \underbrace{(1 - t)a + tb}_{x} + \underbrace{(1 - t)c + td}_{y}$$

Because intervals are convex we have $x \in X$ and $y \in Y$ and hence $z \in X + Y$.

The remaining two proofs are left for the problem sheet.

■

We want to implement floating point variants of these operations that are guaranteed to contain the true set arithmetic operations. We note that if we round the bottom of an interval down and the top of an interval up, then the actual interval is guaranteed to lie inside the resulting interval with rounded endpoints. We can implement this idea using rounded floating point arithmetic:

Definition 11 (floating point interval arithmetic)**.** For intervals $A = [a, b]$ and $B = [c, d]$ satisfying $0 < a \leq b$ and $0 < c \leq d$, and $n > 0$, define:

$$\begin{aligned} [a, b] \oplus [c, d] &:= [\mathrm{fl}^{\mathrm{down}}(a + c), \mathrm{fl}^{\mathrm{up}}(b + d)] \\ [a, b] \ominus [c, d] &:= [\mathrm{fl}^{\mathrm{down}}(a - d), \mathrm{fl}^{\mathrm{up}}(b - c)] \\ [a, b] \oslash n &:= [\mathrm{fl}^{\mathrm{down}}(a/n), \mathrm{fl}^{\mathrm{up}}(b/n)] \\ [a, b] \otimes [c, d] &:= [\mathrm{fl}^{\mathrm{down}}(ac), \mathrm{fl}^{\mathrm{up}}(bd)] \end{aligned}$$

We now explore this idea in pen-and-paper computations, in particular, to compute the exponential, in order to build understanding of how it works on a computer.

Example 10 (small sum)**.** consider evaluating the first few terms in the Taylor series of the exponential at $x = 1$ using interval arithmetic with half-precision F_{16} arithmetic. The first three terms are exact since all numbers involved are exactly floats, in particular if we evaluate $1 + x + x^2/2$ with $x = 1$ we get

$$1 + 1 + 1/2 \in 1 \oplus [1, 1] \oplus ([1, 1] \otimes [1, 1]) \oslash 2 = [5/2, 5/2]$$

Noting that

$$1/6 = (1/3)/2 = 2^{-3}(1.01010101\ldots)_2$$

we can extend the computation to another term:

$$\begin{aligned} 1 + 1 + 1/2 + 1/6 \in [5/2, 5/2] \oplus ([1, 1] \oslash 6) \\ &= [2(1.01)_2, 2(1.01)_2] \oplus 2^{-3}[(1.0101010101)_2, (1.0101010110)_2] \\ &= [\mathrm{fl}^{\mathrm{down}}(2(1.01010101010101)_2), \mathrm{fl}^{\mathrm{up}}(2(1.0101010101011)_2)] \\ &= [2(1.0101010101)_2, 2(1.0101010110)_2] \\ &= [2.666015625, 2.66796875] \end{aligned}$$

Example 11 (exponential with intervals)**.** Consider computing $\exp(x)$ for $0 \leq x \leq 1$ from the Taylor series approximation:

$$\exp(x) = \sum_{k=0}^{n} \frac{x^k}{k!} + \underbrace{\exp(t)\frac{x^{n+1}}{(n+1)!}}_{\delta_{x,n}}$$

where we can bound the error by (using the fact that $\mathrm{e} = 2.718\ldots \leq 3$, an inequality whose proof we leave as an exercise)

$$|\delta_{x,n}| \leq \frac{\exp(1)}{(n+1)!} \leq \frac{3}{(n+1)!}.$$

Put another way: $\delta_{x,n} \in \left[-\frac{3}{(n+1)!}, \frac{3}{(n+1)!}\right]$. We can use this to adjust the bounds derived from interval arithmetic for the interval arithmetic expression:

$$\exp(X) \subseteq \left(\bigoplus_{k=0}^{n} X \oslash k \oslash k!\right) \oplus \left[\mathrm{fl}^{\mathrm{down}}\left(-\frac{3}{(n+1)!}\right), \mathrm{fl}^{\mathrm{up}}\left(\frac{3}{(n+1)!}\right)\right]$$

For example, with $n = 3$ we have $|\delta_{1,2}| \leq 3/4! = 1/2^3$. Thus we can prove that:

$$\begin{aligned}
\mathrm{e} &= 1 + 1 + 1/2 + 1/6 + \delta_x \\
&\in [2(1.0101010101)_2, 2(1.0101010110)_2] \oplus [-1/2^3, 1/2^3] \\
&= [2(1.0100010101)_2, 2(1.0110010110)_2] = [2.541015625, 2.79296875]
\end{aligned}$$

II.3.1 Lab and Problem Sheet

In the lab we see how interval arithmetic with floating point numbers can be easily implemented by carefully setting the rounding mode. We also see how special functions like $\exp x$ can be implemented with rigorous bounds by combining computation with intervals with bounds on the Taylor series, turning the pen-and-paper example in these notes into an actual algorithm. As a fun example we compute $\exp 1 \equiv \mathrm{e}$ with rigorous bounds with as many as 1000 digits. This is a baby example of computer-assisted proofs, a concept that is increasingly important in pure mathematics and has been used for many important problems, including the proof of Kepler's conjecture, which was unsolved with non-computer-based techniques for almost 400 years.

In the problem sheet we complete the proofs of arithmetic with intervals and explore the rigorous computation of special functions like $\sin 1$ using interval arithmetic by-hand. This helps to build understanding on what the computer is doing in the labs.

Chapter III

Numerical Linear Algebra

Linear equations, especially ordinary and partial differential equations, are everywhere in applied mathematics, physics, and engineering. This is especially true in data science, eg., linear and polynomial regression for approximating data sets. Moreover, neural networks are built on top of linear algebra, with the basic layers being described in terms of matrix-vector products.

Numerical methods for linear equations invariably result in (finite-dimensional) linear systems that must be solved numerically on a computer: the dimensions of the problems are often in the 1000s, millions, or even billions. One would certainly not want to tackle that with Gaussian elimination by hand! In this chapter we discuss algorithms, and in particular matrix factorisations, that are computed using floating point operations.

In particular we discuss:

1. III.1 Structured Matrices: We discuss special structured matrices such as triangular and tridiagonal matrices, and how this structure can be used for better complexity matrix-vector multiplication and triangular solves.

2. III.2 LU and PLU Factorisations: We see that Gaussian elimination can be recast as computing a factorisation of a square matrix as a product of a lower and upper triangular matrix, potentially with a permutation matrix corresponding to the case where row pivoting is required.

3. III.3 Cholesky Factorisation: In the special case where the matrix is symmetric positive definite the LU factorisation has a special form. Hidden in this is an algorithm to prove positive definiteness.

4. III.4 Orthogonal Matrices: For rectangular problems (as in polynomial regression) we need an alternative to LU factorisation, built on orthogonal matrices. We discuss different types of orthogonal matrices, which will be used to simplify rectangular least squares problems.

5. III.5 QR Factorisation: We introduce an algorithm to compute a factorisation of a rectangular matrix as a product of an orthogonal and upper triangular matrix, thereby giving an algorirthm for solving least squares problems.

Here we are constructing underlying computational tools that are important in applications, such as solving differential equations and data regression, which we discuss later.

III.1 Structured Matrices

We have seen how algebraic operations (`+`, `-`, `*`, `/`) are defined exactly in terms of rounding ($\oplus$, $\ominus$, $\otimes$, $\oslash$) for floating point numbers. Now we see how this allows us to do (approximate) linear algebra operations on matrices.

A matrix can be stored in different formats, in particular it is important for large scale simulations that we take advantage of *sparsity*: if we know a matrix has entries that are guaranteed to be zero we can implement faster algorithms. We shall see that this comes up naturally in numerical methods for solving differential equations.

In particular, we will discuss some basic types of structure in matrices:

1. *Dense*: This can be considered unstructured, where we need to store all entries in a vector or matrix. Matrix-vector multiplication reduces directly to standard algebraic operations. Solving linear systems with dense matrices will be discussed later.

2. *Triangular*: If a matrix is upper or lower triangular, multiplication requires roughly half the number of operations. Crucially, we can solve linear systems involving triangular matrices using forward- or back-substitution.

3. *Banded*: If a matrix is zero apart from entries a fixed distance from the diagonal it is called banded and matrix-vector multiplication has a lower *complexity*: the number of operations scales linearly with the dimension (instead of quadratically). We discuss three cases: diagonal, tridiagonal and bidiagonal matrices.

Remark For those who took the first half of the module, there was an important emphasis on working with *linear operators* rather than *matrices*. That is, there was an emphasis on basis-independent mathematical techniques, which is critical for extension of results to infinite-dimensional spaces (which might not have a complete basis). However, in terms of practical computation we need to work with some representation of an operator and the most natural is a matrix. And indeed we will see in the next chapter how infinite-dimensional differential equations can be solved by reduction to finite-dimensional matrices. (Restricting attention to matrices is also important as some of the students have not taken the first half of the module.)

III.1.1 Dense matrices

A basic operation is matrix-vector multiplication. For a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$, or this can be relaxed to be a ring), consider a matrix and vector whose entries are in $\mathbb{F}$:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{a}_1 | \cdots | \boldsymbol{a}_n \end{bmatrix} \in \mathbb{F}^{m\times n}, \qquad \boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{F}^n.$$

where $\boldsymbol{a}_j = A\boldsymbol{e}_j \in \mathbb{F}^m$ are the columns of A. Recall the usual definition of matrix multiplication:

$$A\boldsymbol{x} := \begin{bmatrix} \sum_{j=1}^n a_{1j}x_j \\ \vdots \\ \sum_{j=1}^n a_{mj}x_j \end{bmatrix}.$$

When we are working with floating point numbers $A \in F^{m\times n}$ we obtain an approximation:

$$A\boldsymbol{x} \approx \begin{bmatrix} \bigoplus_{j=1}^n (a_{1j} \otimes x_j) \\ \vdots \\ \bigoplus_{j=1}^n (a_{mj} \otimes x_j) \end{bmatrix}.$$

This actually encodes an algorithm for computing the entries.

This algorithm uses $O(mn)$ floating point operations (see the appendix if you are unaware of Big-O notation, here our complexities are implicitly taken to be when m or n tends to ∞): each of the m entries consists of n multiplications and $n-1$ additions, hence we have a total of $2n-1 = O(n)$ operations per row for a total of $m(2n-1) = O(mn)$ operations. For a square matrix this is $O(n^2)$ operations which grows quadratically with n. In the problem sheet we see how the floating point error can be bounded in terms of norms, thus reducing the problem to a purely mathematical concept.

Sometimes there are multiple ways of implementing numerical algorithms. We have an alternative formula where we multiply by columns:

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n.$$

The floating point formula for this is exactly the same as the previous algorithm and the number of operations is the same. Just the order of operations has changed. Suprisingly, this latter version is significantly faster, which is explored in the lab.

Remark Floating point operations are sometimes called FLOPs, which are a standard measurement of speed of CPUs. However, FLOP sometimes uses an alternative definitions that combines an addition and multiplication as a single FLOP. In the lab we give an example showing that counting the precise number of operations is somewhat of a fools errand: algorithms such as the two approaches for matrix multiplication with the exact same number of operations can have wildly different speeds. We will therefore only be concerned with *complexity*; the asymptotic growth (Big-O) of operations as $n \to \infty$, in which case the difference between FLOPs and operations is immaterial.

III.1.2 Triangular matrices

The simplest sparsity case is being triangular: where all entries above or below the diagonal are zero. We consider upper and lower triangular matrices:

$$U = \begin{bmatrix} u_{11} & \cdots & u_{1n} \\ & \ddots & \vdots \\ & & u_{nn} \end{bmatrix}, \qquad L = \begin{bmatrix} \ell_{11} & & \\ \vdots & \ddots & \\ \ell_{n1} & \cdots & \ell_{nn} \end{bmatrix}.$$

Matrix multiplication can be modified to take advantage of the zero pattern of the matrix. Eg., if $L \in \mathbb{F}^{n\times n}$ is lower triangular we have:

$$L\boldsymbol{x} = \begin{bmatrix} \ell_{1,1}x_1 \\ \sum_{j=1}^2 \ell_{2j}x_j \\ \vdots \\ \sum_{j=1}^n \ell_{nj}x_j \end{bmatrix}.$$

When implemented in floating point this uses roughly half the number of multiplications: $1 + 2 + \ldots + n = n(n+1)/2$ multiplications. (It is also about twice as fast in practice.) The complexity is still quadratic: $O(n^2)$ operations.

Triangularity allows us to also invert systems using forward- or back-substitution. In particular if $\boldsymbol{x}$ solves $L\boldsymbol{x} = \boldsymbol{b}$ then we have:

$$x_k = \frac{b_k - \sum_{j=1}^{k-1} \ell_{kj} x_j}{\ell_{kk}}$$

Thus we can compute $x_1, x_2, \ldots, x_n$ in sequence.

III.1.3 Banded matrices

A *banded matrix* is zero off a prescribed number of diagonals. We call the number of (potentially) non-zero diagonals the *bandwidths*:

Definition 12 (bandwidths)**.** A matrix A has *lower-bandwidth* l if $a_{kj} = 0$ for all $k - j > l$ and *upper-bandwidth* u if $a_{kj} = 0$ for all $j - k > u$. We say that it has *strictly lower-bandwidth* l if it has lower-bandwidth l and there exists a j such that $a_{j+l,j} \neq 0$. We say that it has *strictly upper-bandwidth* u if it has upper-bandwidth u and there exists a k such that $a_{k,k+u} \neq 0$.

A square banded matrix has the sparsity pattern:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1,u+1} & & & \\ \vdots & a_{22} & \ddots & a_{2,u+2} & & \\ a_{l+1,1} & \ddots & \ddots & \ddots & \ddots & \\ & a_{l+2,2} & \ddots & \ddots & \ddots & a_{n-u,n} \\ & & \ddots & \ddots & \ddots & \vdots \\ & & & a_{n,n-l} & \cdots & a_{nn} \end{bmatrix}$$

A banded matrix has better complexity for matrix multiplication and solving linear systems: we can multiply square banded matrices in linear complexity: $O(n)$ operations. We consider two cases in particular (in addition to diagonal): bidiagonal and tridiagonal.

Definition 13 (Bidiagonal)**.** If a square matrix has bandwidths $(l, u) = (1, 0)$ it is *lower-bidiagonal* and if it has bandwidths $(l, u) = (0, 1)$ it is *upper-bidiagonal*.

For example, if

$$L = \begin{bmatrix} \ell_{11} & & & \\ \ell_{21} & \ell_{22} & & \\ & \ddots & \ddots & \\ & & \ell_{n,n-1} & \ell_{nn} \end{bmatrix}$$

then lower-bidiagonal multiplication becomes

$$L\boldsymbol{x} = \begin{bmatrix} \ell_{1,1} x_1 \\ \ell_{21} x_1 + \ell_{22} x_2 \\ \vdots \\ \ell_{n,n-1} x_{n-1} + \ell_{nn} x_n \end{bmatrix}.$$

This requires $O(1)$ operations per row (at most 2 multiplications and 1 addition) and hence the total is only $O(n)$ operations. A bidiagonal matrix is always triangular and we can also invert in $O(n)$ operations: if $L\boldsymbol{x} = \boldsymbol{b}$ then $x_1 = b_1/\ell_{11}$ and for $k = 2, \ldots, n$ we can compute

$$x_k = \frac{b_k - \ell_{k-1,k} x_{k-1}}{\ell_{kk}}.$$

Definition 14 (Tridiagonal)**.** If a square matrix has bandwidths $l = u = 1$ it is *tridiagonal*.

For example,

$$A = \begin{bmatrix} a_{11} & a_{12} & & & \\ a_{21} & a_{22} & a_{23} & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-1,n-2} & a_{n-1,n-1} & a_{n-1,n} \\ & & & a_{n,n-1} & a_{nn} \end{bmatrix}$$

is tridiagonal. Matrix multiplication is clearly $O(n)$ operations: each row has $O(1)$ non-zeros and there are n rows. But so is solving linear systems, which we shall see later.

III.1.4 Lab and Problem Sheet

In the lab we see how matrices and vectors can be constructed in Julia in multiple ways. We also see that there are different types to represent different structured matrices, including dense, diagonal, tridiagonal and bidiagonal, as well as a "lazy" transposes and range vectors (which can't be modified). Simple algorithms like computing matrix-vector multiplications are easily implemented, though there are some surprises: the order we access memory has a significant impact on the performance! Finally, we see how we can create our own types for representing structured matrices, in particular, we implement a type to represent an upper-tridiagonal matrix, including optimal complexity matrix-vector multiplication and linear solves.

The content of this section is largely explored in the lab. In the problem sheet, we look at a single problem investigating the effect of rounding error when using floating point arithmetic in matrix-vector multiplication. This results in a very nice formula for the error in terms of a matrix norm. This is important as it allows us to understand the impact of floating point errors in terms of fundamental mathematical concepts (like norms).

III.2 LU and PLU factorisations

One of the most fundamental problems in linear algebra is solving linear systems. For a field $\mathbb{F}$ (for us either $\mathbb{R}$ or $\mathbb{C}$), given invertible matrix $A \in \mathbb{F}^{n\times n}$ and vector $\boldsymbol{b} \in \mathbb{F}^n$, find $\boldsymbol{x} \in \mathbb{F}^n$ such that

$$A\boldsymbol{x} = \boldsymbol{b}.$$

This can of course be done via Gaussian elimination, using row swaps (or *pivoting*) if a zero is encountered on the diagonal, which can be viewed as an algorithm that can be implemented on a computer. However, a basic observation makes the practical implementation more straightforward and easier to apply to multiple right-hand sides, and connects with fundamental aspects in matrix analysis.

In particular, Gaussian elimination is equivalent to computing an *LU factorisation*:

$$A = LU$$

where L is lower triangular and U is upper triangular. Thus if we compute L and U we can deduce

$$\boldsymbol{x} = A^{-1}\boldsymbol{b} = U^{-1}L^{-1}\boldsymbol{b}$$

where $\boldsymbol{c} = L^{-1}\boldsymbol{b}$ can be computed using forward-substitution and $U^{-1}\boldsymbol{c}$ using back-substitution.

On the other hand, Gaussian elimination with pivoting (row-swapping) is equivalent to a *PLU factorisation*:

$$A = P^\top LU$$

where P is a permutation matrix (see appendix). Thus if we can compute P, L and U we can deduce

$$\boldsymbol{x} = A^{-1}\boldsymbol{b} = U^{-1}L^{-1}P\boldsymbol{b}$$

where multiplication by P is a simple swap of entries of $\boldsymbol{b}$ and L and U are again invertible via forward- and back-substitution.

III.2.1 Outer products

In what follows we will use outer products extensively:

Definition 15 (outer product)**.** Given $\boldsymbol{x} \in \mathbb{F}^m$ and $\boldsymbol{y} \in \mathbb{F}^n$ the *outer product* is:

$$\boldsymbol{x}\boldsymbol{y}^\top := [\boldsymbol{x}y_1|\cdots|\boldsymbol{x}y_n] = \begin{bmatrix} x_1y_1 & \cdots & x_1y_n \\ \vdots & \ddots & \vdots \\ x_my_1 & \cdots & x_my_n \end{bmatrix} \in \mathbb{F}^{m\times n}.$$

Note this is equivalent to matrix-matrix multiplication if we view $\boldsymbol{x}$ as a $m \times 1$ matrix and $\boldsymbol{y}^\top$ as a $1 \times n$ matrix.

Proposition 4 (rank-1)**.** *A matrix $A \in \mathbb{F}^{m\times n}$ has rank 1 if and only if there exists $\boldsymbol{x} \in \mathbb{F}^m$ and $\boldsymbol{y} \in \mathbb{F}^n$ such that*

$$A = \boldsymbol{x}\boldsymbol{y}^\top.$$

Proof If $A = \boldsymbol{x}\boldsymbol{y}^\top$ then all columns are multiples of $\boldsymbol{x}$, that is the column span has dimension 1. On the other hand, if A has rank-1 then its columns span a one-dimensional subspace: there exists $\boldsymbol{x} \in \mathbb{F}^m$

$$\operatorname{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n) = \{c\boldsymbol{x} : c \in \mathbb{F}\}.$$

Thus there exist $y_k \in \mathbb{F}$ such that $\boldsymbol{a}_k = y_k\boldsymbol{x}$ and we have

$$A = \boldsymbol{x}\underbrace{\begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}}_{\boldsymbol{y}^\top}.$$

■

III.2.2 LU factorisation

Gaussian elimination can be interpreted as an LU factorisation. Write a matrix $A \in \mathbb{F}^{n\times n}$ as follows:

$$A = \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ \boldsymbol{v} & K \end{bmatrix}$$

where $\alpha = a_{11}$, $\boldsymbol{v} = A[2:n,1]$ and $\boldsymbol{w} = A[1,2:n]$ (that is, $\boldsymbol{v} \in \mathbb{F}^{n-1}$ is a vector whose entries are the 2nd through last row of the first column of A whilst $\boldsymbol{w} \in \mathbb{F}^{n-1}$ is a vector containing the 2nd through last column of the first row of A). Gaussian elimination consists of taking the first row, dividing by α and subtracting from all other rows. That is equivalent to multiplying by a lower triangular matrix:

$$\underbrace{\begin{bmatrix} 1 & \\ -\boldsymbol{v}/\alpha & I \end{bmatrix}}_{L_1^{-1}} A = \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & K - \boldsymbol{v}\boldsymbol{w}^\top/\alpha \end{bmatrix}$$

where $A_2 := K - \boldsymbol{v}\boldsymbol{w}^\top/\alpha$ happens to be a rank-1 perturbation of K. We can write this another way:

$$A = \underbrace{\begin{bmatrix} 1 & \\ \boldsymbol{v}/\alpha & I \end{bmatrix}}_{L_1} \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & A_2 \end{bmatrix}$$

Now assume we continue this process and manage to deduce an LU factorisation $A_2 = \tilde{L}\tilde{U}$. Then

$$A = L_1 \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & \tilde{L}\tilde{U} \end{bmatrix} = \underbrace{L_1 \begin{bmatrix} 1 & \\ & \tilde{L} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & \tilde{U} \end{bmatrix}}_{U}$$

Note we can multiply through to find

$$L = \begin{bmatrix} 1 & \\ \boldsymbol{v}/\alpha & \tilde{L} \end{bmatrix}.$$

Noting that if $A \in \mathbb{F}^{1\times 1}$ then it has a trivial LU factorisation we can use the above construction to proceed recursively until we arrive at the trivial case.

Rather than a recursive definition, we can view the above as an inductive procedure:

$$\begin{aligned} A &= L_1 \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & A_2 \end{bmatrix} = L_1 \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & L_2 \begin{bmatrix} \alpha_2 & \boldsymbol{w}_2^\top \\ & A_3 \end{bmatrix} \end{bmatrix} \\ &= L_1 \begin{bmatrix} 1 & \\ & L_2 \end{bmatrix} \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & \begin{bmatrix} \alpha_2 & \boldsymbol{w}_2^\top \\ & L_3 \begin{bmatrix} \alpha_3 & \boldsymbol{w}_3^\top \\ & A_4 \end{bmatrix} \end{bmatrix} \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1 & \\ \boldsymbol{v}_1/\alpha_1 & \begin{bmatrix} 1 & \\ \boldsymbol{v}_2/\alpha_2 & \begin{bmatrix} 1 & \\ \boldsymbol{v}_3/\alpha_3 & \ddots \end{bmatrix} \end{bmatrix} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & \begin{bmatrix} \alpha_2 & \boldsymbol{w}_2^\top \\ & \begin{bmatrix} \alpha_3 & \boldsymbol{w}_3^\top \\ & \ddots \end{bmatrix} \end{bmatrix} \end{bmatrix}}_{U}. \end{aligned}$$

We can see this procedure clearer in the following example.

Example 12 (LU by-hand)**.** Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ 1 & 4 & 9 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & & \\ 2 & 1 & \\ 1 & & 1 \end{bmatrix}}_{L_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 6 \\ 0 & 3 & 8 \end{bmatrix}$$

In more detail, for $\alpha_1 := a_{11} = 1$, $\boldsymbol{v}_1 := A[2:3,1] = [2,1]^\top$, $\boldsymbol{w}_1 = A[1,2:3] = [1,1]^\top$ and

$$K_1 := A[2:3,2:3] = \begin{bmatrix} 4 & 8 \\ 4 & 9 \end{bmatrix}$$

we have

$$A_2 := K_1 - \boldsymbol{v}_1 \boldsymbol{w}_1^\top / \alpha_1 = \begin{bmatrix} 4 & 8 \\ 4 & 9 \end{bmatrix} - \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 3 & 8 \end{bmatrix}.$$

We then repeat the process and determine (with $\alpha_2 := A_2[1,1] = 2$, $\boldsymbol{v}_2 := A_2[2:2,1] = [3]$, $\boldsymbol{w}_2 := A_2[1,2:2] = [6]$ and $K_2 := A_2[2:2,2:2] = [8]$):

$$A_2 = \begin{bmatrix} 2 & 6 \\ 3 & 8 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \\ 3/2 & 1 \end{bmatrix}}_{L_2} \begin{bmatrix} 2 & 6 \\ & -1 \end{bmatrix}$$

The last "matrix" is 1 x 1 so we get the trivial factorisation:

$$A_3 := K_2 - \boldsymbol{v}_2 \boldsymbol{w}_2^\top / \alpha_2 = [-1] = \underbrace{[1]}_{L_3} [-1]$$

Putting everything together and placing the j-th column of L_j inside the j-th column of L we have

$$A = \underbrace{\begin{bmatrix} 1 & & \\ 2 & 1 & \\ 1 & 3/2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ & 2 & 6 \\ & & -1 \end{bmatrix}}_{U}$$

III.2.3 PLU factorisation

We learned in first year linear algebra that if a diagonal entry is zero when doing Gaussian elimination one has to *row pivot* (i.e., swap rows). For stability, in implementation one may wish to pivot even if the diagonal entry is nonzero: swap the largest in magnitude entry for the entry on the diagonal turns out to be significantly more stable than standard LU.

This is equivalent to a PLU factorisation. Here we use a *permutation matrix*, whose action on a vector permutes its entries, as discussed in the appendix. That is, consider a permutation which we identify with a vector $\sigma = [\sigma_1, \ldots, \sigma_n]$ containing the integers $1, \ldots, n$ exactly once. The permutation operator represents the action of permuting the entries in a vector:

$$P_\sigma(\boldsymbol{v}) := \boldsymbol{v}[\sigma] = \begin{bmatrix} v_{\sigma_1} \\ \vdots \\ v_{\sigma_n} \end{bmatrix}$$

This is a linear operator, and hence we can identify it with a *permutation matrix* $P_\sigma \in \mathbb{R}^{n \times n}$ (more precisely the entries of P_σ are either 1 or 0). Importantly, products of permutation

matrices are also permutation matrices and permutation matrices are orthogonal, that is, $P_\sigma^\top P_\sigma = I$, or in other words $P_\sigma^{-1} = P_\sigma^\top$.

Every invertible matrix has a PLU factorisation:

Theorem 5 (PLU)**.** *A matrix $A \in \mathbb{C}^{n\times n}$ is invertible if and only if it has a PLU factorisation:*

$$A = P^\top LU$$

where the diagonal of L are all equal to 1 and the diagonal of U are all non-zero, and P is a permutation matrix.

Proof

If we have a PLU factorisation of this form then L and U are invertible and hence the inverse is simply $A^{-1} = U^{-1}L^{-1}P$. Hence we consider the orther direction.

If $A \in \mathbb{C}^{1\times 1}$ we trivially have an LU factorisation $A = [1] * [a_{11}]$ as all 1×1 matrices are triangular. We now proceed by induction: assume all invertible matrices of lower dimension have a PLU factorisation. As A is invertible not all entries in the first column are zero. Therefore there exists a permutation P_1 so that $\alpha := (P_1A)[1,1] \neq 0$. Hence we write

$$P_1A = \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ \boldsymbol{v} & K \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \\ \boldsymbol{v}/\alpha & I \end{bmatrix}}_{L_1} \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & K - \boldsymbol{v}\boldsymbol{w}^\top/\alpha \end{bmatrix}$$

We deduce that $A_2 := K - \boldsymbol{v}\boldsymbol{w}^\top/\alpha$ is invertible because A and L_1 are invertible (Exercise).

By assumption we can write $A_2 = \tilde{P}^\top \tilde{L}\tilde{U}$. Thus we have:

$$\begin{aligned}
\underbrace{\begin{bmatrix} 1 & \\ & \tilde{P} \end{bmatrix} P_1}_{P} A &= \begin{bmatrix} 1 & \\ & \tilde{P} \end{bmatrix} L_1 \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & A_2 \end{bmatrix} = \begin{bmatrix} 1 & \\ & \tilde{P} \end{bmatrix} L_1 \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & \tilde{P}^\top \tilde{L}\tilde{U} \end{bmatrix} \\
&= \begin{bmatrix} 1 & \\ \tilde{P}\boldsymbol{v}/\alpha & \tilde{P} \end{bmatrix} \begin{bmatrix} 1 & \\ & \tilde{P}^\top \tilde{L} \end{bmatrix} \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & \tilde{U} \end{bmatrix} \\
&= \underbrace{\begin{bmatrix} 1 & \\ \tilde{P}\boldsymbol{v}/\alpha & \tilde{L} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & \tilde{U} \end{bmatrix}}_{U}.
\end{aligned}$$

■

For stability one uses the permutation that always puts the largest in magnitude entry in the top row, eg., by a simple swap with the row corresponding to the diagonal. One could try to justify this by considering floating point rounding, but actually there is no guaranteed this will produce accurate results and indeed in the lab we given an example of a "bad matrix" where large errors are still produced.

Again, the above recursive proof encodes an inductive procedure, which we see in the following example.

Example 13 (PLU by-hand)**.** Consider the matrix

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 6 & 2 \\ 1 & -1 & 5 \end{bmatrix}$$

The largest entry in the first column is 2 in the second row, hence we swap these rows then factor:

$$\underbrace{\begin{bmatrix} 0 & 1 & \\ 1 & 0 & \\ & & 1 \end{bmatrix}}_{P_1} A = \begin{bmatrix} 2 & 6 & 2 \\ 0 & 2 & 1 \\ 1 & -1 & 5 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 1/2 & 0 & 1 \end{bmatrix}}_{L_1} \begin{bmatrix} 2 & 6 & 2 \\ 0 & 2 & 1 \\ 0 & -4 & 4 \end{bmatrix}$$

Even though

$$A_2 := \begin{bmatrix} 2 & 1 \\ -4 & 4 \end{bmatrix}$$

is non-singular, we still permute the largest entry to the diagonal (this is helpful on a computer for stability). So we permute again to get:

$$\underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{P_2} A_2 = \begin{bmatrix} -4 & 4 \\ 2 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \\ -1/2 & 1 \end{bmatrix}}_{L_2} = \underbrace{\begin{bmatrix} -4 & 4 \\ & 3 \end{bmatrix}}_{U_2}$$

Putting it together we have

$$\begin{aligned} A &= P_1^\top L_1 \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & A_2 \end{bmatrix} = P_1^\top L_1 \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & P_2^\top L_2 U_2 \end{bmatrix} \\ &= P_1^\top \begin{bmatrix} 1 & \\ \boldsymbol{v}_1/\alpha_1 & I \end{bmatrix} \begin{bmatrix} 1 & \\ & P_2^\top L_2 \end{bmatrix} \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & U_2 \end{bmatrix} = P_1^\top \begin{bmatrix} 1 & \\ & P_2^\top \end{bmatrix} \begin{bmatrix} 1 & \\ P_2 \boldsymbol{v}_1/\alpha_1 & L_2 \end{bmatrix} \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & U_2 \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{P^\top} \underbrace{\begin{bmatrix} 1 & & \\ 1/2 & 1 & \\ 0 & -1/2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 6 & 2 \\ & -4 & 4 \\ & & 3 \end{bmatrix}}_{U}. \end{aligned}$$

III.2.4 Lab and Problem Sheet

In the problem sheet we see some pen-and-paper examples of LU and PLU factorisations, to emphasise the relationship with Gaussian elimination with or without pivoting. In the lab we focus on the practical usage of LU and PLU factorisations using the inbuilt commands, observing that the PLU factorisation is what is used in practice. We also investigate a special "bad matrix" where PLU factorisation surprisingly fails. This might raise the question why we use PLU factorisation despite the chance of faiulure, but as we see in he lab the probability of encountering such a matrix is extremely small. The biggest open problem in numerical linear algebra is proving this observation rigorously.

III.3 Cholesky factorisation

In the special case where A is a real square *symmetric positive definite* (SPD, that is $A \in \mathbb{R}^{n\times n}$ such that $A^\top = A$ and $\boldsymbol{x}^\top A\boldsymbol{x} > 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{x} \neq 0$) matrix the LU factorisation has a special form called the *Cholesky factorisation*:

$$A = LL^\top,$$

i.e., $U = L^\top$, but now L does not necessarily have 1s on the diagonal. This provides an algorithmic way to *prove* that a matrix is symmetric positive definite, and is roughly twice as fast as the LU factorisation to compute.

Definition 16 (positive definite)**.** A square matrix $A \in \mathbb{R}^{n\times n}$ is *positive definite* if for all $\boldsymbol{x} \in \mathbb{R}^n, x \neq 0$ we have

$$\boldsymbol{x}^\top A\boldsymbol{x} > 0$$

First we establish some basic properties of positive definite matrices:

Proposition 5 (conjugating positive definite)**.** *If $A \in \mathbb{R}^{n\times n}$ is positive definite and $V \in \mathbb{R}^{n\times n}$ is non-singular then*

$$V^\top AV$$

is positive definite.

Proof

For all $\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{x} \neq 0$, define $\boldsymbol{y} = V\boldsymbol{x} \neq 0$ (since V is non-singular). Thus we have

$$\boldsymbol{x}^\top V^\top AV\boldsymbol{x} = \boldsymbol{y}^\top A\boldsymbol{y} > 0.$$

■

Proposition 6 (diag positivity)**.** *If $A \in \mathbb{R}^{n\times n}$ is positive definite then its diagonal entries are positive: $a_{kk} > 0$.*

Proof

$$a_{kk} = \boldsymbol{e}_k^\top A\boldsymbol{e}_k > 0.$$

■

Lemma 4 (subslice positive definite)**.** *If $A \in \mathbb{R}^{n\times n}$ is positive definite then $A[2:n, 2:n] \in \mathbb{R}^{(n-1)\times(n-1)}$ is also positive definite.*

Proof For all $\boldsymbol{x} \in \mathbb{R}^{n-1}$, define $\boldsymbol{y} := [0, \boldsymbol{x}]$. Then we have

$$\boldsymbol{x}^\top A[2:n, 2:n]\boldsymbol{x} = \boldsymbol{y}^\top A\boldsymbol{y} > 0.$$

■

Here is the key result:

Theorem 6 (Cholesky and SPD)**.** *A matrix A is symmetric positive definite if and only if it has a Cholesky factorisation*

$$A = LL^\top$$

where L is lower triangular with positive diagonal entries.

Proof If A has a Cholesky factorisation it is symmetric ($A^\top = (LL^\top)^\top = A$) and for $\boldsymbol{x} \neq 0$ we have

$$\boldsymbol{x}^\top A\boldsymbol{x} = (L^\top\boldsymbol{x})^\top L^\top\boldsymbol{x} = \|L^\top\boldsymbol{x}\|^2 > 0$$

where we use the fact that L is non-singular.

For the other direction we will prove it by induction, with the 1×1 case being trivial. Assume all lower dimensional symmetric positive definite matrices have Cholesky decompositions. Modifying the LU factorisation slightly we write

$$A = \begin{bmatrix} \alpha & \boldsymbol{v}^\top \\ \boldsymbol{v} & K \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{\alpha} & \\ \frac{\boldsymbol{v}}{\sqrt{\alpha}} & I \end{bmatrix}}_{L_1} \begin{bmatrix} 1 & \\ & K - \frac{\boldsymbol{v}\boldsymbol{v}^\top}{\alpha} \end{bmatrix} \underbrace{\begin{bmatrix} \sqrt{\alpha} & \frac{\boldsymbol{v}^\top}{\sqrt{\alpha}} \\ & I \end{bmatrix}}_{L_1^\top}.$$

Note that $A_2 := K - \frac{\boldsymbol{v}\boldsymbol{v}^\top}{\alpha}$ is a subslice of $L_1^{-1}AL_1^{-\top}$, hence by combining the previous propositions is itself SPD. Thus we can write

$$A_2 = K - \frac{\boldsymbol{v}\boldsymbol{v}^\top}{\alpha} = L_2 L_2^\top$$

and hence $A = LL^\top$ for

$$L = L_1 \begin{bmatrix} 1 & \\ & L_2 \end{bmatrix} = \begin{bmatrix} \sqrt{\alpha} & \\ \frac{\boldsymbol{v}}{\sqrt{\alpha}} & L_2 \end{bmatrix}.$$

■

Example 14 (Cholesky by hand)**.** Consider the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

Then $\alpha_1 = 2$, $\boldsymbol{v}_1 = [1, 1, 1]$, and

$$A_2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}.$$

Continuing, we have $\alpha_2 = 3/2$, $\boldsymbol{v}_2 = [1/2, 1/2]$, and

$$A_3 = \frac{1}{2} \left(\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \right) = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

Next, $\alpha_3 = 4/3$, $\boldsymbol{v}_3 = [1/3]$, and

$$A_4 = [4/3 - 3/4 * (1/3)^2] = [5/4]$$

i.e. $\alpha_4 = 5/4$.

Thus we get

$$L = \begin{bmatrix} \sqrt{\alpha_1} & & & \\ \frac{\boldsymbol{v}_1[1]}{\sqrt{\alpha_1}} & \sqrt{\alpha_2} & & \\ \frac{\boldsymbol{v}_1[2]}{\sqrt{\alpha_1}} & \frac{\boldsymbol{v}_2[1]}{\sqrt{\alpha_2}} & \sqrt{\alpha_3} & \\ \frac{\boldsymbol{v}_1[3]}{\sqrt{\alpha_1}} & \frac{\boldsymbol{v}_2[2]}{\sqrt{\alpha_2}} & \frac{\boldsymbol{v}_3[1]}{\sqrt{\alpha_3}} & \sqrt{\alpha_4} \end{bmatrix} = \begin{bmatrix} \sqrt{2} & & & \\ \frac{1}{\sqrt{2}} & \sqrt{\frac{3}{2}} & & \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{3}} & \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{12}} & \frac{\sqrt{5}}{2} \end{bmatrix}$$

III.3.1 Lab and Problem Sheet

In the problem sheet we see how the Cholesky factorisation can be used to prove positive definiteness of both specific examples of small matrices, but also special families of arbitrarily large matrices. We also see the *reverse* Cholesky factorisation of the form $A = UU^\top$. In the lab we investigate the implementation of the Cholesky factorisation, and see how bandedness can be incorporated into the implementation to achieve better complexity.

III.4 Orthogonal and Unitary Matrices

PLU factorisations are an effective scheme for inverting systems, however, we saw in the lab that for very special matrices it can fail to be accurate. In the next two sections we introduce an alternative approach that is guaranteed to be stable: factorise a matrix as

$$A = QR$$

where Q is an orthogonal/unitary matrix and R is a *right-triangular matrix*, which for square matrices is another name for upper-triangular.

This factorisation is valid for rectangular matrices $A \in \mathbb{C}^{m\times n}$, where now *right-triangular* is a rectangular version of upper-triangular. For rectangular systems we can no longer solve linear systems of the form $A\boldsymbol{x} = \boldsymbol{b}$ (unless $\boldsymbol{b}$ lies in the column span of A) but instead we want to solve $A\boldsymbol{x} \approx \boldsymbol{b}$, where $\boldsymbol{x} \in \mathbb{C}^n$ and $\boldsymbol{b} \in \mathbb{C}^m$. More precisely, we can use a QR factorisation to solve *least squares* problems, find $\boldsymbol{x}$ that minimises the 2-norm:

$$\|A\boldsymbol{x} - \boldsymbol{b}\|.$$

Before we discuss the computation of a QR factorisation and its role in solving least-squares problems, we introduce orthogonal and unitary matrices. In particular we will discuss reflections and rotations, which can be used to represent more general orthogonal matrices.

Definition 17 (orthogonal/unitary matrix)**.** A square real matrix is *orthogonal* if its inverse is its transpose:

$$O(n) = \{Q \in \mathbb{R}^{n\times n} : Q^\top Q = I\}$$

A square complex matrix is *unitary* if its inverse is its adjoint:

$$U(n) = \{Q \in \mathbb{C}^{n\times n} : Q^\star Q = I\}.$$

Here the adjoint is the same as the conjugate-transpose: $Q^\star := \bar{Q}^\top$.

Note that $O(n) \subset U(n)$ as for real matrices $Q^\star = Q^\top$. Because in either case $Q^{-1} = Q^\star$ we also have $QQ^\star = I$ (which for real matrices is $QQ^\top = I$). These matrices are particularly important for numerical linear algebra for a number of reasons (we'll explore these properties in the problem sheets):

1. They are norm-preserving: for any vector $\boldsymbol{x} \in \mathbb{C}^n$ and $Q \in U(n)$ we have $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$ where $\|\boldsymbol{x}\|^2 := \sum_{k=1}^n x_k^2$ (i.e. the 2-norm).
2. All eigenvalues have absolute value equal to 1.
3. For $Q \in O(n)$, $\det Q = \pm 1$.
4. They are trivially invertible (just take the adjoint).
5. They are generally "stable": errors due to rounding when multiplying a vector by Q are controlled.
6. They are *normal matrices*: they commute with their adjoint ($QQ^\star = QQ^\star$).
7. Both $O(n)$ and $U(n)$ are groups, in particular, they are closed under multiplication. Though this is only approximately true

when floating point arithmetic is used.

On a computer there are multiple ways of representing orthogonal/unitary matrices. The obvious way is to store entries as a dense matrix, however, this is very inefficient. In the appendices we have seen permutation matrices, which are a special type of orthogonal matrices where we only store the order the entries are permuted as a vector.

More generally, we will use the group structure: represent general orthogonal/unitary matrices as products of simpler elements of the group. In particular we will use two building blocks:

1. *Rotations*: Rotations are equivalent to special orthogonal matrices $SO(2)$ and correspond to rotations in 2D.

2. *Reflections*: Reflections are elements of $U(n)$ that are defined in terms of a single unit vector $\boldsymbol{v} \in \mathbb{C}^n$ which is reflected.

We remark a related concept to orthogonal/unitary matrices are rectangular matrices with orthonormal columns, e.g.

$$U = [\boldsymbol{u}_1 | \cdots | \boldsymbol{u}_n] \in \mathbb{C}^{m \times n}$$

where $m \geq n$ such that $U^\star U = I_n$ (the $n \times n$ identity matrix). In the case where $m > n$ we must have $UU^\star \neq I_m$ as the rank of U is $n < m$.

III.4.1 Rotations

We begin with the definition of a group of orthogonal matrices:

Definition 18 (Special Orthogonal and Rotations)**.** *Special Orthogonal Matrices* are

$$SO(n) := \{Q \in O(n) | \det Q = 1\}.$$

And (simple) *rotations* are $SO(2)$.

In what follows we use the following for writing the angle of a vector:

Definition 19 (two-arg arctan)**.** The two-argument arctan function gives the angle θ through the point $[a, b]^\top$, i.e.,

$$\sqrt{a^2 + b^2} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

It can be defined in terms of the standard arctan as follows (this definition is non-examinable):

$$\operatorname{atan}(b, a) := \begin{cases} \operatorname{atan}\frac{b}{a} & a > 0 \\ \operatorname{atan}\frac{b}{a} + \pi & a < 0 \text{ and } b > 0 \\ \operatorname{atan}\frac{b}{a} - \pi & a < 0 \text{ and } b < 0 \\ \pi/2 & a = 0 \text{ and } b > 0 \\ -\pi/2 & a = 0 \text{ and } b < 0 \end{cases}$$

We show $SO(2)$ are exactly equivalent to standard rotations:

Proposition 7 (simple rotation)**.** *A 2×2* rotation matrix *through angle θ is*

$$Q_\theta := \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

We have $Q \in SO(2)$ if and only if $Q = Q_\theta$ for some $\theta \in \mathbb{R}$.

Proof

First assume Q_θ is of that form and write $c = \cos\theta$ and $s = \sin\theta$. Then we have

$$Q_\theta^\top Q_\theta = \begin{pmatrix} c & s \\ -s & c \end{pmatrix} \begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} c^2 + s^2 & 0 \\ 0 & c^2 + s^2 \end{pmatrix} = I$$

and $\det Q_\theta = c^2 + s^2 = 1$ hence $Q_\theta \in SO(2)$.

Now suppose $Q = [\boldsymbol{q}_1, \boldsymbol{q}_2] \in SO(2)$. $Q^\top Q = I$ tells us that its columns have norm 1, i.e. $\|\boldsymbol{q}_k\| = 1$, and are orthogonal. Write $\boldsymbol{q}_1 = [c, s]$ where we know $c = \cos\theta$ and $s = \sin\theta$ for $\theta = \operatorname{atan}(s, c)$. Since $\boldsymbol{q}_1 \cdot \boldsymbol{q}_2 = 0$ we can deduce $\boldsymbol{q}_2 = \pm[-s, c]$. The sign is positive as $\det Q = \pm(c^2 + s^2) = \pm 1$.

■

We can rotate an arbitrary vector in $\mathbb{R}^2$ to the unit axis using rotations, which are useful in linear algebra decompositions. Interestingly it only requires basic algebraic functions (no trigonometric functions):

Proposition 8 (rotation of a vector)**.** *The matrix*

$$Q = \frac{1}{\sqrt{a^2 + b^2}} \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$

is a rotation matrix ($Q \in SO(2)$) satisfying

$$Q \begin{bmatrix} a \\ b \end{bmatrix} = \sqrt{a^2 + b^2} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Proof

The last equation is trivial so the only question is that it is a rotation matrix. This follows immediately:

$$Q^\top Q = \frac{1}{a^2 + b^2} \begin{bmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{bmatrix} = I$$

and $\det Q = 1$.

■

Example 15 (rotating a vector)**.** Consider the vector

$$\boldsymbol{x} = \begin{bmatrix} -1 \\ -\sqrt{3} \end{bmatrix}.$$

We can use the proposition above to deduce the rotation matrix that rotates this vector to the positive real axis is:

$$\frac{1}{\sqrt{1+3}} \begin{bmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}.$$

Alternatively, we could determine the matrix by computing the angle of the vector via:

$$\theta = \operatorname{atan}(-\sqrt{3}, -1) = \operatorname{atan}(\sqrt{3}) - \pi = -\frac{2\pi}{3}.$$

We thus compute:

$$Q_{-\theta} = \begin{bmatrix} \cos(2\pi/3) & -\sin(2\pi/3) \\ \sin(2\pi/3) & \cos(2\pi/3) \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}.$$

More generally, we can consider rotations that operate on two entries of a vector at a time. This will be explored in the problem sheet/lab.

III.4.2 Reflections

In addition to rotations, another type of orthogonal/unitary matrix are reflections. These are specified by a single vector which is reflected, with every vector orthogonal to the defining vector left fixed.

Definition 20 (reflection matrix)**.** Given a unit vector $\boldsymbol{v} \in \mathbb{C}^n$ (satisfying $\|\boldsymbol{v}\| = 1$), define the corresponding *reflection matrix* as:

$$Q_{\boldsymbol{v}} := I - 2\boldsymbol{v}\boldsymbol{v}^\star$$

These are indeed reflections in the direction of $\boldsymbol{v}$. We can show this as follows:

Proposition 9 (Householder properties)**.** *$Q_{\boldsymbol{v}}$ satisfies:*

1. *Symmetry: $Q_{\boldsymbol{v}} = Q_{\boldsymbol{v}}^\star$.*
2. *Orthogonality: $Q_{\boldsymbol{v}} \in U(n)$.*
3. *The vector $\boldsymbol{v}$ is an eigenvector of $Q_{\boldsymbol{v}}$ with eigenvalue -1.*
4. *For the dimension $n-1$ space $W := \{\boldsymbol{w} : \boldsymbol{w}^\star\boldsymbol{v} = 0\}$, all vectors $\boldsymbol{w} \in W$ satisfy $Q_{\boldsymbol{v}}\boldsymbol{w} = \boldsymbol{w}$.*
5. *Not a rotation: $\det Q_{\boldsymbol{v}} = -1$ so $Q_{\boldsymbol{v}} \notin SO(n)$.*

Proof

Property 1 follows immediately. Property 2 follows from

$$Q_{\boldsymbol{v}}^\star Q_{\boldsymbol{v}} = Q_{\boldsymbol{v}}^2 = I - 4\boldsymbol{v}\boldsymbol{v}^\star + 4\boldsymbol{v}\boldsymbol{v}^\star\boldsymbol{v}\boldsymbol{v}^\star = I.$$

Property 3 follows since

$$Q_{\boldsymbol{v}}\boldsymbol{v} = \boldsymbol{v} - 2\boldsymbol{v}(\boldsymbol{v}^\star\boldsymbol{v}) = -\boldsymbol{v}.$$

Property 4 follows from:

$$Q_{\boldsymbol{v}}\boldsymbol{w} = \boldsymbol{w} - 2\boldsymbol{v}(\boldsymbol{w}^\star\boldsymbol{v}) = \boldsymbol{w}$$

Property 5 then follows: Property 4 tells us that 1 is an eigenvalue with multiplicity $n-1$. Since -1 is an eigenvalue with multiplicity 1, the determinant, which is product of the eigenvalues, is -1.

■

Example 16 (reflection through 2-vector)**.** Consider reflection through $\boldsymbol{x} = [1, 2]^\top$. We first need to normalise $\boldsymbol{x}$:

$$\boldsymbol{v} = \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

The reflection matrix is:

$$Q_{\boldsymbol{v}} = I - 2\boldsymbol{v}\boldsymbol{v}^\top = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} - \frac{2}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ -4 & -3 \end{bmatrix}$$

Indeed it is symmetric, and orthogonal. It sends $\boldsymbol{x}$ to $-\boldsymbol{x}$:

$$Q_{\boldsymbol{v}}\boldsymbol{x} = \frac{1}{5}\begin{bmatrix} 3 - 8 \\ -4 - 6 \end{bmatrix} = -\boldsymbol{x}$$

Any vector orthogonal to $\boldsymbol{x}$, like $\boldsymbol{y} = [-2, 1]^\top$, is unchanged:

$$Q_{\boldsymbol{v}}\boldsymbol{y} = \frac{1}{5}\begin{bmatrix} -6 - 4 \\ 8 - 3 \end{bmatrix} = \boldsymbol{y}$$

Note that *building* the matrix $Q_{\boldsymbol{v}}$ will be expensive ($O(n^2)$ operations), but we can *apply* $Q_{\boldsymbol{v}}$ to a vector in $O(n)$ operations using the expression:

$$Q_{\boldsymbol{v}}\boldsymbol{x} = \boldsymbol{x} - 2\boldsymbol{v}(\boldsymbol{v}^\star\boldsymbol{x}) = \boldsymbol{x} - 2\boldsymbol{v}(\boldsymbol{v} \cdot \boldsymbol{x}).$$

Householder reflections

Just as rotations can be used to rotate vectors to be aligned with coordinate axes, so can reflections, but in this case it works for vectors in $\mathbb{C}^n$, not just $\mathbb{R}^2$. We begin with the real case:

Definition 21 (Householder reflection, real case)**.** For a given vector $\boldsymbol{x} \in \mathbb{R}^n$, define the Householder reflection

$$Q_{\boldsymbol{x}}^{\pm,\mathrm{H}} := Q_{\boldsymbol{w}}$$

for $\boldsymbol{y} = \mp\|\boldsymbol{x}\|\boldsymbol{e}_1 + \boldsymbol{x}$ and $\boldsymbol{w} = \frac{\boldsymbol{y}}{\|\boldsymbol{y}\|}$. The default choice in sign is:

$$Q_{\boldsymbol{x}}^{\mathrm{H}} := Q_{\boldsymbol{x}}^{-\mathrm{sign}(x_1),\mathrm{H}}.$$

Lemma 5 (Householder reflection maps to axis)**.** *For* $\boldsymbol{x} \in \mathbb{R}^n$*,*

$$Q_{\boldsymbol{x}}^{\pm,\mathrm{H}}\boldsymbol{x} = \pm\|\boldsymbol{x}\|\boldsymbol{e}_1$$

Proof Note that

$$\begin{aligned} \|\boldsymbol{y}\|^2 &= 2\|\boldsymbol{x}\|^2 \mp 2\|\boldsymbol{x}\|x_1, \\ \boldsymbol{y}^\top\boldsymbol{x} &= \|\boldsymbol{x}\|^2 \mp \|\boldsymbol{x}\|x_1 \end{aligned}$$

where $x_1 = \boldsymbol{e}_1^\top\boldsymbol{x}$. Therefore:

$$Q_{\boldsymbol{x}}^{\pm,\mathrm{H}}\boldsymbol{x} = (I - 2\boldsymbol{w}\boldsymbol{w}^\top)\boldsymbol{x} = \boldsymbol{x} - 2\frac{\boldsymbol{y}\|\boldsymbol{x}\|}{\|\boldsymbol{y}\|^2}(\|\boldsymbol{x}\| \mp x_1) = \boldsymbol{x} - \boldsymbol{y} = \pm\|\boldsymbol{x}\|\boldsymbol{e}_1.$$

■

Remark Why do we choose the the opposite sign of x_1 for the default reflection? For stability, but we won't discuss this in more detail.

We can extend this definition for complex vectors. In this case the choice of the sign is delicate and so we only generalise the default choice using a complex-analogue of the sign fuunction.

Definition 22 (Householder reflection, complex case)**.** For a given vector $\boldsymbol{x} \in \mathbb{C}^n$, define the Householder reflection as

$$Q_{\boldsymbol{x}}^{\mathrm{H}} := Q_{\boldsymbol{w}}$$

for $\boldsymbol{y} = \mathrm{csign}(x_1)\|\boldsymbol{x}\|\boldsymbol{e}_1 + \boldsymbol{x}$ and $\boldsymbol{w} = \frac{\boldsymbol{y}}{\|\boldsymbol{y}\|}$, for $\mathrm{csign}(z) = \mathrm{e}^{\mathrm{i}\arg z}$.

Lemma 6 (Householder reflection maps to axis, complex case)**.** *For* $\boldsymbol{x} \in \mathbb{C}^n$,

$$Q_{\boldsymbol{x}}^{\mathrm{H}}\boldsymbol{x} = -\mathrm{csign}(x_1)\|\boldsymbol{x}\|\boldsymbol{e}_1$$

Proof Denote $\alpha := \mathrm{csign}(x_1)$. Note that $\bar{\alpha}x_1 = \mathrm{e}^{-\mathrm{i}\arg x_1}x_1 = |x_1|$. Now we have

$$\begin{aligned}
\|\boldsymbol{y}\|^2 &= (\alpha\|\boldsymbol{x}\|\boldsymbol{e}_1 + \boldsymbol{x})^\star(\alpha\|\boldsymbol{x}\|\boldsymbol{e}_1 + \boldsymbol{x}) = |\alpha|\|\boldsymbol{x}\|^2 + \|\boldsymbol{x}\|\alpha\bar{x}_1 + \bar{\alpha}x_1\|\boldsymbol{x}\| + \|\boldsymbol{x}\|^2 \\
&= 2\|\boldsymbol{x}\|^2 + 2|x_1|\|\boldsymbol{x}\| \\
\boldsymbol{y}^\star\boldsymbol{x} &= \bar{\alpha}x_1\|\boldsymbol{x}\| + \|\boldsymbol{x}\|^2 = \|\boldsymbol{x}\|^2 + |x_1|\|\boldsymbol{x}\|
\end{aligned}$$

Therefore:

$$Q_{\boldsymbol{x}}^{\mathrm{H}}\boldsymbol{x} = (I - 2\boldsymbol{w}\boldsymbol{w}^\star)\boldsymbol{x} = \boldsymbol{x} - 2\frac{\boldsymbol{y}}{\|\boldsymbol{y}\|^2}(\|\boldsymbol{x}\|^2 + |x_1|\|\boldsymbol{x}\|) = \boldsymbol{x} - \boldsymbol{y} = -\alpha\|\boldsymbol{x}\|\boldsymbol{e}_1.$$

■

III.4.3 Lab and Problem Sheet

In the problem sheet we explore some examples of constructing simple rotations and Householder reflections. We then prove basic properties of orthogonal matrices that we will use: (1) they preserve norm, (2) all eigenvalues are on the complex unit circle, (3) the determinant is ± 1, (4) they are *normal*, that is, they commute with their adjoint, and (5) using the spectral theorem (normal matrices are diagonalisable with unitary eigenvectors) we show that $Q = I$ if and only if all eigenvalues are 1.

In the lab we see how special types can be constructed to represent a rotation and products of rotations, as well as Householder reflections. Using special types is important to ensure that multiplication has optimal complexity: a Householder reflection can be applied in $O(n)$ operations, whereas using a standard dense matrix would take $O(n^2)$ operations.

III.5 QR Factorisation

Let $A \in \mathbb{C}^{m\times n}$ be a rectangular or square matrix such that $m \geq n$ (i.e. more rows then columns). In this section we consider two closely related factorisations, which can be viewed in some sense as a competitor to the PLU factorisation, but one that extends to rectangular matrices and allows for the solution of least squares problems.

The QR factorisation consists of writing A as a product of a (square) *orthogonal* and a (rectangular) *right triangular* matrix:

Definition 23 (QR factorisation)**.** The *QR factorisation* is

$$A = QR = \underbrace{\begin{bmatrix}\boldsymbol{q}_1 | \cdots | \boldsymbol{q}_m\end{bmatrix}}_{Q \in U(m)} \underbrace{\begin{bmatrix} \times & \cdots & \times \\ & \ddots & \vdots \\ & & \times \\ & & 0 \\ & & \vdots \\ & & 0 \end{bmatrix}}_{R \in \mathbb{C}^{m \times n}}$$

where Q is unitary (i.e., $Q \in U(m)$, satisfying $Q^\star Q = I$, with columns $\boldsymbol{q}_j \in \mathbb{C}^m$) and R is *right triangular*, which means it is only nonzero on or to the right of the diagonal ($r_{kj} = 0$ if $k > j$).

It is often more convenient to work with an alternative version known as the reduced QR factorisation, which is a product of a (rectangular) matrix with orthonormal columns and a (square) upper triangular matrix:

Definition 24 (Reduced QR factorisation)**.** The *reduced QR factorisation*

$$A = \hat{Q}\hat{R} = \underbrace{\begin{bmatrix}\boldsymbol{q}_1 | \cdots | \boldsymbol{q}_n\end{bmatrix}}_{\hat{Q} \in \mathbb{C}^{m \times n}} \underbrace{\begin{bmatrix} \times & \cdots & \times \\ & \ddots & \vdots \\ & & \times \end{bmatrix}}_{\hat{R} \in \mathbb{C}^{n \times n}}$$

where $\hat{Q}$ has orthonormal columns ($\hat{Q}^\star \hat{Q} = I$, $\boldsymbol{q}_j \in \mathbb{C}^m$) and $\hat{R}$ is upper triangular.

Note for a square matrix the reduced QR factorisation is equivalent to the QR factorisation, in which case R is *upper triangular*. The importance of these factorisation for square matrices is that their component pieces are easy to invert:

$$A = QR \qquad \Rightarrow \qquad A^{-1}\boldsymbol{b} = R^{-1}Q^\top \boldsymbol{b}$$

and we saw previously that triangular and orthogonal matrices are easy to invert when applied to a vector $\boldsymbol{b}$. On the other hand, in the rectangular case the QR factorisation contains within it the reduced QR factorisation:

$$A = QR = \begin{bmatrix}\hat{Q} | \boldsymbol{q}_{n+1} | \cdots | \boldsymbol{q}_m\end{bmatrix} \begin{bmatrix} \hat{R} \\ \boldsymbol{0}_{m-n \times n} \end{bmatrix} = \hat{Q}\hat{R}.$$

For rectangular matrices we will see that the QR factorisation leads to efficient solutions to the *least squares problem*: find $\boldsymbol{x}$ that minimizes the 2-norm $\|A\boldsymbol{x} - \boldsymbol{b}\|$.

In this section we discuss the following:

1. Reduced QR and Gram–Schmidt: The Reduced QR factorisation is equivalent to the Gram–Schmidt procedure, which you may have seen in 1st year or in the 1st half of the module.

2. Householder reflections and QR: Alternatively, the QR factorisation can be computed using a sequence of Householder reflections. This process mimics that of the LU factorisation, but using orthogonal matrices in-place of lower triangular ones to introduces zeros in column-by-column. This is a more accurate approach for computing QR factorisations than Gram–Schmidt as applying orthogonal matrices is stable.

3. QR and least squares: We discuss the QR factorisation and its usage in solving least squares problems.

III.5.1 Reduced QR and Gram–Schmidt

How do we compute the QR factorisation? We begin with a method you may have seen before in another guise. Write

$$A = \begin{bmatrix} \boldsymbol{a}_1 | \cdots | \boldsymbol{a}_n \end{bmatrix}$$

where $\boldsymbol{a}_k \in \mathbb{C}^m$ and assume they are linearly independent (A has full column rank).

Proposition 10 (Column spaces match)**.** *Suppose $A = \hat{Q}\hat{R}$ where $\hat{Q} = [\boldsymbol{q}_1 | \ldots | \boldsymbol{q}_n]$ has orthonormal columns and $\hat{R}$ is upper-triangular, and A has full rank. Then the first j columns of $\hat{Q}$ span the same space as the first j columns of A:*

$$span(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_j) = span(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_j).$$

Proof

Because A has full rank we know $\hat{R}$ is invertible, i.e. its diagonal entries do not vanish: $r_{jj} \neq 0$. If $\boldsymbol{v} \in \text{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_j)$ we have for $\boldsymbol{c} \in \mathbb{C}^j$

$$\boldsymbol{v} = \begin{bmatrix} \boldsymbol{a}_1 | \cdots | \boldsymbol{a}_j \end{bmatrix} \boldsymbol{c} = \begin{bmatrix} \boldsymbol{q}_1 | \cdots | \boldsymbol{q}_j \end{bmatrix} \hat{R}[1:j, 1:j]\boldsymbol{c} \in \text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_j)$$

while if $\boldsymbol{w} \in \text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_j)$ we have for $\boldsymbol{d} \in \mathbb{R}^j$

$$\boldsymbol{w} = \begin{bmatrix} \boldsymbol{q}_1 | \cdots | \boldsymbol{q}_j \end{bmatrix} \boldsymbol{d} = \begin{bmatrix} \boldsymbol{a}_1 | \cdots | \boldsymbol{a}_j \end{bmatrix} \hat{R}[1:j, 1:j]^{-1}\boldsymbol{d} \in \text{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_j).$$

■

It is possible to find $\hat{Q}$ and $\hat{R}$ using the *Gram–Schmidt algorithm.* We construct it column-by-column. For $j = 1, 2, \ldots, n$ define

$$\begin{aligned} \boldsymbol{v}_j &:= \boldsymbol{a}_j - \sum_{k=1}^{j-1} \underbrace{\boldsymbol{q}_k^\star \boldsymbol{a}_j}_{r_{kj}} \boldsymbol{q}_k, \\ r_{jj} &:= \|\boldsymbol{v}_j\|, \\ \boldsymbol{q}_j &:= \frac{\boldsymbol{v}_j}{r_{jj}}. \end{aligned}$$

Theorem (Gram–Schmidt and reduced QR) Define $\boldsymbol{q}_j$ and r_{kj} as above (with $r_{kj} = 0$ if $k > j$). Then a reduced QR factorisation is given by:

$$A = \underbrace{\begin{bmatrix} \boldsymbol{q}_1 | \cdots | \boldsymbol{q}_n \end{bmatrix}}_{\hat{Q} \in \mathbb{C}^{m \times n}} \underbrace{\begin{bmatrix} r_{11} & \cdots & r_{1n} \\ & \ddots & \vdots \\ & & r_{nn} \end{bmatrix}}_{\hat{R} \in \mathbb{C}^{n \times n}}$$

Proof

We first show that $\hat{Q}$ has orthonormal columns. Assume that $\boldsymbol{q}_\ell^\star \boldsymbol{q}_k = \delta_{\ell k}$ for $k, \ell < j$. For $\ell < j$ we then have

$$\boldsymbol{q}_\ell^\star \boldsymbol{v}_j = \boldsymbol{q}_\ell^\star \boldsymbol{a}_j - \sum_{k=1}^{j-1} \boldsymbol{q}_\ell^\star \boldsymbol{q}_k \boldsymbol{q}_k^\star \boldsymbol{a}_j = 0$$

hence $\boldsymbol{q}_\ell^\star \boldsymbol{q}_j = 0$ and indeed $\hat{Q}$ has orthonormal columns. Further: from the definition of $\boldsymbol{v}_j$ we find

$$\boldsymbol{a}_j = \boldsymbol{v}_j + \sum_{k=1}^{j-1} r_{kj} \boldsymbol{q}_k = \sum_{k=1}^{j} r_{kj} \boldsymbol{q}_k = \hat{Q}\hat{R}\boldsymbol{e}_j$$

■

III.5.2 Householder reflections and QR

As an alternative, we will consider using Householder reflections to introduce zeros below the diagonal. Thus, if Gram–Schmidt is a process of *triangular orthogonalisation* (using triangular matrices to orthogonalise), Householder reflections is a process of *orthogonal triangularisation* (using orthogonal matrices to triangularise).

Consider multiplication by the Householder reflection corresponding to the first column, that is, for

$$Q_1 := Q^{\mathrm{H}}_{\boldsymbol{a}_1},$$

consider

$$Q_1 A = \begin{bmatrix} \times & \times & \cdots & \times \\ & \times & \cdots & \times \\ & \vdots & \ddots & \vdots \\ & \times & \cdots & \times \end{bmatrix} = \begin{bmatrix} \alpha_1 & \boldsymbol{w}_1^\top \\ & A_2 \end{bmatrix}$$

where

$$\alpha_1 := -\mathrm{csign}(a_{11})\|\boldsymbol{a}_1\|, \boldsymbol{w}_1 = (Q_1 A)[1, 2:n] \qquad \text{and} \qquad A_2 = (Q_1 A)[2:m, 2:n],$$

where as before $\mathrm{csign}(z) := \mathrm{e}^{\mathrm{i}\arg z}$. That is, we have made the first column triangular. In terms of an algorithm, we then introduce zeros into the first column of A_2, leaving an A_3, and so-on. But we can wrap this iterative algorithm into a simple proof by induction, reminisicent of our proofs for the PLU and Cholesky factorisations:

Theorem 7 (QR)**.** *Every matrix $A \in \mathbb{C}^{m\times n}$ has a QR factorisation:*

$$A = QR$$

where $Q \in U(m)$ and $R \in \mathbb{C}^{m\times n}$ is right triangular.

Proof

First assume $m \geq n$. If $A = [\boldsymbol{a}_1] \in \mathbb{C}^{m\times 1}$ then we have for the Householder reflection $Q_1 = Q^{\mathrm{H}}_{\boldsymbol{a}_1}$

$$Q_1 A = \alpha \boldsymbol{e}_1$$

which is right triangular, where $\alpha = -\mathrm{csign}(a_{11})\|\boldsymbol{a}_1\|$. In other words

$$A = \underbrace{Q_1}_{Q} \underbrace{\alpha \boldsymbol{e}_1}_{R}.$$

For $n > 1$, assume every matrix with less columns than n has a QR factorisation. For $A = [\boldsymbol{a}_1 | \dots | \boldsymbol{a}_n] \in \mathbb{C}^{m\times n}$, let $Q_1 = Q^{\mathrm{H}}_{\boldsymbol{a}_1}$ so that

$$Q_1 A = \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & A_2 \end{bmatrix}.$$

By assumption $A_2 = \tilde{Q}\tilde{R}$. Thus we have (recalling that $Q_1^{-1} = Q_1^\star = Q_1$):

$$\begin{aligned} A &= Q_1 \begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & \tilde{Q}\tilde{R} \end{bmatrix} \\ &= \underbrace{Q_1 \begin{bmatrix} 1 & \\ & \tilde{Q} \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} \alpha & \boldsymbol{w}^\top \\ & \tilde{R} \end{bmatrix}}_{R}. \end{aligned}$$

If $m < n$, i.e., A has more columns then rows, write

$$A = \begin{bmatrix} \tilde{A} & B \end{bmatrix}$$

where $\tilde{A} \in \mathbb{C}^{m\times m}$. From above we know we can write $\tilde{A} = Q\tilde{R}$. We thus have

$$A = Q \underbrace{\begin{bmatrix} \tilde{R} & Q^\star B \end{bmatrix}}_{R}$$

where R is right triangular.

■

Example 17 (QR by hand)**.** We will now do an example by hand. Consider finding the QR factorisation where the diagonal of R is positive for the 4×3 matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

For the first column, since the entry $a_{11} > 0$ on a computer we would want to choose the Householder reflection that makes this negative, but in this case we want R to have a positive diagonal (partly because the numbers involved become very complicated otherwise!). So instead we choose the "wrong" sign and leave it positive. Since $\|\boldsymbol{a}_1\| = 2$ we have

$$\boldsymbol{y}_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} \Rightarrow \boldsymbol{w}_1 = \frac{\boldsymbol{y}_1}{\|\boldsymbol{y}_1\|} = \frac{1}{2}\begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}.$$

Hence

$$Q_1 := I - \frac{1}{2}\begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}\begin{bmatrix} -1 & -1 & -1 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$$

so that

$$Q_1 A = \begin{bmatrix} 2 & 1 & 0 \\ & 0 & 0 \\ & -1 & -1 \\ & 0 & -1 \end{bmatrix}$$

For the second column we have a zero entry so on a computer we can either send it to positive or negative sign, but in this case we are told to make it positive. Thus we have

$$\boldsymbol{y}_2 := [0, -1, 0] - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \boldsymbol{w}_2 = \frac{\boldsymbol{y}_2}{\|\boldsymbol{y}_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

Thus we have

$$Q_2 := I - \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

so that

$$\tilde{Q}_2 Q_1 A = \begin{bmatrix} 2 & 1 & 0 \\ & 1 & 1 \\ & & 0 \\ & & -1 \end{bmatrix}$$

The final vector is

$$\boldsymbol{y}_3 := \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \Rightarrow \boldsymbol{w}_3 = -\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Hence

$$Q_3 := I - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

so that

$$\tilde{Q}_3 \tilde{Q}_2 Q_1 A = \begin{bmatrix} 2 & 1 & 0 \\ & 1 & 1 \\ & & 1 \\ & & 0 \end{bmatrix} =: R$$

and

$$Q := Q_1 \tilde{Q}_2 \tilde{Q}_3 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 \\ -1 & 1 & -1 & -1 \end{bmatrix}.$$

III.5.3 QR and least squares

We consider rectangular matrices with more rows than columns. Given $A \in \mathbb{C}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{C}^m$, a least squares problem consists of finding a vector $\boldsymbol{x} \in \mathbb{C}^n$ that minimises the 2-norm: $\|A\boldsymbol{x} - \boldsymbol{b}\|$. There is a lot of theory around least squares, however, we focus on a simple computational aspect: we can solve least squares problems using the QR factorisation.

Theorem 8 (least squares via QR)**.** *Suppose $A \in \mathbb{C}^{m \times n}$ with $m \geq n$ has full rank and a QR factorisation $A = QR$ (which includes within it a reduced QR factorisation $A = \hat{Q}\hat{R}$). The vector*

$$\boldsymbol{x} = \hat{R}^{-1} \hat{Q}^\star \boldsymbol{b}$$

minimises $\|A\boldsymbol{x} - \boldsymbol{b}\|$.

Proof

The norm-preserving property ($\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$) of unitary matrices tells us

$$\|A\boldsymbol{x} - \boldsymbol{b}\| = \|QR\boldsymbol{x} - \boldsymbol{b}\| = \|Q(R\boldsymbol{x} - Q^\star\boldsymbol{b})\| = \|R\boldsymbol{x} - Q^\star\boldsymbol{b}\| = \left\| \begin{bmatrix} \hat{R} \\ \mathbf{0}_{m-n\times n} \end{bmatrix} \boldsymbol{x} - \begin{bmatrix} \hat{Q}^\star \\ \boldsymbol{q}_{n+1}^\star \\ \vdots \\ \boldsymbol{q}_m^\star \end{bmatrix} \boldsymbol{b} \right\|$$

Now note that the rows $k > n$ are independent of $\boldsymbol{x}$ and are a fixed contribution. Thus to minimise this norm it suffices to drop them and minimise:

$$\|\hat{R}\boldsymbol{x} - \hat{Q}^\star\boldsymbol{b}\|.$$

This norm is minimised if it is zero. Provided the column rank of A is full, $\hat{R}$ will be invertible.

III.5.4 Lab and Problem Sheet

In the lab we see how Householder reflection can be implemented by an iterative algorithm. To achieve optimal complexity it is important to take advantage of the structure of the Householder reflections, which is explored in the problems. For Tridiagonal matrices it is possible to use rotations in-place of reflections to upper-triangularise a matrix, in which case the resulting matrix is upper-tridiagonal. We investigate implementing this, giving an $O(n)$ algorithm for computing the QR factorisation of a tridiagonal matrix. (This can also be done with sparse Householder reflections, and thereby extended to general banded matrices). Finally, we look at the relationship between least squares and the QR factorisation.

In the problem sheet we see a simple example of a computing the QR factorisation using Householder reflections by-hand. (It's actually very hard to come up with examples where it's reasonable to do it by hand: it's more of a computer-based algorithm than one meant for pen-and-pencil calculations. Therefore this is not examinable but is there to help facilitate understanding.) We also explore some basic properties of QR factorisation such as uniqueness.

Chapter IV

Linear Algebra Applications

Numerical linear algebra underlies many numerical methods in applications, from simulating fluids, to understanding data and neural networks. Here we briefly investigate some applications, allowing us to go beyond our preliminary numerical algorithms from Chapter I.

1. IV.1 Polynomial Interpolation and Regression: Often in data science one needs to approximate data by a polynomial. We discuss polynomial interpolation and see how it can be used to compute integrals with higher accuracy than rectangular or trapezium rules. We also discuss regression, where more data is used than the degree of the polynomial, leading to a robust approach produced by solving a rectangular least squares problem.

2. IV.2 Singular Value Decomposition and Compression: We discuss the singular value decomposition, which gives the 2-norm of a matrix. We also discuss how it leads to an effective approach to compression, in particular, it allows for the computation of the best low-rank approximation to a matrix, in the sense that the matrix 2-norm is minimised.

IV.1 Polynomial Interpolation and Regression

Polynomial interpolation is the process of finding a polynomial that equals data at a precise set of points. In this section we see how an interpolant can be constructed by either solving a linear system involving the Vandermonde matrix, or directly in terms of the Lagrange basis for polynomials. We also investigate an application of polynomial interpolation to computing integrals, giving an alternative to the rectangular and triangular rules from the first chapter.

A more robust scheme that overcomes some of the issues with naive polynomial interpolation is *polynomial regression*, where we use more data than the degrees of freedom in the polynomial. We can determine such a polynomial by solving a *least squares problem*: instead of insisting that the polynomial matches data exactly, we find the polynomial whose samples at the points are as close as possible to the data, as measured in the 2-norm.

IV.1.1 Polynomial interpolation

Our prelimary goal is given a set of points and data at those points, usually samples of a function $f_j = f(x_j)$, find a polynomial that interpolates the data at the points:

Definition 25 (interpolatory polynomial)**.** Given *distinct* points $\boldsymbol{x} = [x_1, \ldots, x_n]^\top \in \mathbb{C}^n$ and *data* $\boldsymbol{f} = [f_1, \ldots, f_n]^\top \in \mathbb{C}^n$, a degree $n-1$ *interpolatory polynomial* $p(x)$ satisfies

$$p(x_j) = f_j$$

The easiest way to solve this problem is to invert the Vandermonde system:

Definition 26 (Vandermonde)**.** The *Vandermonde matrix* associated with $\boldsymbol{x} \in \mathbb{C}^m$ is the matrix

$$V_{\boldsymbol{x},n} := \begin{bmatrix} 1 & x_1 & \cdots & x_1^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & \cdots & x_m^{n-1} \end{bmatrix} \in \mathbb{C}^{m \times n}.$$

When it is clear from context we omit the subscripts $\boldsymbol{x}, n$.

Writing the coefficients of a polynomial

$$p(x) = \sum_{k=0}^{n-1} c_k x^k$$

as a vector $\boldsymbol{c} = [c_0, \ldots, c_{n-1}]^\top \in \mathbb{C}^n$, we note that V encodes the linear map from coefficients to values at a grid, that is,

$$V\boldsymbol{c} = \begin{bmatrix} c_0 + c_1 x_1 + \cdots + c_{n-1} x_1^{n-1} \\ \vdots \\ c_0 + c_1 x_m + \cdots + c_{n-1} x_m^{n-1} \end{bmatrix} = \begin{bmatrix} p(x_1) \\ \vdots \\ p(x_m) \end{bmatrix}.$$

In the square case (where $m = n$), the coefficients of an interpolatory polynomial are given by $\boldsymbol{c} = V^{-1}\boldsymbol{f}$, so that

$$\begin{bmatrix} p(x_1) \\ \vdots \\ p(x_n) \end{bmatrix} = V\boldsymbol{c} = VV^{-1}\boldsymbol{f} = \begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix}.$$

This inversion is justified by the following:

Proposition 11 (interpolatory polynomial uniqueness)**.** *Interpolatory polynomials are unique and therefore square Vandermonde matrices are invertible.*

Proof Suppose p and $\tilde{p}$ are both interpolatory polynomials of the same function. Then $p(x) - \tilde{p}(x)$ vanishes at n distinct points x_j. By the fundamental theorem of algebra it must be zero, i.e., $p = \tilde{p}$.

For the second part, if $V\boldsymbol{c} = 0$ for $\boldsymbol{c} = [c_0, \ldots, c_{n-1}]^\top \in \mathbb{C}^n$ then for $q(x) = c_0 + \cdots + c_{n-1}x^{n-1}$ we have

$$q(x_j) = \boldsymbol{e}_j^\top V\boldsymbol{c} = 0$$

hence q vanishes at n distinct points and is therefore 0, i.e., $\boldsymbol{c} = 0$.

■

We can invert square Vandermonde matrix numerically in $O(n^3)$ operations using the PLU factorisation. But it turns out we can also construct the interpolatory polynomial directly, and evaluate the polynomial in only $O(n^2)$ operations. We will use the following polynomials which equal 1 at one grid point and zero at the others:

Definition 27 (Lagrange basis polynomial)**.** The *Lagrange basis polynomial* is defined as

$$\ell_k(x) := \prod_{j \neq k} \frac{x - x_j}{x_k - x_j} = \frac{(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}$$

Plugging in the grid points verifies that $\ell_k(x_j) = \delta_{kj}$.

We can use the Lagrange basis to directly construct the interpolatory polynomial:

Theorem 9 (Lagrange interpolation)**.** *The unique interpolation polynomial is:*

$$p(x) = f_1 \ell_1(x) + \cdots + f_n \ell_n(x)$$

Proof It follows from inspection:

$$p(x_j) = \sum_{k=1}^{n} f_k \ell_k(x_j) = f_j.$$

■

Example 18 (interpolating an exponential)**.** We can interpolate $\exp(x)$ at the points $0, 1, 2$. That is, our data is $\boldsymbol{f} = [1, \mathrm{e}, \mathrm{e}^2]^\top$ and the interpolatory polynomial is

$$\begin{aligned} p(x) &= \ell_1(x) + \mathrm{e}\ell_2(x) + \mathrm{e}^2 \ell_3(x) = \frac{(x-1)(x-2)}{(-1)(-2)} + \mathrm{e}\frac{x(x-2)}{(-1)} + \mathrm{e}^2 \frac{x(x-1)}{2} \\ &= (1/2 - \mathrm{e} + \mathrm{e}^2/2)x^2 + (-3/2 + 2\mathrm{e} - \mathrm{e}^2/2)x + 1 \end{aligned}$$

Remark Interpolating at evenly spaced points is a really *bad* idea as it is inherently ill-conditioned. The lab explores this issue experimentally. Another serious issue is that monomials are a horrible basis for interpolation. This is intuitive: when n is large x^n is basically zero near the origin and hence x_j^n numerically lose linear independence, that is, on a computer they appear to be linearly dependent (up to rounding errors). Use alternative sets of points and bases entirely overcomes this issue.

IV.1.2 Interpolatory quadrature rules

Interpolation leads naturally to quadrature rules where one integrates the interpolatory polynomial exactly. This can be viewed as an extension of one-panel Rectangular Rules (which are degree 0 interpolants at a single point) and Trapezium Rules (which are degree 1 interpolants at two points). Using the Lagrange basis for interpolation we can write general interpolatory quadrature rules as a simple weighted sum:

Definition 28 (interpolatory quadrature rule)**.** Given a set of points $\boldsymbol{x} = [x_1, \ldots, x_n]^\top$ the interpolatory quadrature rule is:

$$\Sigma_n^{w,\boldsymbol{x}}[f] := \sum_{j=1}^{n} w_j f(x_j)$$

where

$$w_j := \int_a^b \ell_j(x) w(x) \mathrm{d}x.$$

The convergence of such a scheme is explored in the lab. But an important feature is that it is exact for all low degree polynomials:

Proposition 12 (interpolatory quadrature is exact for polynomials)**.** *Interpolatory quadrature is exact for all degree $n-1$ polynomials p:*

$$\int_a^b p(x) w(x) \mathrm{d}x = \Sigma_n^{w,\boldsymbol{x}}[p]$$

Proof The result follows since, by uniqueness of interpolatory polynomial, if p is a polynomial then

$$p(x) = \sum_{j=1}^n p(x_j) \ell_j(x)$$

Hence

$$\int_a^b p(x) w(x) \mathrm{d}x = \sum_{j=1}^n p(x_j) \int_a^b \ell_j(x) w(x) \mathrm{d}x = \Sigma_n^{w,\boldsymbol{x}}[p].$$

■

Example 19 (3-point interpolatory quadrature)**.** We find the interpolatory quadrature rule for $w(x) = 1$ on $[0,1]$ with points $[x_1, x_2, x_3] = [0, 1/4, 1]$. We have:

$$\begin{aligned} w_1 &= \int_0^1 w(x) \ell_1(x) \mathrm{d}x = \int_0^1 \frac{(x-1/4)(x-1)}{(-1/4)(-1)} \mathrm{d}x = -1/6 \\ w_2 &= \int_0^1 w(x) \ell_2(x) \mathrm{d}x = \int_0^1 \frac{x(x-1)}{(1/4)(-3/4)} \mathrm{d}x = 8/9 \\ w_3 &= \int_0^1 w(x) \ell_3(x) \mathrm{d}x = \int_0^1 \frac{x(x-1/4)}{3/4} \mathrm{d}x = 5/18 \end{aligned}$$

That is we have

$$\Sigma_n^{w,\boldsymbol{x}}[f] = -\frac{f(0)}{6} + \frac{8f(1/4)}{9} + \frac{5f(1)}{18}.$$

This is indeed exact for polynomials up to degree 2 (and no more):

$$\Sigma_n^{w,\boldsymbol{x}}[1] = 1, \Sigma_n^{w,\boldsymbol{x}}[x] = 1/2, \Sigma_n^{w,\boldsymbol{x}}[x^2] = 1/3, \Sigma_n^{w,\boldsymbol{x}}[x^3] = 7/24 \neq 1/4.$$

IV.1.3 Polynomial regression

In many settings interpolation is not an accurate or appropriate tool. Data is often on an evenly spaced grid in which case (as seen in the lab) interpolation breaks down catastrophically. Or the data is noisy and one ends up over resolving: approximating the noise rather than the signal. A simple solution is *polynomial regression*: use more sample points than the degrees of freedom in the polynomial. The special case of an affine polynomial is called *linear regression.*

More precisely, for $\boldsymbol{x} \in \mathbb{C}^m$ and for $n < m$ we want to find a degree $n-1$ polynomial

$$p(x) = \sum_{k=0}^{n-1} c_k x^k$$

such that

$$\begin{bmatrix} p(x_1) \\ \vdots \\ p(x_m) \end{bmatrix} \approx \underbrace{\begin{bmatrix} f_1 \\ \vdots \\ f_m \end{bmatrix}}_{\boldsymbol{f}}.$$

Mapping between coefficients $\boldsymbol{c} \in \mathbb{C}^n$ to polynomial values on a grid can be accomplished via rectangular Vandermonde matrices. In particular, our goal is to choose $\boldsymbol{c} \in \mathbb{C}^n$ so that

$$V\boldsymbol{c} = \begin{bmatrix} p(x_1) \\ \vdots \\ p(x_m) \end{bmatrix} \approx \boldsymbol{f}.$$

We do so by solving the *least squares* system: given $V \in \mathbb{C}^{m\times n}$ and $\boldsymbol{f} \in \mathbb{C}^m$ we want to find $\boldsymbol{c} \in \mathbb{C}^n$ such that

$$\|V\boldsymbol{c} - \boldsymbol{f}\|$$

is minimal. Note interpolation is a special case where this norm is precisely zero (which is indeed minimal), but in general this norm may be rather large. We will discuss the numerical solution of least squares problems in the next few sections.

Remark Using regression instead of interpolation can overcome the issues with evenly spaced grids. However, the monomial basis is still very problematic.

IV.1.4 Lab and Problem Sheet

In the lab we explore the practical implementation of interpolation, but observe that issues are encountered with interpolation at evenly spaced points. Using special points that are defined trigonometrically to cluster at the endpoints will over come this issue. We also explore interpolatory rules, finding that we can achieve much higher accuracy than the simple rectangular and trapezium rules introduced in the first week. We also see how least squares can provide a means of overcoming issues with interpolation. Later in the module, we will see orthogonal polynomials provide a more robust basis for very accurate and reliable interpolation.

In the problem sheet we see some simple examples of interpolation, regression, and the explicit calculation of interpolatory quadrature rules. We also discuss the relationship between the Vandermonde matrix and its LU factorisation.

IV.2 Singular Value Decomposition and Matrix Compression

In the previous section we saw an application of least squares to regression, where the best 2-norm fit to a vector of function samples was computed. But what if the data is a matrix, eg. corresponding to a 2D function? Here we consider finding the best approximation to a matrix in the 2-norm sense by by a matrix with a lower rank. This concept has numerous applications in compressing matrices, including Principle Component Analysis (PCA) and Machine Learning (where one might want to compress the “weight” matrices to minimise degrees of freedom).

We will use induced matrix norms (see appendix), in particular the 2-norm of a matrix $A \in \mathbb{C}^{m\times n}$ is defined via

$$\|A\| := \sup_{\boldsymbol{v}\in\mathbb{C}^m : \|\boldsymbol{v}\|_X = 1} \|A\boldsymbol{v}\| = \sup_{\boldsymbol{x}\in\mathbb{C}^m} \frac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|}$$

The matrix 1- and ∞-norms have simple definitions in terms of column/row sums. On the other hand, the 2-norm has no simple formula.

In order to define the 2-norm we will introduce the *Singular Value Decomposition (SVD)*: a matrix factorisation that encodes how much a matrix "stretches" vectors:

Definition 29 (singular value decomposition)**.** For $A \in \mathbb{C}^{m\times n}$ with rank $r > 0$, the *(reduced) singular value decomposition (SVD)* is

$$A = U\Sigma V^\star$$

where $U \in \mathbb{C}^{m\times r}$ and $V \in \mathbb{C}^{n\times r}$ have orthonormal columns and $\Sigma \in \mathbb{R}^{r\times r}$ is diagonal whose diagonal entries, which which we call *singular values*, are all positive and non-increasing: $\sigma_1 \geq \cdots \geq \sigma_r > 0$. The *full singular value decomposition (SVD)* is

$$A = \tilde{U}\tilde{\Sigma}\tilde{V}^\star$$

where $\tilde{U} \in U(m)$ and $\tilde{V} \in U(n)$ are unitary matrices and $\tilde{\Sigma} \in \mathbb{R}^{m\times n}$ has only diagonal non-zero entries, i.e., if $m > n$,

$$\tilde{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \\ & & 0 \\ & & \vdots \\ & & 0 \end{bmatrix}$$

and if $m < n$,

$$\tilde{\Sigma} = \begin{bmatrix} \sigma_1 & & & & & \\ & \ddots & & & & \\ & & \sigma_m & 0 & \cdots & 0 \end{bmatrix}$$

where $\sigma_k = 0$ if $k > r$.

In particular, we discuss:

1. Existence of the SVD: we show that an SVD exists by relating it to the eigenvalue Decomposition of $A^\star A$ and $AA^\star$.
2. 2-norm and SVD: the 2-norm of a matrix is equal to the largest singular value σ_1.
3. Best rank-k approximation and compression: the best approximation of a matrix by a smaller rank matrix can be constructed using the SVD, which gives an effective way to compress matrices.

What we do not discuss is computation of the SVD. There are reliable and efficient iterative algorithms for computing the SVD, similar to computing eigen-decompositions, but this is beyond the scope of this module.

IV.2.1 Existence

To show the SVD exists we first establish some properties of a *Gram matrix* ($A^\star A$), which has within its eigendecomposition part of the SVD. The Gram matrix is best viewed as the matrix of inner products of the columns of a matrix: if $A = \begin{bmatrix} \boldsymbol{a}_1 | \cdots | \boldsymbol{a}_n \end{bmatrix} \in \mathbb{C}^{m\times n}$ then the Gram matrix is the Hermitian matrix

$$A^\star A = \begin{bmatrix} \boldsymbol{a}_1^\star \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_1^\star \boldsymbol{a}_n \\ \vdots & \ddots & \vdots \\ \boldsymbol{a}_n^\star \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_n^\star \boldsymbol{a}_n \end{bmatrix} \in \mathbb{C}^{n\times n}.$$

We first establish that the kernels match:

Proposition 13 (Gram matrix kernel)**.** *The kernel of A equals the kernel of $A^\star A$.*

Proof If $\boldsymbol{x} \in \ker(A)$, i.e., $A\boldsymbol{x} = 0$, then clearly $A^\star A\boldsymbol{x} = 0 = A^\star 0 = 0$. On the other hand, if $\boldsymbol{x} \in \ker(A^\star A)$ so that $A^\star A\boldsymbol{x} = 0$ then we have

$$0 = \boldsymbol{x}^\star A^\star A\boldsymbol{x} = \|A\boldsymbol{x}\|^2$$

which means $A\boldsymbol{x} = 0$ and $\boldsymbol{x} \in \ker(A)$. ■

As mentioned in PS6, the spectral theorem states that any normal matrix is unitarily diagonalisable: if A is normal then $A = Q\Lambda Q^\star$ where $Q \in U(n)$ and Λ is diagonal. In the special case where A is symmetric/Hermitian you would have seen a proof of this in first year. We can use this to ensure that the Gram matrix is diagonalisable:

Proposition 14 (Gram matrix diagonalisation)**.** *The Gram matrix satisfies*

$$A^\star A = Q\Lambda Q^\star \in \mathbb{C}^{n\times n}$$

is a Hermitian matrix where $Q \in U(n)$ and the eigenvalues λ_k are real and non-negative. If $A \in \mathbb{R}^{m\times n}$ then we can choose $Q \in O(n)$.

Proof $A^\star A$ is Hermitian so we appeal to the spectral theorem for the existence of the decomposition. To see that the eigenvalues are real and positive note for the corresponding (orthonormal) eigenvector $\boldsymbol{q}_k$ we have

$$\lambda_k = \lambda_k \boldsymbol{q}_k^\star \boldsymbol{q}_k = \boldsymbol{q}_k^\star A^\star A\boldsymbol{q}_k = \|A\boldsymbol{q}_k\|^2 \geq 0.$$

The fact that real A implies $Q \in O(n)$ (i.e. has real entries) follows since if $\boldsymbol{q}_k \in \mathbb{C}^n$ is an eigenvector than so is $\Re\boldsymbol{q}_k$:

$$A^\top A\Re\boldsymbol{q}_k = \Re A^\top A\boldsymbol{q}_k = \Re\lambda_k\boldsymbol{q}_k = \lambda_k\Re\boldsymbol{q}_k.$$

■

This connection allows us to prove existence:

Theorem 10 (SVD existence)**.** *Every $A \in \mathbb{C}^{m\times n}$ has an SVD.*

Proof Consider

$$A^\star A = Q\Lambda Q^\star.$$

Assume (as usual) that the eigenvalues are sorted in decreasing modulus, and so $\lambda_1, \ldots, \lambda_r$ are an enumeration of the non-zero eigenvalues and

$$V := \left[\boldsymbol{q}_1 | \cdots | \boldsymbol{q}_r\right]$$

the corresponding (orthonormal) eigenvectors, with the columns

$$K = \left[\boldsymbol{q}_{r+1} | \cdots | \boldsymbol{q}_n\right]$$

spanning the kernel of $A^\star A$ (and hence A). Define

$$\Sigma := \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_r} \end{bmatrix}$$

Now define $U := AV\Sigma^{-1}$. Since $A^\star AV = V\Sigma^2$ we can verify that U has orthonormal columns:

$$U^\star U = \Sigma^{-1} V^\star A^\star A V \Sigma^{-1} = I.$$

Thus we have

$$U\Sigma V^\star = AVV^\star = A \underbrace{\left[V|K\right]}_{Q} \underbrace{\begin{bmatrix} V^\star \\ K^\star \end{bmatrix}}_{Q^\star} = A$$

since $QQ^\star = I$, and where we use the fact that $AK = 0$ so that concatenating K does not change the value.

■

IV.2.2 The matrix 2-norm

Somewhat surprisingly, the 2-norm for matrices does not have a simple formula but instead, as we shall show, can be defined in terms of the singular values. We begin with two cases where we do have simple formulæ:

Proposition 15 (diagonal/orthogonal column 2-norms)**.** *If Λ is diagonal with entries λ_k then $\|\Lambda\| = \max_k |\lambda_k|$. If U has orthonormal columns then $\|U\| = \|U^\star\| = 1$.*

Proof

The first property follows from

$$\|\Lambda \boldsymbol{x}\| = \sqrt{\sum_{k=1}^{n} |\lambda_k|^2 |x_k|^2} \leq \max_k |\lambda_k| \|\boldsymbol{x}\|$$

hence $\|\Lambda\| \leq \max_k |\lambda_k|$. If k is an index where this maximum is achieved we see this upper bound is achieved by $\|\Lambda \boldsymbol{e}_k\| = |\lambda_k|$.

U with orthonormal columns has unit norm because:

$$\|U\boldsymbol{x}\|^2 = \boldsymbol{x}^\star \underbrace{U^\star U}_{I} \boldsymbol{x} = \|\boldsymbol{x}\|^2.$$

To show $U^\star$ also has unit norm is more challenging. We first note from the spectral theorem we know we can diagonalise

$$UU^\star = Q\Lambda Q^\star$$

and therefore

$$\|UU^\star\| \leq \|Q\|\|\Lambda\|\|Q^\star\| = \|\Lambda\|.$$

We further note this is a projection, that is, it equals its square:

$$(UU^\star)^2 = U\underbrace{U^\star U}_{I}U^\star = UU^\star.$$

Thus the eigenvalues are either 0 or 1 since:

$$\lambda_j = \lambda_j \boldsymbol{q}_j^\star \boldsymbol{q}_j = \boldsymbol{q}_j^\star UU^\star \boldsymbol{q}_j = \boldsymbol{q}_j^\star (UU^\star)^2 \boldsymbol{q}_j = \lambda_j^2$$

and thus $\|UU^\star\| \leq \|\Lambda\| \leq 1$. We thus have (using the fact that multiplication by U preserves norm)

$$\|U^\star \boldsymbol{x}\| = \|UU^\star \boldsymbol{x}\| \leq \|UU^\star\|\|\boldsymbol{x}\| \leq \|\boldsymbol{x}\|$$

hence $\|U^\star\| \leq 1$. But this bound is achieved using any column of U since, eg. for $\boldsymbol{u}_1 = U\boldsymbol{e}_1$ we have

$$\sup_{\|\boldsymbol{v}\|=1} \|U^\star \boldsymbol{v}\| \geq \|U^\star \boldsymbol{u}_1\| = \|\boldsymbol{e}_1\| = 1.$$

■

These two facts allow us to deduce the 2-norm from the SVD of a matrix:

Corollary 3 (singular values and norm)**.**

$$\|A\| = \sigma_1$$

and if $A \in \mathbb{C}^{n\times n}$ is invertible, then

$$\|A^{-1}\| = \sigma_n^{-1}$$

Proof

First we establish the upper-bound using the fact for induced norms that $\|AB\| \leq \|A\|\|B\|$ (see appendix):

$$\|A\| \leq \|U\|\|\Sigma\|\|V^\star\| = \|\Sigma\| = \sigma_1$$

This is attained using the first right singular vector: for $\boldsymbol{v}_1 = V\boldsymbol{e}_1$ we have

$$\|A\boldsymbol{v}_1\| = \|\Sigma V^\star \boldsymbol{v}_1\| = \|\Sigma \boldsymbol{e}_1\| = \sigma_1$$

The inverse result follows since the inverse has SVD

$$A^{-1} = V\Sigma^{-1}U^\star = (VW)(W\Sigma^{-1}W)(WU)^\star$$

is the SVD of A^{-1}, i.e. $VW \in U(n)$ are the left singular vectors and WU are the right singular vectors, where

$$W := P_\sigma = \begin{bmatrix} & & 1 \\ & \iddots & \\ 1 & & \end{bmatrix}$$

is the permutation that reverses the entries of a vector, that is, σ has Cauchy notation

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ n & n-1 & \cdots & 1 \end{pmatrix}.$$

■

IV.2.3 Best rank-k approximation and compression

One of the main usages for SVDs is low-rank compression: approximating a (possibly full rank) matrix $A \in \mathbb{C}^{m\times n}$ by a matrix with a much smaller rank $k \ll m, n$.

Theorem 11 (best low rank approximation)**.** *The matrix*

$$A_k := \underbrace{\begin{bmatrix}\boldsymbol{u}_1|\cdots|\boldsymbol{u}_k\end{bmatrix}}_{=:U_k\in\mathbb{C}^{m\times k}} \underbrace{\begin{bmatrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_k\end{bmatrix}}_{=:\Sigma_k\in\mathbb{C}^{k\times k}} \underbrace{\begin{bmatrix}\boldsymbol{v}_1|\cdots|\boldsymbol{v}_k\end{bmatrix}^\star}_{=:V_k^\star\in\mathbb{C}^{k\times n}}$$

is the best 2-norm approximation of A by a rank k matrix, that is, for all rank-k matrices B, we have

$$\sigma_{k+1} = \|A - A_k\| \le \|A - B\|.$$

Proof We have

$$A - A_k = U \begin{bmatrix} 0 & & & & & \\ & \ddots & & & & \\ & & 0 & & & \\ & & & \sigma_{k+1} & & \\ & & & & \ddots & \\ & & & & & \sigma_r \end{bmatrix} V^\star = \begin{bmatrix}\boldsymbol{u}_{k+1}|\cdots|\boldsymbol{u}_r\end{bmatrix} \begin{bmatrix}\sigma_{k+1} & & \\ & \ddots & \\ & & \sigma_r\end{bmatrix} \begin{bmatrix}\boldsymbol{v}_{k+1}|\cdots|\boldsymbol{v}_r\end{bmatrix}^\star$$

hence $\|A - A_k\| = \sigma_{k+1}$.

Suppose a rank-k matrix B has

$$\|A - B\| < \|A - A_k\| = \sigma_{k+1}.$$

For all $\boldsymbol{w} \in \ker(B)$ we have

$$\|A\boldsymbol{w}\| = \|(A - B)\boldsymbol{w}\| \le \|A - B\|\|\boldsymbol{w}\| < \sigma_{k+1}\|\boldsymbol{w}\|$$

But for all $\boldsymbol{x} \in \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{k+1})$, that is, $\boldsymbol{x} = V[:, 1 : k+1]\boldsymbol{c}$ for some $\boldsymbol{c} \in \mathbb{C}^{k+1}$ we have

$$\|A\boldsymbol{x}\|^2 = \|U\Sigma_k\boldsymbol{c}\|^2 = \|\Sigma_k\boldsymbol{c}\|^2 = \sum_{j=1}^{k+1}(\sigma_j c_j)^2 \ge \sigma_{k+1}^2\|\boldsymbol{c}\|^2,$$

i.e., $\|A\boldsymbol{x}\| \ge \sigma_{k+1}\|\boldsymbol{x}\|$, where we use the fact that

$$\|\boldsymbol{x}\|^2 = \|V[:, 1 : k+1]\boldsymbol{c}\|^2 = \boldsymbol{c}^\star V[:, 1 : k+1]^\star V[:, 1 : k+1]\boldsymbol{c} = \boldsymbol{c}^\star\boldsymbol{c} = \|\boldsymbol{c}\|^2.$$

Thus $\boldsymbol{w}$ cannot be in this span.

The dimension of the span of $\ker(B)$ is at least $n-k$, but the dimension of $\operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{k+1})$ is at least $k+1$. Since these two spaces cannot intersect (apart from at 0) we have a contradiction, since $(n-k) + (k+1) = n+1 > n$. ■

IV.2.4 Lab and Problem Sheet

In the lab we explore the relationship between smoothness of a function and the decay in the singular values. We also implement low rank compression and see for samples of some smooth functions that we can massively reduce the data needed to represent the matrix. We also look at the case of the Hilbert matrix

$$H_n := \begin{bmatrix} 1 & 1/2 & 1/3 & \cdots & 1/n \\ 1/2 & 1/3 & 1/4 & \cdots & 1/(n+1) \\ 1/3 & 1/4 & 1/5 & \cdots & 1/(n+2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1/n & 1/(n+1) & 1/(n+2) & \cdots & 1/(2n-1) \end{bmatrix},$$

a famous matrix introduced by Hilbert which has a number of applications in approximation theory.

In the problem sheet we see that the SVD can be used to define the *Moore–Penrose pseudoinverse*, a way of constructing a notion if an inverse for a rectangular or non-invertible matrix that is consistent with least sqaures.

Chapter V

Numerical Fourier series

We have seen intuitive numerical methods for computing derivatives and integrals, primarily based on representing functions by their values at a grid of points. But by using more sophisticated mathematical tools, we can achieve much more accurate and reliable numerical methods. In particular, we can effectively use Fourier series for computing very accurately with periodic functions, which will later be adapted in the form of orthogonal polynomials for non-periodic functions that are smooth within an interval.

1. V.1 Fourier Expansions: we discuss Fourier series and their usage in approximating periodic functions, using the Trapezium rule to compute the Fourier coefficients.

2. V.2 Discrete Fourier Transform: The Trapezium rule approximation can be recast as a unitary matrix, known as the Discrete Fourier Transform (DFT). This is used to prove interpolation properties.

3. V.3 Convolutions: Convolutions are natural ways of representing operators that commute with translations, and underly Convolutional Neural Networks which have advantages for problems with natural translation symmetry (for example, number recognisition which should be invariant to where the number is in an image). We discuss how discrete convolutions can be diagonalised using the DFT leading to fast convolution when implemented via the Fast Fourier Transform (FFT).

V.1 Fourier Expansions

Fourier series are a powerful tool in wide areas of mathematics, including solving partial differential equations, signal processing, and elsewhere. They are also very useful in computational methods, particularly for problems that have periodicity. Periodicity arises naturally when solving problems in radial coordinates, or when approximating a problem on the real line by a periodic problem with a large period. Fourier series are also related to orthogonal polynomials, which can be used for non-periodic problems.

The most fundamental basis is (complex) Fourier: we have $\mathrm{e}^{\mathrm{i}k\theta}$ are orthogonal with respect to the inner product

$$\langle f, g\rangle := \frac{1}{2\pi}\int_0^{2\pi} \bar{f}(\theta)g(\theta)\mathrm{d}\theta,$$

where we conjugate the first argument to be consistent with the vector inner product $\boldsymbol{x}^\star \boldsymbol{y}$. We will use the notation $\mathbb{T} := [0, 2\pi)$ (typically this has the topology of a circle attached but we do not need to worry about that here). We can (typically) expand functions in this basis:

Definition 30 (Fourier)**.** A function f has a Fourier expansion if

$$f(\theta) = \sum_{k=-\infty}^{\infty} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta}$$

where

$$\hat{f}_k := \langle \mathrm{e}^{\mathrm{i}k\theta}, f \rangle = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}k\theta} f(\theta) \mathrm{d}\theta$$

A basic observation is if a Fourier expansion has no negative terms it is equivalent to a Taylor series in disguise:

Definition 31 (Fourier-Taylor)**.** A function f has a Fourier–Taylor expansion if

$$f(\theta) = \sum_{k=0}^{\infty} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta} = \sum_{k=0}^{\infty} \hat{f}_k z^k$$

where $\hat{f}_k := \langle \mathrm{e}^{\mathrm{i}k\theta}, f \rangle$, and $z = \mathrm{e}^{\mathrm{i}\theta}$.

In numerical analysis we try to build on the analogy with linear algebra as much as possible. Therefore we can write this this as:

$$f(\theta) = \underbrace{[1|\mathrm{e}^{\mathrm{i}\theta}|\mathrm{e}^{2\mathrm{i}\theta}|\cdots]}_{T(\theta)} \underbrace{\begin{bmatrix} \hat{f}_0 \\ \hat{f}_1 \\ \hat{f}_2 \\ \vdots \end{bmatrix}}_{\hat{\boldsymbol{f}}}.$$

Essentially, expansions in bases are viewed as a way of turning *functions* into (infinite) *vectors*. And (differential) *operators* into *matrices*.

V.1.1 Convergence of Fourier series

In analysis one typically works with continuous functions and relates results to continuity. In numerical analysis we inherently have to work with *vectors*, so it is more natural to focus on the case where the *Fourier coefficients* $\hat{f}_k$ are *absolutely convergent*:

Definition 32 (absolute convergent)**.** We write $\hat{\boldsymbol{f}} \in \ell^1$ if it is absolutely convergent, or in otherwords, the 1-norm of $\hat{\boldsymbol{f}}$ is bounded:

$$\|\hat{\boldsymbol{f}}\|_1 := \sum_{k=-\infty}^{\infty} |\hat{f}_k| < \infty.$$

We first state a basic results (whose proof is beyond the scope of this module):

Theorem 12 (Fourier series equivalence)**.** *If $f, g : \mathbb{T} \to \mathbb{C}$ are periodic and continuous and $\hat{f}_k = \hat{g}_k$ for all $k \in \mathbb{Z}$ then $f = g$.*

Proof See Körner 2022 (Theorem 2.4). ■

This allows us to prove the following:

Theorem 13 (Absolute converging Fourier series)**.** *If* $\hat{\boldsymbol{f}} \in \ell^1$ *then*

$$f(\theta) = \sum_{k=-\infty}^{\infty} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta},$$

which converges uniformly.

Proof

Note that

$$g_n(\theta) := \sum_{k=-n}^{n} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta}$$

is uniformly-absolutely convergent as $n \to \infty$, that is,

$$\sum_{k=-n}^{n} |\hat{f}_k \mathrm{e}^{\mathrm{i}k\theta}| = \sum_{k=-n}^{n} |\hat{f}_k| \to \|\hat{\boldsymbol{f}}\|_1.$$

This guarantees that $g_n(\theta)$ converges uniformly to a continuous function $g(\theta)$. We have for $n > k$, that the k-th Fourier coefficient of $g_n(\theta)$ equals $\hat{f}_k$. Thus, by the properties of uniform convergence,

$$\hat{f}_k = \lim_{n\to\infty} \hat{f}_k = \lim_{n\to\infty} \frac{1}{2\pi} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}k\theta} g_n(\theta) \mathrm{d}\theta = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}k\theta} \lim_{n\to\infty} g_n(\theta) \mathrm{d}\theta = \hat{g}_k.$$

Since f and g are continuous and share the same Fourier coefficients, they are equal.

■

When does a function have absolutely convergent Fourier coefficients? We can deduce it from periodic differentiability of the function:

Lemma 7 (differentiability and absolutely convergence)**.** *If* $f : \mathbb{T} \to \mathbb{C}$ *and* f' *are periodic and* f'' *is uniformly bounded, then* $\hat{\boldsymbol{f}} \in \ell^1$.

Proof Integrate by parts twice using the fact that $f(0) = f(2\pi)$, $f'(0) = f'(2\pi)$:

$$\begin{aligned}
2\pi \hat{f}_k &= \int_0^{2\pi} f(\theta) \mathrm{e}^{-\mathrm{i}k\theta} \mathrm{d}\theta = \left[f(\theta) \frac{\mathrm{e}^{-\mathrm{i}k\theta}}{-\mathrm{i}k} \right]_0^{2\pi} + \frac{1}{\mathrm{i}k} \int_0^{2\pi} f'(\theta) \mathrm{e}^{-\mathrm{i}k\theta} \mathrm{d}\theta \\
&= \left[f'(\theta) \frac{\mathrm{e}^{-\mathrm{i}k\theta}}{(-\mathrm{i}k)^2} \right]_0^{2\pi} - \frac{1}{k^2} \int_0^{2\pi} f''(\theta) \mathrm{e}^{-\mathrm{i}k\theta} \mathrm{d}\theta \\
&= -\frac{1}{k^2} \int_0^{2\pi} f''(\theta) \mathrm{e}^{-\mathrm{i}k\theta} \mathrm{d}\theta.
\end{aligned}$$

Thus uniform boundedness of f'' guarantees $|\hat{f}_k| \leq M|k|^{-2}$ for some M, and we have

$$\sum_{k=-\infty}^{\infty} |\hat{f}_k| \leq |\hat{f}_0| + 2M \sum_{k=1}^{\infty} |k|^{-2} < \infty$$

using the dominant convergence test.

■

This condition can be weakened to Lipschitz continuity but the proof is beyond the scope of this module. Of more practical importance is the other direction: the more times differentiable a function the faster the coefficients decay, and thence the faster Fourier expansions converge. In fact, if a function is smooth and 2π-periodic its Fourier coefficients decay faster than algebraically: they decay like $O(k^{-\lambda})$ for any λ. This will be explored in the problem sheet.

V.1.2 Trapezium rule and discrete Fourier coefficients

For periodic functions the Trapezium rule simplies and has a lot of nice properties (some of which we saw in the first lab). For concreteness we define a specific normalisation for the Trapezium rule as follows:

Definition 33 (Periodic Trapezium Rule)**.** Let $\theta_j = 2\pi j/n$ for $j = 0, 1, \ldots, n$ denote $n+1$ evenly spaced points over $[0, 2\pi]$. Recall that the *Trapezium rule* over $[0, 2\pi]$ is the approximation:

$$\int_0^{2\pi} f(\theta)\mathrm{d}\theta \approx \frac{2\pi}{n}\left[\frac{f(0)}{2} + \sum_{j=1}^{n-1} f(\theta_j) + \frac{f(2\pi)}{2}\right]$$

But if f is periodic we have $f(0) = f(2\pi)$ and we get the *periodic Trapezium rule*:

$$\frac{1}{2\pi}\int_0^{2\pi} f(\theta)\mathrm{d}\theta \approx \underbrace{\frac{1}{n}\sum_{j=0}^{n-1} f(\theta_j)}_{=:\Sigma_n[f]}$$

We know that $\mathrm{e}^{\mathrm{i}k\theta}$ are orthogonal with respect to the continuous inner product. The following says that this property is maintained (up to "aliasing") when we replace the continuous integral with a trapezium rule approximation:

Lemma 8 (Discrete orthogonality)**.** *We have:*

$$\sum_{j=0}^{n-1} \mathrm{e}^{\mathrm{i}k\theta_j} = \begin{cases} n & k = \ldots, -2n, -n, 0, n, 2n, \ldots \\ 0 & \textit{otherwise} \end{cases}$$

In other words,

$$\Sigma_n[\mathrm{e}^{\mathrm{i}(k-\ell)\theta}] = \begin{cases} 1 & k - \ell = \ldots, -2n, -n, 0, n, 2n, \ldots \\ 0 & \textit{otherwise} \end{cases}.$$

Proof

Consider $\omega := \mathrm{e}^{\mathrm{i}\theta_1} = \mathrm{e}^{\frac{2\pi\mathrm{i}}{n}}$. This is an n-th root of unity: $\omega^n = 1$. Note that $\mathrm{e}^{\mathrm{i}\theta_j} = \mathrm{e}^{\frac{2\pi\mathrm{i}j}{n}} = \omega^j$.

(Case 1: $k = pn$ for an integer p) We have

$$\sum_{j=0}^{n-1} \mathrm{e}^{\mathrm{i}k\theta_j} = \sum_{j=0}^{n-1} \omega^{kj} = \sum_{j=0}^{n-1} (\omega^{pn})^j = \sum_{j=0}^{n-1} 1 = n$$

(Case 2: $k \neq pn$ for an integer p) Recall that (via a telescoping sum argument)

$$\sum_{j=0}^{n-1} z^j = \frac{z^n - 1}{z - 1}.$$

Then we have

$$\sum_{j=0}^{n-1} \mathrm{e}^{\mathrm{i}k\theta_j} = \sum_{j=0}^{n-1} (\omega^k)^j = \frac{\omega^{kn} - 1}{\omega^k - 1} = 0.$$

where we use the fact that k is not a multiple of n to guarantee that $\omega^k \neq 1$.

■

V.1.3 Convergence of Approximate Fourier expansions

To perform computations with Fourier expansions we need to compute the Fourier coefficients, and the periodic Trapezium rule provides a natural way to do so. We will now use the Trapezium rule to approximate Fourier coefficients and expansions:

Definition 34 (Discrete Fourier coefficients)**.** Define the Trapezium rule approximation to the Fourier coefficients by:

$$\hat{f}_k^n := \Sigma_n[\mathrm{e}^{-ik\theta} f(\theta)] = \frac{1}{n} \sum_{j=0}^{n-1} \mathrm{e}^{-ik\theta_j} f(\theta_j)$$

A remarkable fact is that the discrete Fourier coefficients can be expressed as a sum of the true Fourier coefficients:

Theorem 14 (discrete Fourier coefficients)**.** *If* $\hat{\boldsymbol{f}} \in \ell^1$ *then*

$$\hat{f}_k^n = \cdots + \hat{f}_{k-2n} + \hat{f}_{k-n} + \hat{f}_k + \hat{f}_{k+n} + \hat{f}_{k+2n} + \cdots$$

Proof

$$\begin{aligned} \hat{f}_k^n = \Sigma_n[f(\theta)\mathrm{e}^{-\mathrm{i}k\theta}] &= \sum_{\ell=-\infty}^{\infty} \hat{f}_\ell \Sigma_n[\mathrm{e}^{\mathrm{i}(\ell-k)\theta}] \\ &= \sum_{\ell=-\infty}^{\infty} \hat{f}_\ell \begin{cases} 1 & \ell - k = \ldots, -2n, -n, 0, n, 2n, \ldots \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

■

Example 20 (Taylor coefficients via Geometric series)**.** Consider the function

$$f(\theta) = \frac{2}{2 - \mathrm{e}^{\mathrm{i}\theta}}$$

Under the change of variables $z = \mathrm{e}^{\mathrm{i}\theta}$ we know for z on the unit circle this becomes (using the geometric series with $z/2$)

$$\frac{2}{2 - z} = \sum_{k=0}^{\infty} \frac{z^k}{2^k}$$

i.e., $\hat{f}_k = 1/2^k$ which is absolutely summable:

$$\sum_{k=0}^{\infty} |\hat{f}_k| = f(0) = 2.$$

If we use an n point discretisation we get for $0 \leq k \leq n-1$ (using the geoemtric series with 2^{-n})

$$\hat{f}_k^n = \hat{f}_k + \hat{f}_{k+n} + \hat{f}_{k+n} + \cdots = \sum_{p=0}^{\infty} \frac{1}{2^{k+pn}} = \frac{2^{n-k}}{2^n - 1}$$

Note that as $n \to \infty$, we have $\hat{f}_k^n \to \hat{f}_k$.

Note that there is redundancy, in that higher order discrete Fourier coefficients are exactly the same as certain low order ones. This phenomena is called *aliasing*:

Corollary 4 (aliasing)**.** *For all $p \in \mathbb{Z}$, $\hat{f}_k^n = \hat{f}_{k+pn}^n$.*

Proof Follows immediately:

$$\hat{f}_{k+pn}^n = \sum_{j=-\infty}^{\infty} \hat{f}_{k+(p+j)n} = \sum_{j=-\infty}^{\infty} \hat{f}_{k+jn} = \hat{f}_k^n.$$

■

In other words if we know $\hat{f}_0^n, \ldots, \hat{f}_{n-1}^n$, we know $\hat{f}_k^n$ for all k via a permutation, for example if $n = 2m+1$ we have

$$\begin{bmatrix} \hat{f}_{-m}^n \\ \vdots \\ \hat{f}_{-1}^n \\ \hat{f}_0^n \\ \vdots \\ \hat{f}_m^n \end{bmatrix} = \underbrace{\begin{bmatrix} & & & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \\ 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \end{bmatrix}}_{P_\sigma} \begin{bmatrix} \hat{f}_0^n \\ \vdots \\ \hat{f}_m^n \\ \hat{f}_{m+1}^n \\ \vdots \\ \hat{f}_{n-1}^n \end{bmatrix}$$

where σ has Cauchy notation (*Careful*: we are using 1-based indexing here):

$$\begin{pmatrix} 1 & 2 & \cdots & m & m+1 & m+2 & \cdots & n \\ m+2 & m+3 & \cdots & n & 1 & 2 & \cdots & m+1 \end{pmatrix}.$$

We can prove *convergence* of an approximate Fourier series using the discrete Fourier coefficients whenever f has absolutely summable coefficients. We will prove the result here in the special case where the negative coefficients are zero. That is, $\hat{f}_0^n, \ldots, \hat{f}_{n-1}^n$ are approximations of the Fourier–Taylor coefficients.

Theorem 15 (Approximate Fourier-Taylor expansions converge)**.** *If $0 = \hat{f}_{-1} = \hat{f}_{-2} = \cdots$ and $\hat{\boldsymbol{f}}$ is absolutely convergent then*

$$f_n(\theta) = \sum_{k=0}^{n-1} \hat{f}_k^n \mathrm{e}^{\mathrm{i}k\theta}$$

converges uniformly to $f(\theta)$.

Proof

$$|f(\theta) - f_n(\theta)| = |\sum_{k=0}^{n-1} (\hat{f}_k - \hat{f}_k^n)\mathrm{e}^{\mathrm{i}k\theta} + \sum_{k=n}^{\infty} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta}| = |\sum_{k=n}^{\infty} \hat{f}_k (\mathrm{e}^{\mathrm{i}k\theta} - \mathrm{e}^{\mathrm{i}\mathrm{mod}(k,n)\theta})| \leq 2 \sum_{k=n}^{\infty} |\hat{f}_k|$$

which goes to zero as $n \to \infty$. ■

For the general case we need to choose a range of coefficients that includes roughly an equal number of negative and positive coefficients (preferring negative over positive in a tie as a convention):

$$f_n(\theta) = \sum_{k=-\lceil n/2 \rceil}^{\lfloor n/2 \rfloor} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta}$$

In the problem sheet we will prove this converges provided the coefficients are absolutely convergent.

V.1.4 Lab and Problem Sheet

In the problem sheet we investigate some examples of Fourier and discrete Fourier coefficients and explore the relationship between smoothness, decay in Fourier coefficients, and convergence of approximate Fourier expansions. We also extend the convergence and interpolation results to the non-Taylor case, that is, where the negative Fourier coefficients may be zero.

In the lab we explore numerical computation of Fourier coefficients and convergence of Fourier expansions. We also observe the *Gibb's phenomena*, where approximate Fourier expansions overshoot near discontinuities.

V.2 Discrete Fourier Transform

In the previous section we explored using the trapezium rule for approximating Fourier coefficients. This is a linear map from function values to coefficients and thus can be reinterpreted as a matrix-vector product, called the the Discrete Fourier Transform (DFT). It turns out the matrix is unitary which leads to important properties including interpolation.

Remark A clever way of decomposing the DFT leads to a fast way of applying and inverting it, which is one of the most influencial algorithms of the 20th century: the Fast Fourier Transform (FFT). But this is beyond the scope of this module.

Definition 35 (DFT)**.** The *Discrete Fourier Transform (DFT)* is defined as:

$$\begin{aligned} Q_n &:= \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \mathrm{e}^{-\mathrm{i}\theta_1} & \mathrm{e}^{-\mathrm{i}\theta_2} & \cdots & \mathrm{e}^{-\mathrm{i}\theta_{n-1}} \\ 1 & \mathrm{e}^{-\mathrm{i}2\theta_1} & \mathrm{e}^{-\mathrm{i}2\theta_2} & \cdots & \mathrm{e}^{-\mathrm{i}2\theta_{n-1}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \mathrm{e}^{-\mathrm{i}(n-1)\theta_1} & \mathrm{e}^{-\mathrm{i}(n-1)\theta_2} & \cdots & \mathrm{e}^{-\mathrm{i}(n-1)\theta_{n-1}} \end{bmatrix} \\ &= \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega^{-1} & \omega^{-2} & \cdots & \omega^{-(n-1)} \\ 1 & \omega^{-2} & \omega^{-4} & \cdots & \omega^{-2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{-(n-1)} & \omega^{-2(n-1)} & \cdots & \omega^{-(n-1)^2} \end{bmatrix} \end{aligned}$$

for the n-th root of unity $\omega = \mathrm{e}^{2\pi\mathrm{i}/n}$.

Note that

$$Q_n^\star = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \mathrm{e}^{\mathrm{i}\theta_1} & \mathrm{e}^{\mathrm{i}2\theta_1} & \cdots & \mathrm{e}^{\mathrm{i}(n-1)\theta_1} \\ 1 & \mathrm{e}^{\mathrm{i}\theta_2} & \mathrm{e}^{\mathrm{i}2\theta_2} & \cdots & \mathrm{e}^{\mathrm{i}(n-1)\theta_2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \mathrm{e}^{\mathrm{i}\theta_{n-1}} & \mathrm{e}^{\mathrm{i}2\theta_{n-1}} & \cdots & \mathrm{e}^{\mathrm{i}(n-1)\theta_{n-1}} \end{bmatrix}$$

$$= \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega^1 & \omega^2 & \cdots & \omega^{(n-1)} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{(n-1)} & \omega^{2(n-1)} & \cdots & \omega^{(n-1)^2} \end{bmatrix}$$

Hence we have

$$\underbrace{\begin{bmatrix} \hat{f}_0^n \\ \vdots \\ \hat{f}_{n-1}^n \end{bmatrix}}_{\hat{\boldsymbol{f}}^n} = \frac{1}{\sqrt{n}} Q_n \underbrace{\begin{bmatrix} f(\theta_0) \\ \vdots \\ f(\theta_{n-1}) \end{bmatrix}}_{\boldsymbol{f}^n}$$

The choice of normalisation constant is motivated by the following:

Proposition 1 (DFT is Unitary) $Q_n \in U(n)$, that is, $Q_n^\star Q_n = Q_n Q_n^\star = I$.

Proof

$$Q_n Q_n^\star = \begin{bmatrix} \Sigma_n[1] & \Sigma_n[\mathrm{e}^{\mathrm{i}\theta}] & \cdots & \Sigma_n[\mathrm{e}^{\mathrm{i}(n-1)\theta}] \\ \Sigma_n[\mathrm{e}^{-\mathrm{i}\theta}] & \Sigma_n[1] & \cdots & \Sigma_n[\mathrm{e}^{\mathrm{i}(n-2)\theta}] \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_n[\mathrm{e}^{-\mathrm{i}(n-1)\theta}] & \Sigma_n[\mathrm{e}^{-\mathrm{i}(n-2)\theta}] & \cdots & \Sigma_n[1] \end{bmatrix} = I$$

■

In other words, Q_n is easily inverted and we also have a map from discrete Fourier coefficients back to values:

$$\sqrt{n} Q_n^\star \hat{\boldsymbol{f}}^n = \boldsymbol{f}^n$$

Example 21 (Computing sum)**.** Define the following infinite sum (which has no name apparently, according to Mathematica):

$$S_n(k) := \sum_{p=0}^{\infty} \frac{1}{(k+pn)!}.$$

We can use the DFT to compute $S_n(0), \ldots, S_n(n-1)$. Consider

$$f(\theta) = \exp(\mathrm{e}^{\mathrm{i}\theta}) = \sum_{k=0}^{\infty} \frac{\mathrm{e}^{\mathrm{i}k\theta}}{k!}$$

where we know the Fourier coefficients from the Taylor series of e^z. The discrete Fourier coefficients satisfy for $0 \leq k \leq n-1$:

$$\hat{f}_k^n = \hat{f}_k + \hat{f}_{k+n} + \hat{f}_{k+2n} + \cdots = S_n(k)$$

Thus we have

$$\begin{bmatrix} S_n(0) \\ \vdots \\ S_n(n-1) \end{bmatrix} = \frac{1}{\sqrt{n}} Q_n \begin{bmatrix} \mathrm{e} \\ \exp(\mathrm{e}^{2\mathrm{i}\pi/n}) \\ \vdots \\ \exp(\mathrm{e}^{2\mathrm{i}(n-1)\pi/n}) \end{bmatrix}$$

V.2.1 Trigonometric Interpolation

We investigated interpolation and least squares using polynomials at evenly spaced points, observing that there were issues with stability. We now show that the DFT actually gives coefficients that interpolate using Fourier expansions. As the DFT is a unitary matrix multiplication is "stable", i.e. it preserves norms and hence we know it cannot cause the same huge blow-up we saw for polynomials. That is: whilst polynomials are bad for interpolation at evenly spaced points, trigonometric polynomials are great.

The following guarantees that our approximate Fourier series actually interpolates the data:

Corollary 5 (Interpolation)**.**

$$f_n(\theta) := \sum_{k=0}^{n-1} \hat{f}_k^n \mathrm{e}^{\mathrm{i}k\theta}$$

interpolates f at θ_j:

$$f_n(\theta_j) = f(\theta_j).$$

Proof We have

$$f_n(\theta_j) = \sum_{k=0}^{n-1} \hat{f}_k^n \mathrm{e}^{\mathrm{i}k\theta_j} = \sqrt{n}\boldsymbol{e}_{j+1}^\top Q_n^\star \hat{\boldsymbol{f}}^n = \boldsymbol{e}_{j+1}^\top Q_n^\star Q_n \boldsymbol{f}^n = f(\theta_j).$$

■

Example 22 (DFT versus Lagrange)**.** Consider interpolating $\exp z$ by a polynomial at the points $1, \mathrm{i}, -1, -\mathrm{i}$. We can use Lagrange polynomials:

$$\begin{aligned}
\ell_1(z) &= \frac{(z-\mathrm{i})(z+1)(z+\mathrm{i})}{2(1-\mathrm{i})(1+\mathrm{i})} = \frac{z^3+z^2+z+1}{4} \\
\ell_2(z) &= \frac{(z-1)(z+1)(z+\mathrm{i})}{(\mathrm{i}-1)(\mathrm{i}+1)2\mathrm{i}} = \frac{\mathrm{i}z^3-z^2-\mathrm{i}z+1}{4} \\
\ell_3(z) &= \frac{(z-1)(z-\mathrm{i})(z+\mathrm{i})}{-2(-1-\mathrm{i})(-1+\mathrm{i})} = \frac{-z^3+z^2-z+1}{4} \\
\ell_4(z) &= \frac{(z-1)(z-\mathrm{i})(z+1)}{(-\mathrm{i}-1)(-2\mathrm{i})(-\mathrm{i}+1)} = \frac{-\mathrm{i}z^3-z^2+\mathrm{i}z+1}{4}
\end{aligned}$$

So we get the interpolant:

$$\begin{aligned}
&\mathrm{e}\ell_1(z) + \mathrm{e}^{\mathrm{i}}\ell_2(z) + \mathrm{e}^{-1}\ell_3(z) + \mathrm{e}^{-\mathrm{i}}\ell_4(z) \\
&= \frac{\mathrm{e}+\mathrm{e}^{\mathrm{i}}+\mathrm{e}^{-1}+\mathrm{e}^{-\mathrm{i}}}{4} + \frac{\mathrm{e}-\mathrm{i}\mathrm{e}^{\mathrm{i}}-\mathrm{e}^{-1}+\mathrm{i}\mathrm{e}^{-\mathrm{i}}}{4} z + \frac{\mathrm{e}-\mathrm{e}^{\mathrm{i}}+\mathrm{e}^{-1}-\mathrm{i}\mathrm{e}^{-\mathrm{i}}}{4} z^2 + \frac{\mathrm{e}+\mathrm{i}\mathrm{e}^{\mathrm{i}}-\mathrm{e}^{-1}-\mathrm{i}\mathrm{e}^{-\mathrm{i}}}{4} z^3
\end{aligned}$$

Alternatively we could have deduced this directly from the DFT. In particular, we know the coefficients of the interpolating polynomial must be, for $\omega = \mathrm{i}$ and $f(\theta) = \exp(\mathrm{e}^{\mathrm{i}\theta})$,

$$\begin{bmatrix} \hat{f}_0^4 \\ \hat{f}_1^4 \\ \hat{f}_2^4 \\ \hat{f}_3^4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{i} & -1 & \mathrm{i} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{i} & -1 & -\mathrm{i} \end{bmatrix} \begin{bmatrix} \mathrm{e} \\ \mathrm{e}^{\mathrm{i}} \\ \mathrm{e}^{-1} \\ \mathrm{e}^{-\mathrm{i}} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} \mathrm{e} + \mathrm{e}^{\mathrm{i}} + \mathrm{e}^{-1} + \mathrm{e}^{-\mathrm{i}} \\ \mathrm{e} - \mathrm{i}\mathrm{e}^{\mathrm{i}} - \mathrm{e}^{-1} + \mathrm{i}\mathrm{e}^{-\mathrm{i}} \\ \mathrm{e} - \mathrm{e}^{\mathrm{i}} + \mathrm{e}^{-1} - \mathrm{e}^{-\mathrm{i}} \\ \mathrm{e} + \mathrm{i}\mathrm{e}^{\mathrm{i}} - \mathrm{e}^{-1} - \mathrm{i}\mathrm{e}^{-\mathrm{i}} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \cosh 1 + \cos 1 \\ \sinh 1 + \sin 1 \\ \cosh 1 - \cos 1 \\ \sinh 1 - \sin 1 \end{bmatrix}.$$

V.2.2 Lab and Problem Sheet

In the problem sheet we see how the DFT can be extended to a cosine expansion, in the form of the Discrete Cosine Transform (DCT). We also show that the interpolation property also applies to the approximation

$$f_n(\theta) = \sum_{k=-\lceil n/2 \rceil}^{\lfloor n/2 \rfloor} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta}$$

for general Fourier series. In the lab, we see how the FFT implements the DFT in a fast manner, and see how the approximate cosine expansion can be constructed numerically.

V.3 Discrete convolutions

Convolutions are a fundamental concept in mathematics that are intrinsically linked to Fourier series and transforms. At the continuous level you have likely seen convolutions of the form

$$(a \star f)(x) := \int_{-\infty}^{\infty} a(x-t) f(t) \mathrm{d}t.$$

This was probably introduced in the context of random variables: if X has a probabilty density function (PDF) given by a and Y has a PDF give by f then the PDF of $X + Y$ is given by $a \star f$. But this can also be viewed as a linear operator acting on f with the basic property that it commutes with translation: if we shift the point evaluation by τ, $f_\tau(x) = f(x - \tau)$, then we have

$$(a \star f_\tau)(x) = \int_{-\infty}^{\infty} a(x-t) f(t-\tau) \mathrm{d}t = \int_{-\infty}^{\infty} a(x - (s+\tau)) f(s) \mathrm{d}s = (a \star f)(x - \tau).$$

All operators coming from physics commute with translations: the physics at a point x are exactly the same as the physics at the point $x - \tau$. Common examples are differentiation ($f'_\tau(x) = f'(x - \tau)$) or even the identity operator. Thus this property is fundamental.

In this section we are interested in a discrete analogue: matrices which commute with translations (in a periodic sense), which we shall show must be of the form of a *discrete* convolution. In the context of machine learning, this property underlies *Convolutional Neural Networks*: often we want to incorporate translational invariance into training. For example, if we are constructing a Neural Network to recognise numbers from their pixels, we want to the answer to be the same even if we shift the number up or down. To achieve this it is beneficial to have intermediate layers of the Neural Network commute with translations, and this is exactly what convolutional layers do.

There is also a natural link between continuous convolutions and Fourier transforms: Fourier transforms turn convolutions into pointwise multiplication. Here we will derive a natural link between discrete convolutions and the DFT, exactly mimicking the continuous version.

V.3.1 Matrices that commute with translations

The starting point is a discrete version of a translation, which are give by the periodic (right) shift operator:

$$S := \begin{bmatrix} 0 & & & & 1 \\ 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & 0 \end{bmatrix} = P_\sigma$$

for the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ n & 1 & 2 & \cdots & n-1 \end{pmatrix}.$$

These shift vectors to the right in a periodic way:

$$S \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{n-1} \\ f_n \end{bmatrix} = \begin{bmatrix} f_n \\ f_1 \\ f_2 \\ \vdots \\ f_{n-1} \end{bmatrix}$$

Thus $S\boldsymbol{f}$ is a discrete, periodic version of the translation $f(x-\tau)$, which can be made precise if we think of $\boldsymbol{f}$ as samples of f on an evenly spaced grid $\theta_j = 2\pi j/n$:

$$S \begin{bmatrix} f(\theta_0) \\ \vdots \\ f(\theta_{n-2}) \\ f(\theta_{n-1}) \end{bmatrix} = \begin{bmatrix} f(\theta_{n-1}) \\ f(\theta_0) \\ \vdots \\ f(\theta_{n-2}) \end{bmatrix},$$

thus it is equivalent to sampling the function $f(\theta - \tau)$ with shift $\tau = 2\pi/n$.

What matrices commute with such discrete translations? We can categorise all possible matrices in terms of a *circluant matrix*:

Definition 36 (circulant matrix)**.** Given

$$\boldsymbol{a} = \begin{bmatrix} a_0 \\ \vdots \\ a_{n-1} \end{bmatrix} \in \mathbb{C}^n$$

the associated *circulant matrix* is

$$C[\boldsymbol{a}] := \begin{bmatrix} a_0 & a_{n-1} & \cdots & a_2 & a_1 \\ a_1 & a_0 & a_{n-1} & \cdots & \vdots \\ \vdots & a_1 & a_0 & \ddots & \vdots \\ a_{n-2} & \cdots & \ddots & \ddots & a_{n-1} \\ a_{n-1} & a_{n-2} & \cdots & a_1 & a_0 \end{bmatrix} = a_0 I + a_1 S + \cdots + a_{n-1} S^{n-1} = a(S),$$

for the polynomial (which we call the *symbol*)

$$a(z) = a_0 + a_1 z + \ldots + a_{n-1} z^{n-1}.$$

Every operator that commutes with (periodic) translations is a circulant matrix:

Lemma 9 (operators which commute with translations)**.** *A matrix $A \in \mathbb{C}^{n\times n}$ satisfies $AS = SA$ if and only if it is a circulant matrix.*

Proof

Circulant matrices follow this property because we have

$$SC[\boldsymbol{a}] = \underbrace{\begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & a_0 \\ a_0 & a_{n-1} & \cdots & a_2 & a_1 \\ \vdots & a_0 & a_{n-1} & \ddots & \vdots \\ a_{n-3} & \cdots & \ddots & \ddots & a_{n-2} \\ a_{n-2} & a_{n-3} & \cdots & a_0 & a_{n-1} \end{bmatrix}}_{C[S\boldsymbol{a}]} = C[\boldsymbol{a}]S.$$

Now writing any square matrix as

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

we want

$$\underbrace{\begin{bmatrix} a_{n1} & \cdots & a_{nn} \\ a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n-1,1} & \cdots & a_{n-1n} \end{bmatrix}}_{SA} = \underbrace{\begin{bmatrix} a_{12} & \cdots & a_{1n} & a_{11} \\ a_{22} & \cdots & a_{2n} & a_{21} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n2} & \cdots & a_{nn} & a_{n1} \end{bmatrix}}_{AS}$$

These conditions require that A is circulant. In particular, we have for $k = 1, \ldots, n$

$$\boldsymbol{e}_k^\top SA\boldsymbol{e}_j = \boldsymbol{e}_k^\top AS\boldsymbol{e}_j \Leftrightarrow \boldsymbol{e}_{\mathrm{mod}(k-1,n)+1}^\top A\boldsymbol{e}_j = \boldsymbol{e}_k^\top A\boldsymbol{e}_{\mathrm{mod}(j+1,n)+1} \Leftrightarrow a_{k-1,j} = a_{k,j+1},$$

where we interpret this in a periodic way: i.e., for $k, j \notin \{1, \ldots n\}$ we define $a_{k,j} := a_{\mathrm{mod}(k-1,n)+1,\mathrm{mod}(j-1,n)+1}$. In other words:

$$\begin{aligned} a_0 &:= a_{11} = a_{22} = \cdots = a_{nn}, \\ a_1 &:= a_{21} = \cdots = a_{n,n-1} = \underbrace{a_{n+1,n}}_{=a_{1,n}}, \\ &\vdots \\ a_{n-1} &:= a_{n1} = \underbrace{a_{n+1,2}}_{=a_{12}} = a_{23} = \cdots = a_{n-1,n}. \end{aligned}$$

■

A cirulant matrix defines a discrete convolution:

Definition 37 (discrete convolution)**.** The *discrete convolution* of two vectors $\boldsymbol{a}, \boldsymbol{f} \in \mathbb{C}^n$ is given by

$$\boldsymbol{a} \star \boldsymbol{f} := C[\boldsymbol{a}]\boldsymbol{f}.$$

The entries of the discrete convolution have a similar formula to that of continuous convolution but with a summation instead of an integral. In particular, if we write

$$\boldsymbol{a} = \begin{bmatrix} a_0 \\ \vdots \\ a_{n-1} \end{bmatrix}, \boldsymbol{f} = \begin{bmatrix} f_0 \\ \vdots \\ f_{n-1} \end{bmatrix}$$

then we have

$$(\boldsymbol{a} * \boldsymbol{f})[k+1] = \sum_{j=0}^{n-1} a_{\mathrm{mod}(k-j,n)} f_j.$$

The analogue between this discrete convolution and the continuous convolution is immediate, and this can be thought of as a discretisation of the continuous convolution using a trapezium rule.

V.3.2 Diagonalising convolutions

Since a circlulant matrix is defined in terms of powers of S, diagonalising S allows us to diagonalise any circulant matrix. A nice property is that the DFT gives us the eigenvectors of S:

Proposition 16 (diagonalising S)**.**

$$S = Q_n^\star \begin{bmatrix} 1 & & & \\ & \omega^{-1} & & \\ & & \ddots & \\ & & & \omega^{1-n} \end{bmatrix} Q_n$$

where Q_n is the DFT matrix and $\omega = \mathrm{e}^{2\pi \mathrm{i}/n}$.

Proof

We can verify that the columns of $Q_n^\star$ are eigenvectors of S:

$$S \begin{bmatrix} 1 \\ \omega^k \\ \vdots \\ \omega^{k(n-2)} \\ \omega^{k(n-1)} \end{bmatrix} = \begin{bmatrix} \omega^{k(n-1)} \\ \omega^k \\ 1 \\ \vdots \\ \omega^{k(n-2)} \end{bmatrix} = \omega^{-k} \begin{bmatrix} 1 \\ \omega^k \\ \vdots \\ \omega^{k(n-1)} \end{bmatrix},$$

since we have $\omega^{-k} = \omega^{k(n-1)}$.

■

This immediately gives us a diagonalisation of circulant matrices:

Theorem 16 (diagonalising circulants)**.**

$$C[\boldsymbol{a}] = \sqrt{n} Q_n^\star \mathrm{diag}(Q_n \boldsymbol{a}) Q_n.$$

Proof

We first see that

$$\begin{aligned} C[\boldsymbol{a}] &= \sum_{k=0}^{n-1} a_k S^k = \sum_{k=0}^{n-1} a_k Q_n^\star \begin{bmatrix} 1 & & \\ & \ddots & \\ & & \omega^{k(1-n)} \end{bmatrix} Q_n \\ &= Q_n^\star \begin{bmatrix} \sum_{k=0}^{n-1} a_k & & \\ & \ddots & \\ & & \sum_{k=0}^{n-1} a_k \omega^{k(1-n)} \end{bmatrix} Q_n = \begin{bmatrix} a(1) & & & \\ & a(\omega^{-1}) & & \\ & & \ddots & \\ & & & a(\omega^{1-n}) \end{bmatrix}. \end{aligned}$$

But we can further observe that

$$a(\omega^{-j}) = \sum_{k=0}^{n-1} a_k \omega^{-kj} = \left[1, \omega^{-j}, \ldots, \omega^{-(n-1)j}\right]^\top \boldsymbol{a} = \sqrt{n} \boldsymbol{e}_{j+1}^\top Q_n \boldsymbol{a}.$$

■

The impact of this is that we can apply a convolution in $O(n \log n)$ when we use the FFT to apply the DFT. That is, we have

$$\boldsymbol{a} \star \boldsymbol{f} = C[\boldsymbol{a}]\boldsymbol{f} = \sqrt{n} Q_n^\star \mathrm{diag}(Q_n \boldsymbol{a}) Q_n \boldsymbol{f}.$$

In other words, in direct analogy to continuous Fourier transforms, we can convolve two vectors by taking their DFT, multiplying entrywise, and taking an inverse DFT. This plays an important role in accelerating Convolutional Neural Networks when the *convolutional kernel* (essentially, our $\boldsymbol{a}$) is large.

V.3.3 Lab and Problem Sheet

In the lab we investigate computing convolutions using the FFT and representing circulant matrices. We also explore a relationship with discrete random variables: if we have two random variables X, Y who takes values of $0, \ldots, n-1$ with probabilities given by vectors $\boldsymbol{f}$ and $\boldsymbol{g}$ then the probabilities of $\mathrm{mod}(X+Y, n)$ are given by $\boldsymbol{f} \star \boldsymbol{g}$. Thus the DFT is important in discrete probability just as Fourier transforms are important in studying continuous PDFs.

In the problem sheet we discuss some of the basic properties of discrete convolution such as commutativity and associativity. We also investigate how discrete convolutions provide a way of approximating continuous convolutions of periodic functions, and relate the errors to those of the Fourier coefficients of the convolved functions. Finally, we discuss how a discrete cross-correlation

$$\sum_{j=0}^{n-1} a_j f_{\mathrm{mod}(j+k,n)},$$

which is what Convolutional Neural Networks are usually defined in terms of, is equivalent to a convolution.

Chapter VI

Orthogonal Polynomials

Fourier series are very powerful for approximating periodic functions. If periodicity is lost, however, the Fourier coefficients are no longer in ℓ^1 and uniform convergence is lost. In this chapter we introduce *Orthogonal Polynomials (OPs)* which are more effective for computing with non-periodic (but still continuous/differentiable) functions. That is we consider expansions of the form

$$f(x) = \sum_{k=0}^{\infty} c_k p_k(x)$$

where $p_k(x)$ are special families of polynomials and c_k are expansion coefficients. We can approximate the coefficients $c_k \approx c_k^n$ using quadrature, just as we did with Fourier series.

Why not use monomials *á la* Taylor series? Hidden in the discussion on Fourier series was that we could effectively compute Taylor coefficients by evaluating on the unit circle in the complex plane, only if the radius of convergence was 1. Many functions are smooth on say $[-1, 1]$ but have non-convergent Taylor series, e.g. *Runge's function*:

$$\frac{1}{25x^2 + 1},$$

the function we have seen in Lab 7 is difficult to interpolate. While orthogonal polynomials span the same space as monomials, orthogonal polynomials are much more stable and can effectively and accurately approximate such functions. In particular, where we saw that interpolation by monomials at evenly spaced points performed horribly in practice we can use orthogonal polynomials with specially chosen points to get reliable interpolation of functions.

Due to time constraints we will only discuss one, extremely important, family of orthogonal polynomials:

1. VI.1 Chebyshev polynomials: We begin with Chebyshev polynomials, a canonical example of orthogonal polynomials that are in fact Fourier series in disguise. These demonstrate many of the essential properties of general orthogonal polynomials.

Fourier and orthogonal polynomial expansions also lead to very effective schemes for solving differential equations and many other applications. In addition to numerics, OPs play an important role in many mathematical areas including functional analysis, integrable systems, singular integral equations, complex analysis, and random matrix theory.

VI.1 Chebyshev Polynomials

How can we approximate functions that are not periodic? The answer is to use orthogonal polynomials, but we will begin with a special case that are, in fact, Fourier series in disguise. Consider a real smooth function $f : [-1, 1] \to \mathbb{R}$. We can turn it into a smooth 2π-periodic function via

$$g(\theta) := f(\cos\theta).$$

First note this is a periodic ($g(\theta) = g(\theta + 2\pi)$) and even ($g(\theta) = g(-\theta) = g(2\pi - \theta)$) function. This tells us, using the change-of-variables $\varphi = 2\pi - \theta$,

$$\hat{g}_k = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{e}^{-\mathrm{i}k\theta} g(\theta)\mathrm{d}\theta = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{e}^{\mathrm{i}k\varphi} g(2\pi - \varphi)\mathrm{d}\varphi = \hat{g}_{-k}.$$

We can thus expand

$$g(\theta) = \sum_{k=-\infty}^{\infty} \hat{g}_k \mathrm{e}^{\mathrm{i}k\theta} = \hat{g}_0 + \sum_{k=1}^{\infty} \hat{g}_k \underbrace{(\mathrm{e}^{\mathrm{i}k\theta} + \mathrm{e}^{-\mathrm{i}k\theta})}_{2\cos k\theta}.$$

But using $\theta = \operatorname{acos} x$ this turns into a *Chebyshev expansion*:

$$f(x) = \sum_{k=0}^{\infty} \check{f}_k T_k(x)$$

where we define

$$T_k(x) := \cos k \operatorname{acos} x$$

and then using the change-of-variables

$$x = \cos\theta \qquad \mathrm{d}x = -\sin\theta\mathrm{d}\theta = -\sqrt{1 - x^2}\mathrm{d}\theta$$

we define (using the evenness of g again)

$$\begin{aligned}
\check{f}_0 :&= \hat{g}_0 = \frac{1}{2\pi}\int_0^{2\pi} g(\theta)\mathrm{d}\theta = \frac{1}{\pi}\int_0^{\pi} g(\theta)\mathrm{d}\theta = \frac{1}{\pi}\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\mathrm{d}x, \\
\check{f}_k :&= 2\hat{g}_k = \hat{g}_k + \hat{g}_{-k} = \frac{1}{\pi}\int_0^{2\pi} g(\theta)\cos k\theta\mathrm{d}\theta = \frac{2}{\pi}\int_0^{\pi} g(\theta)\cos k\theta\mathrm{d}\theta = \frac{2}{\pi}\int_{-1}^{1} \frac{f(x)T_k(x)}{\sqrt{1-x^2}}\mathrm{d}x.
\end{aligned}$$

The functions $T_k(x)$ have a number of wonderful properties:

1. They are *classical* orthogonal polynomials, with respect to the weight $1/\sqrt{1-x^2}$ on $[-1, 1]$.

2. They satisfy a simple three-term recurrence (as do all orthogonal polynomials!).

3. Their roots give excellent interpolation points (as do all orthogonal polynomials!).

Here we establish these properties in the case of Chebyshev polynomials via their connection with Fourier series. But these properties can be extended to general families of orthogonal polynomials directly, without building on the link with Fourier series.

VI.1.1 Chebyshev polynomials are orthogonal polynomials

We want to establish the fact that $T_k(x)$ are in fact (1) polynomials and (2) orthogonal. We start by showing that they satisfy a simple three-term recurrence:

Lemma 10 (Chebyshev 3-term recurrence)**.**

$$\begin{aligned} xT_0(x) &= T_1(x) \\ xT_n(x) &= \frac{T_{n-1}(x) + T_{n+1}(x)}{2} \end{aligned}$$

Proof This follows from trigonometry:

$$xT_n(x) = \cos\theta \cos n\theta = \frac{\cos(n-1)\theta + \cos(n+1)\theta}{2} = \frac{T_{n-1}(x) + T_{n+1}(x)}{2}.$$

■

Note that the recurrence gives an easy means for systematically determining the Chebyshev polynomials in a monomial expansion, eg., we can rapidly deduce $T_5(x)$:

$$\begin{aligned} T_0(x) &= 1, \\ T_1(x) &= xT_0(x) = x, \\ T_2(x) &= 2xT_1(x) - T_0(x) = 2x^2 - 1, \\ T_3(x) &= 2xT_2(x) - T_1(x) = 4x^3 - 3x, \\ T_4(x) &= 2xT_3(x) - T_2(x) = 8x^4 - 8x^2 + 1, \\ T_5(x) &= 2xT_4(x) - T_3(x) = 16x^5 - 20x^3 + 5x. \end{aligned}$$

We now show that they are in fact orthogonal with respect to a simple weight:

Theorem 17 (Chebyshev polynomials, 1st kind)**.** *$T_n(x)$ are orthogonal polynomials with respect to the inner product*

$$\int_{-1}^{1} \frac{f(x)g(x)}{\sqrt{1-x^2}} \mathrm{d}x.$$

Proof

The fact that they are polynomials follows from the three-term recurrence. In detail: we know $T_0(x) = 1$ and $T_1(x) = x$. Now assume

$$T_k(x) = 2^{k-1}x^k + O(x^{k-1})$$

are polynomials polynomial for $k = 0, \ldots, n$. Then

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) = 2^n x^{n+1} + O(x^n)$$

is a degree $n+1$ polynomial of the assumed form.

Orthogonality follows under a change of variables:

$$\int_{-1}^{1} \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}} \mathrm{d}x = \int_0^{\pi} \frac{\cos(n\theta)\cos(m\theta)}{\sqrt{1-\cos^2\theta}} \sin\theta \mathrm{d}\theta = \int_0^{\pi} \cos(n\theta)\cos(m\theta)\mathrm{d}\theta = 0$$

if $n \neq m$, which can be verified using complex exponentials.

■

VI.1.2 Interpolation at roots of Chebyshev polynomials

The roots of $T_n(x)$ give an excellent set of points for interpolation, particularly if we represent the interpolatory polynomial in a Chebyshev expansion. In fact, we saw this already in initial experiments in Lab 7. Note that T_n has n roots, which are exactly the points tested in Lab 7:

$$x_j = \cos\left(\pi \frac{j - 1/2}{n}\right)$$

since

$$T_n(x_j) = \cos(\pi(j - 1/2)) = 0.$$

We can construct a Vandermonde-like matrix using Chebyshev polynomials via:

$$V := \begin{bmatrix} T_0(x_1) & \cdots & T_{n-1}(x_1) \\ \vdots & \ddots & \vdots \\ T_0(x_n) & \cdots & T_{n-1}(x_n) \end{bmatrix}.$$

This is a map from expansion coefficients in a Chebyhev expansion and its values at the points x_j.

The magic here is we can construct V^{-1} explicitly:

Theorem 18 (inverse Chebyshev Vandermonde)**.** *We have*

$$V^{-1} = \frac{1}{n} \begin{bmatrix} 1 & & & \\ & 2 & & \\ & & \ddots & \\ & & & 2 \end{bmatrix} V^\top.$$

Proof

Note that, for $\theta_j = \pi \frac{j-1/2}{n}$,

$$V := \begin{bmatrix} 1 & \cos\theta_1 & \cdots & \cos((n-1)\theta_1) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \cos\theta_n & \cdots & \cos((n-1)\theta_n) \end{bmatrix}.$$

But then we can see that

$$\underbrace{\begin{bmatrix} \sqrt{1/n} & & & \\ & \sqrt{2/n} & & \\ & & \ddots & \\ & & & \sqrt{2/n} \end{bmatrix}}_{=:D} V^\top = C_n$$

where C_n is the Discrete Cosine Transform (DCT), i.e., the orthogonal matrix from PS8 Q4(b). Thus we have

$$V^{-1} = (C_n^\top D^{-1})^{-1} = DC_n = D^2 V^\top.$$

■

VI.1.3 Stability of Chebyshev interpolation

This connection between the Vandermonde-like matrix V and the DCT guarantees that we can apply and invert V in $O(n \log n)$ time which is much faster than the $O(n^3)$ time required for a general matrix. Moreover, because the DCT is orthogonal it can be applied and inverted *stably*: we do not see the instability observed with monomials in Lab 7. We can make this precise in terms of the singular values: we can understand how a matrix blows up errors by looking at the norm of its inverse, i.e., if ε represents errors in a vector $\boldsymbol{x}$ we can bound the error in the solution by:

$$\|V^{-1}(\boldsymbol{x} + \varepsilon) - V^{-1}\boldsymbol{x}\| = \|V^{-1}\varepsilon\| \leq \underbrace{\|V^{-1}\|}_{\sigma_n^{-1}} \|\varepsilon\|.$$

But in this case we can directly bound $\|V^{-1}\|$, since D is diagonal and C_n is orthogonal,

$$\|V^{-1}\| \leq \|D\|\|C_n\| = \sqrt{2/n}.$$

(Actually this bound is achieved and we have equality: $\|V^{-1}\| = \sqrt{2/n}$.) Thus the blow up of error in a 2-norm sense actually decreases with n!

This doesn't take into account floating point round-off error. To do so we need to bound the condition number (see appendix), which is given by:

$$\kappa(V) := \underbrace{\|V\|}_{\sigma_1} \underbrace{\|V^{-1}\|}_{\sigma_n^{-1}} \leq \|C_n^\top\|\|D^{-1}\|\sqrt{2/n} = \sqrt{2}.$$

That is: the transform from Chebyshev points to coefficients is guaranteed to only slightly amplify errors even when implemented in floating point! The power and reliability of Chebyshev interpolation is the starting point of what makes them such powerful tools in computation.

VI.1.4 Lab and Problem Sheet

In the problem sheet we investigate a second kind of Chebyshev polynomials defined as

$$U_n(x) = \frac{\sin(n+1)\operatorname{acos} x}{\sqrt{1-x^2}}.$$

These are orthogonal with respect to the *semicircle weight* $\sqrt{1-x^2}$, and we can relate the derivative of T_n to U_n.

In the lab we explore the computation of Chebyshev polynomials, in particular using their three-term recurrence to construct them using only basic floating point operations (avoiding the need for more expensive trigonometry). We also see how their coefficients can be computed using the Discrete Cosine Transform (DCT) from the previous problem sheet/lab. Finally, we explore the usage of Chebyshev U polynomials for differentiation, observing that we can achieve faster than exponential convergence to the true derivative, and avoiding (to some extent) the instability observed in divided differences.

This connection with differentiation underlies spectral methods, which can solve Ordinary and Partial Differential Equations (ODEs and PDEs) to very high accuracy. In particular, the relationship can be viewed as a *banded* matrix, which leads to fast algorithms. See for example my own work with Alex Townsend on the ultraspherical spectral method.

Chapter VII

Future Directions

In these notes we explored some basic numerical algorithms for integration, differentiation, root-finding, and polynomial regression. We further saw that rigorous computations and analysis could be deduced by understanding floating point arithmetic, and its usage in constructing interval arithmetic. We also saw the relationship between these problems and numerical linear algebra, exploring some fundamental algorithms for solving linear systems and least squares problems. We concluded by introducing computational techniques for Fourier series and orthogonal polynomials.

This sets the ground work for many areas of computational, applied and even pure mathematics:

1. Machine Learning (ML): Dual numbers (forward-mode automatic differentiation), Newton's method, and polynomial regression are baby versions of how ML trains neural networks. As an example, consider a neural network trained to recognise whether a picture is a cat or dog. On a basic level the neural network is a function with many parameters that maps from the set of images to "cat" or "dog" (or in practice, a 2-dimensional vector space where e_1 corresponds to cat and e_2 corresponds to dog to turn it into a continuous problem). The process of training is essentially regression: find the right parameters so that our function equals (or approximately equals) the training data at the given "samples". Since this problem is nonlinear its not as simple as least squares, instead, it relies on more sophisticated optimisation techniques, that require automatic differentiation to compute gradients for the parameters.

2. Scientific Machine Learning (SciML): A particular exciting field is the combination of ML and traditional numerical algorithms for differential equations. For example, if one is doing climate modelling it is possible to combine differential equations that capture physics with a neural network trained to fit real-world data. This is an area where Julia has excelled as it allows high-level, differentiable, code that is compiled and hence efficient.

3. Computer-assisted proofs: By combining interval arithmetic with methods for solving differential equations important theorems in dynamical systems and elsewhere have been proven. The synthesis of numerical analysis and pure mathematics is really just at the beginning and will likely become increasingly important as we run into the limits of what theorems humans can prove without computer assistance.

4. Finite Element Methods (FEM): finite elements are powerful techniques for solving

partial differential equations using specially designed bases built from piecewise polynomials. Orthogonal polynomials underly the quadrature rules used to turn PDEs into sparse linear system. Firedrake is a popular software package for PDEs based on FEM that is developed at Imperial.

5. Spectral methods: orthogonal polynomials in higher dimensions and spherical harmonics (which can be viewed as Fourier series on the sphere) underly spectral methods, which are powerful high-accuracy techniques for solving PDEs, used a lot in fluid simulations which need more accuracy than conventional FEM allows. Dedalus is a software package for solving fluids problems using spectral methods. At a more basic level Chebfun and my package ApproxFun.jl give a user friendly way of computing with functions built on orthogonal polynomials.

Appendix A

Asymptotics and Computational Cost

We introduce Big-O, little-o and asymptotic notation and see how they can be used to describe computational cost.

A.1 Asymptotics as $n \to \infty$

Big-O, little-o, and "asymptotic to" are used to describe behaviour of functions at infinity.

Definition 38 (Big-O)**.**

$$f(n) = O(\phi(n)) \qquad (\text{as } n \to \infty)$$

means $\left|\frac{f(n)}{\phi(n)}\right|$ is bounded for sufficiently large n. That is, there exist constants C and N_0 such that, for all $n \geq N_0$, $|\frac{f(n)}{\phi(n)}| \leq C$.

Definition 39 (little-O)**.**

$$f(n) = o(\phi(n)) \qquad (\text{as } n \to \infty)$$

means $\lim_{n\to\infty} \frac{f(n)}{\phi(n)} = 0$.

Definition 40 (asymptotic to)**.**

$$f(n) \sim \phi(n) \qquad (\text{as } n \to \infty)$$

means $\lim_{n\to\infty} \frac{f(n)}{\phi(n)} = 1$.

Example 23 (asymptotics with n)**.** 1.

$$\frac{\cos n}{n^2 - 1} = O(n^{-2})$$

as

$$\left|\frac{\frac{\cos n}{n^2-1}}{n^{-2}}\right| \leq \left|\frac{n^2}{n^2 - 1}\right| \leq 2$$

for $n \geq N_0 = 2$.

2.
$$\log n = o(n)$$
as $\lim_{n\to\infty} \frac{\log n}{n} = 0$.

3.
$$n^2 + 1 \sim n^2$$
as $\frac{n^2+1}{n^2} \to 1$.

Note we sometimes write $f(O(\phi(n)))$ for a function of the form $f(g(n))$ such that $g(n) = O(\phi(n))$.

We have some simple algebraic rules:

Proposition 17 (Big-O rules)**.**

$$O(\phi(n))O(\psi(n)) = O(\phi(n)\psi(n)) \qquad (\textit{as } n \to \infty)$$
$$O(\phi(n)) + O(\psi(n)) = O(|\phi(n)| + |\psi(n)|) \qquad (\textit{as } n \to \infty).$$

Proof See any standard book on asymptotics, eg F.W.J. Olver, Asymptotics and Special Functions. ■

A.2 Asymptotics as $x \to x_0$

We also have Big-O, little-o and "asymptotic to" at a point:

Definition 41 (Big-O)**.**
$$f(x) = O(\phi(x)) \qquad (\text{as } x \to x_0)$$
means $|\frac{f(x)}{\phi(x)}|$ is bounded in a neighbourhood of x_0. That is, there exist constants C and r such that, for all $0 \le |x - x_0| \le r$, $|\frac{f(x)}{\phi(x)}| \le C$.

Definition 42 (little-O)**.**
$$f(x) = o(\phi(x)) \qquad (\text{as } x \to x_0)$$
means $\lim_{x\to x_0} \frac{f(x)}{\phi(x)} = 0$.

Definition 43 (asymptotic to)**.**
$$f(x) \sim \phi(x) \qquad (\text{as } x \to x_0)$$
means $\lim_{x\to x_0} \frac{f(x)}{\phi(x)} = 1$.

Example 24 (asymptotics with x)**.**
$$\exp x = 1 + x + O(x^2) \qquad \text{as } x \to 0$$
since $\exp x = 1 + x + \frac{\exp t}{2}x^2$ for some $t \in [0, x]$ and
$$\left|\frac{\frac{\exp t}{2}x^2}{x^2}\right| \le \frac{3}{2}$$
provided $x \le 1$.

A.3 Computational cost

We will use Big-O notation to describe the computational cost of algorithms. Consider the following simple sum

$$\sum_{k=1}^{n} x_k^2$$

which we might implement as:

```
function sumsq(x)
    n = length(x)
    ret = 0.0
    for k = 1:n
        ret = ret + x[k]^2
    end
    ret
end
```

```
sumsq (generic function with 1 method)
```

Each step of this algorithm consists of one memory look-up (`z = x[k]`), one multiplication (`w = z*z`) and one addition (`ret = ret + w`). We will ignore the memory look-up in the following discussion. The number of CPU operations per step is therefore 2 (the addition and multiplication). Thus the total number of CPU operations is $2n$. But the constant 2 here is misleading: we didn't count the memory look-up, thus it is more sensible to just talk about the asymptotic complexity, that is, the *computational cost* is $O(n)$.

Now consider a double sum like:

$$\sum_{k=1}^{n} \sum_{j=1}^{k} x_j^2$$

which we might implement as:

```
function sumsq2(x)
    n = length(x)
    ret = 0.0
    for k = 1:n
        for j = 1:k
            ret = ret + x[j]^2
        end
    end
    ret
end
```

```
sumsq2 (generic function with 1 method)
```

Now the inner loop is $O(1)$ operations (we don't try to count the precise number), which we do k times for $O(k)$ operations as $k \to \infty$. The outer loop therefore takes

$$\sum_{k=1}^{n} O(k) = O\left(\sum_{k=1}^{n} k\right) = O\left(\frac{n(n+1)}{2}\right) = O(n^2)$$

operations.

Appendix B

Integers

In this appendix we discuss the following:

1. Unsigned integers: how computers represent non-negative integers using only p-bits, via modular arithmetic.
2. Signed integers: how negative integers are handled using the Two's-complement format.

Mathematically, CPUs only act on p-bits at a time, with 2^p possible sequences. That is, essentially all functions f are either of the form $f : \mathbb{Z}_{2^p} \to \mathbb{Z}_{2^p}$ or $f : \mathbb{Z}_{2^p} \times \mathbb{Z}_{2^p} \to \mathbb{Z}_{2^p}$, where we use the following notation:

Definition 44 (finite integers)**.** Denote the set of the first m non-negative integers as $\mathbb{Z}_m := \{0, 1, \ldots, m-1\}$.

To translate between integers and bits we will need to write integers in binary format. That is, as sequence of `0`s and `1`s:

Example 25 (integers in binary)**.** A simple integer example is $5 = 2^2 + 2^0 = (101)_2$. On the other hand, we write $-5 = -(101)_2$. Another example is $258 = 2^8 + 2 = (100000010)_2$.

B.0.1 Unsigned Integers

Computers represent integers by a finite number of p-bits, with 2^p possible combinations of 0s and 1s. Denote these p-bits as $B_{p-1} \ldots B_1 B_0$ where $B_k \in \{0, 1\}$. For *unsigned integers* (non-negative integers) these bits dictate the first p binary digits: $(B_{p-1} \ldots B_1 B_0)_2$. Integers represented with p-bits on a computer are interpreted as representing elements of $\mathbb{Z}_{2^p}$ and integer arithmetic on a computer is equivalent to arithmetic modulo 2^p. We denote modular arithmetic with $m = 2^p$ as follows:

$$
\begin{aligned}
x \oplus_m y &:= (x + y) \pmod{m} \\
x \ominus_m y &:= (x - y) \pmod{m} \\
x \otimes_m y &:= (x * y) \pmod{m}
\end{aligned}
$$

When m is implied by context we just write $\oplus, \ominus, \otimes$. Note that the $(\text{mod } m)$ function simply drops all bits except for the first p-bits when writing a number in binary.

Example 26 (arithmetic with 8-bit unsigned integers)**.** If the result of an operation lies between 0 and $m = 2^8 = 256$ then airthmetic works exactly like standard integer arithmetic. For example,

$$17 \oplus_{256} 3 = 20 \pmod{256} = 20$$
$$17 \ominus_{256} 3 = 14 \pmod{256} = 14$$

Example 27 (overflow with 8-bit unsigned integers)**.** If we go beyond the range the result "wraps around". For example, with true integers we have

$$255 + 1 = (11111111)_2 + (00000001)_2 = (100000000)_2 = 256$$

However, the result is impossible to store in just 8-bits! So as mentioned instead it treats the integers as elements of $\mathbb{Z}_{256}$ by dropping any extra digits:

$$255 \oplus_{256} 1 = 255 + 1 \pmod{256} = (100000000)_2 \pmod{256} = 0.$$

On the other hand, if we go below 0 we wrap around from above:

$$3 \ominus_{256} 5 = -2 \pmod{256} = 254 = (11111110)_2$$

Example 28 (multiplication of 8-bit unsigned integers)**.** Multiplication works similarly: for example,

$$254 \otimes_{256} 2 = 254 * 2 \pmod{256} = (11111110)_2 * 2 \pmod{256} = (111111100)_2 \pmod{256} = 252.$$

Note that multiplication by 2 is the same as shifting the binary digits left by one, just as multiplication by 10 shifts base-10 digits left by 1.

B.0.2 Signed integer

Signed integers use the Two's complemement convention. The convention is if the first bit is 1 then the number is negative: in this case if the bits had represented the unsigned integer $2^p - y$ then the represent the signed integer $-y$. Thus for $p = 8$ we are interpreting 2^7 through $2^8 - 1$ as negative numbers. More precisely:

Definition 45 (signed integers)**.** Denote the finite signed integers as

$$\mathbb{Z}^{\mathrm{s}}_{2^p} := \{-2^{p-1}, \ldots, -1, 0, 1, \ldots, 2^{p-1} - 1\}.$$

Definition 46 (Shifted mod)**.** Define for $y = x \pmod{2^p}$

$$x \pmod[\mathrm{s}]{2^p} := \begin{cases} y & 0 \leq y \leq 2^{p-1} - 1 \\ y - 2^p & 2^{p-1} \leq y \leq 2^p - 1 \end{cases}$$

Note that if $R_p(x) = x \pmod[\mathrm{s}]{2^p}$ then it can be viewed as a map $R_p : \mathbb{Z} \to \mathbb{Z}^{\mathrm{s}}_{2^p}$ or a one-to-one map $R_p : \mathbb{Z}_{2^p} \to \mathbb{Z}^{\mathrm{s}}_{2^p}$ whose inverse is $R_p^{-1}(x) = x \pmod{2^p}$. It can also be viewed as the identity map on signed integers $R_p : \mathbb{Z}^{\mathrm{s}}_{2^p} \to \mathbb{Z}^{\mathrm{s}}_{2^p}$, that is, $R_p(x) = x$ if $x \in \mathbb{Z}^{\mathrm{s}}_{2^p}$.

Arithmetic works precisely the same for signed and unsigned integers up to the mapping R_p, e.g. we have for $m = 2^p$

$$x \oplus^{\mathrm{s}}_m y := (x + y) \pmod[\mathrm{s}]{m}$$
$$x \ominus^{\mathrm{s}}_m y := (x - y) \pmod[\mathrm{s}]{m}$$
$$x \otimes^{\mathrm{s}}_m y := (x * y) \pmod[\mathrm{s}]{m}$$

Example 29 (addition of 8-bit signed integers)**.** Consider `(-1) + 1` in 8-bit arithmetic:

$$-1 \oplus^{\mathrm{s}}_{256} 1 = -1 + 1 \pmod{^{\mathrm{s}}\ 256} = 0$$

On the bit level this computation is exactly the same as unsigned integers. We represent the number -1 using the same bits as the unsigned integer $2^8 - 1 = 255$, that is using the bits `11111111` (i.e., we store it equivalently to $(11111111)_2 = 255$) and the number 1 is stored using the bits `00000001`. When we add this with true integer arithmetic we have

$$\begin{aligned}&(011111111)_2 \ +\\ &(000000001)_2 \ \ =\\ &(100000000)_2\end{aligned}$$

Modular arithmetic drops the leading 1 and we are left with all zeros.

Example 30 (signed overflow with 8-bit signed integers)**.** If we go above $2^{p-1}-1 = 2^7-1 = 127$ we have perhaps unexpected results:

$$127 \oplus^{\mathrm{s}}_{256} 1 = 128 \pmod{^{\mathrm{s}}\ 256} = 128 - 256 = -128.$$

Again on the bit level this computation is exactly the same as unsigned integers. We represent the number 127 using the bits `01111111` and the number 1 is stored using the bits `00000001`. When we add this with true integer arithmetic we have

$$\begin{aligned}&(01111111)_2 \ +\\ &(00000001)_2 \ \ =\\ &(10000000)_2\end{aligned}$$

Because the first bit is `1` we interpret this as a negative number using the formula:

$$(10000000)_2 \pmod{^{\mathrm{s}}\ 256} = 128 (\mathrm{mod}^{\mathrm{s}}\ 256) = -128.$$

Example 31 (multiplication of 8-bit signed integers)**.** Consider computation of `(-2) * 2`:

$$(-2) \otimes^{\mathrm{s}}_{2^p} 2 = -4 \pmod{^{\mathrm{s}}\ 2^p} = -4$$

On the bit level, the bits of -2 (which is one less than -1) are `11111110`. Multiplying by 2 is like multiplying by 10 in base-10, that is, we shift the bits. Hence in true arithmetic we have

$$\begin{aligned}&(011111110)_2 * 2 =\\ &(111111100)_2\end{aligned}$$

We drop the leading 1 due to modular arithmetic. We still have a leading 1 hence the number is viewed as negative. In particular we have

$$\begin{aligned}(111111100)_2 \pmod{^{\mathrm{s}}\ 256} &= (11111100)_2 \pmod{^{\mathrm{s}}\ 256} = 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 \pmod{^{\mathrm{s}}\ 256}\\ &= 252 \pmod{^{\mathrm{s}}\ 256} = -4.\end{aligned}$$

B.0.3 Hexadecimal format

In coding it is often convenient to use base-16 as it is a power of 2 but uses less characters than binary. The digits used are 0 through 9 followed by $a = 10$, $b = 11$, $c = 12$, $d = 13$, $e = 14$, and $f = 15$.

Example 32 (Hexadecimal number)**.** We can interpret a number in format as follows:

$$(a5f2)_{16} = a * 16^3 + 5 * 16^2 + f * 16 + 2 = 10 * 16^3 + 5 * 16^2 + 15 * 16 + 2 = 42,482$$

We will see in the labs that unsigned integers are displayed in base-16.

Appendix C

Permutation Matrices

Permutation matrices are matrices that represent the action of permuting the entries of a vector, that is, matrix representations of the symmetric group S_n, acting on $\mathbb{R}^n$. Recall every $\sigma \in S_n$ is a bijection between $\{1, 2, \ldots, n\}$ and itself. We can write a permutation σ in *Cauchy notation*:

$$\begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ \sigma_1 & \sigma_2 & \sigma_3 & \cdots & \sigma_n \end{pmatrix}$$

where $\{\sigma_1, \ldots, \sigma_n\} = \{1, 2, \ldots, n\}$ (that is, each integer appears precisely once). We denote the *inverse permutation* by σ^{-1}, which can be constructed by swapping the rows of the Cauchy notation and reordering.

We can encode a permutation in vector $\sigma = [\sigma_1, \ldots, \sigma_n]$. This induces an action on a vector (using indexing notation)

$$\boldsymbol{v}[\sigma] = \begin{bmatrix} v_{\sigma_1} \\ \vdots \\ v_{\sigma_n} \end{bmatrix}$$

Example 33 (permutation of a vector)**.** Consider the permutation σ given by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 2 & 5 & 3 \end{pmatrix}$$

We can apply it to a vector:

```
using LinearAlgebra
σ = [1, 4, 2, 5, 3]
v = [6, 7, 8, 9, 10]
v[σ] # we permutate entries of v
```

```
5-element Vector{Int64}:
  6
  9
  7
 10
  8
```

Its inverse permutation σ^{-1} has Cauchy notation coming from swapping the rows of the Cauchy notation of σ and sorting:

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 4 & 3 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{pmatrix}$$

Note that the operator

$$P_\sigma(\boldsymbol{v}) = \boldsymbol{v}[\sigma]$$

is linear in $\boldsymbol{v}$, therefore, we can identify it with a matrix whose action is:

$$P_\sigma \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} v_{\sigma_1} \\ \vdots \\ v_{\sigma_n} \end{bmatrix}.$$

The entries of this matrix are

$$P_\sigma[k,j] = \boldsymbol{e}_k^\top P_\sigma \boldsymbol{e}_j = \boldsymbol{e}_k^\top \boldsymbol{e}_{\sigma_j^{-1}} = \delta_{k,\sigma_j^{-1}} = \delta_{\sigma_k,j}$$

where $\delta_{k,j}$ is the *Kronecker delta*:

$$\delta_{k,j} := \begin{cases} 1 & k = j \\ 0 & \text{otherwise} \end{cases}.$$

This construction motivates the following definition:

Definition 47 (permutation matrix)**.** $P \in \mathbb{R}^{n\times n}$ is a permutation matrix if it is equal to the identity matrix with its rows permuted.

Proposition 18 (permutation matrix inverse)**.** *Let P_σ be a permutation matrix corresponding to the permutation σ. Then*

$$P_\sigma^\top = P_{\sigma^{-1}} = P_\sigma^{-1}$$

That is, P_σ is orthogonal*:*

$$P_\sigma^\top P_\sigma = P_\sigma P_\sigma^\top = I.$$

Proof

We prove orthogonality via:

$$\boldsymbol{e}_k^\top P_\sigma^\top P_\sigma \boldsymbol{e}_j = (P_\sigma \boldsymbol{e}_k)^\top P_\sigma \boldsymbol{e}_j = \boldsymbol{e}_{\sigma_k^{-1}}^\top \boldsymbol{e}_{\sigma_j^{-1}} = \delta_{k,j}$$

This shows $P_\sigma^\top P_\sigma = I$ and hence $P_\sigma^{-1} = P_\sigma^\top$.

■

Appendix D

Norms

In this appendix we discuss matrix and vector norms.

1. Vector norms: we discuss the standard p-norm for vectors in $\mathbb{R}^n$.
2. Matrix norms: we discuss how two vector norms can be used to induce a norm on matrices. These

satisfy an additional multiplicative inequality.

D.0.1 Vector norms

Recall the definition of a (vector-)norm:

Definition 48 (vector-norm)**.** A norm $\|\cdot\|$ on a vector space V (e.g. $\mathbb{R}^n$ or $\mathbb{C}^n$) over a field $\mathbb{F}$ (e.g. $\mathbb{R}$ or $\mathbb{C}$) is a function that satisfies the following, for $\boldsymbol{x}, \boldsymbol{y} \in V$ and $c \in \mathbb{F}$:

1. Triangle inequality: $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$
2. Homogeneity: $\|c\boldsymbol{x}\| = |c|\|\boldsymbol{x}\|$
3. Positive-definiteness: $\|\boldsymbol{x}\| = 0$ implies that $\boldsymbol{x} = 0$.

Consider the following example:

Definition 49 (p-norm)**.** For $1 \leq p < \infty$ and $\boldsymbol{x} \in \mathbb{C}^n$, define the p-norm:

$$\|\boldsymbol{x}\|_p := \left(\sum_{k=1}^{n} |x_k|^p\right)^{1/p}$$

where x_k is the k-th entry of $\boldsymbol{x}$. For $p = \infty$ we define

$$\|\boldsymbol{x}\|_\infty := \max_k |x_k|.$$

The default norm is the 2-norm $\|\boldsymbol{x}\| := \|\boldsymbol{x}\|_2$.

Theorem 19 (p-norm)**.** *$\|\cdot\|_p$ is a norm for $1 \leq p \leq \infty$.*

Proof

We will only prove the case $p = 1, 2, \infty$ as general p is more involved.

Homogeneity and positive-definiteness are straightforward: e.g.,

$$\|c\boldsymbol{x}\|_p = (\sum_{k=1}^{n} |cx_k|^p)^{1/p} = (|c|^p \sum_{k=1}^{n} |x_k|^p)^{1/p} = |c|\|\boldsymbol{x}\|$$

and if $\|\boldsymbol{x}\|_p = 0$ then all $|x_k|^p$ are have to be zero.

For $p = 1, \infty$ the triangle inequality is also straightforward:

$$\|\boldsymbol{x} + \boldsymbol{y}\|_\infty = \max_k(|x_k + y_k|) \leq \max_k(|x_k| + |y_k|) \leq \|\boldsymbol{x}\|_\infty + \|\boldsymbol{y}\|_\infty$$

and

$$\|\boldsymbol{x} + \boldsymbol{y}\|_1 = \sum_{k=1}^{n} |x_k + y_k| \leq \sum_{k=1}^{n}(|x_k| + |y_k|) = \|\boldsymbol{x}\|_1 + \|\boldsymbol{y}\|_1$$

For $p = 2$ it can be proved using the Cauchy–Schwartz inequality:

$$|\boldsymbol{x}^\star \boldsymbol{y}| \leq \|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2$$

That is, we have

$$\|\boldsymbol{x} + \boldsymbol{y}\|^2 = \|\boldsymbol{x}\|^2 + 2\boldsymbol{x}^\top \boldsymbol{y} + \|\boldsymbol{y}\|^2 \leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 = (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)$$

■

D.0.2 Matrix norms

Just like vectors, matrices have norms that measure their "length". The simplest example is the Fröbenius norm:

Definition 50 (Fröbenius norm)**.** For $A \in \mathbb{C}^{m\times n}$ define

$$\|A\|_F := \sqrt{\sum_{k=1}^{m}\sum_{j=1}^{n} |a_{kj}|^2}$$

While this is the simplest norm, it is not the most useful. Instead, we will build a matrix norm from a vector norm:

Definition 51 (matrix-norm)**.** Suppose $A \in \mathbb{C}^{m\times n}$ and consider two norms $\|\cdot\|_X$ on $\mathbb{C}^n$ and $\|\cdot\|_Y$ on $\mathbb{C}^n$. Define the *(induced) matrix norm* as:

$$\|A\|_{X\to Y} := \sup_{\boldsymbol{v}:\|\boldsymbol{v}\|_X=1} \|A\boldsymbol{v}\|_Y$$

Also define

$$\|A\|_X := \|A\|_{X\to X}$$

For the induced p-norm we use the notation $\|A\|_p$ and the default norm is the 2-norm: $\|A\| := \|A\|_2$.

Note an equivalent definition of the induced norm:

$$\|A\|_{X\to Y} = \sup_{\boldsymbol{x}\in\mathbb{R}^n, \boldsymbol{x}\neq 0} \frac{\|A\boldsymbol{x}\|_Y}{\|\boldsymbol{x}\|_X}$$

This follows since we can scale $\boldsymbol{x}$ by its norm so that it has unit norm, that is, $\frac{\boldsymbol{x}}{\|\boldsymbol{x}\|_X}$ has unit norm.

Lemma 11 (matrix norms are norms)**.** *Induced matrix norms are norms, that is for* $\|\cdot\| = \|\cdot\|_{X\to Y}$ *we have:*

1. *Triangle inequality:* $\|A+B\| \leq \|A\| + \|B\|$
2. *Homogeneneity:* $\|cA\| = |c|\|A\|$
3. *Positive-definiteness:* $\|A\| = 0 \Rightarrow A = 0$

In addition, they satisfy the following additional properties:

1. $$\|A\boldsymbol{x}\|_Y \leq \|A\|_{X\to Y}\|\boldsymbol{x}\|_X$$
2. *Multiplicative inequality:* $\|AB\|_{X\to Z} \leq \|A\|_{Y\to Z}\|B\|_{X\to Y}$

Proof

First we show the *triangle inequality*:

$$\|A+B\| \leq \sup_{\boldsymbol{v}:\|\boldsymbol{v}\|_X=1} (\|A\boldsymbol{v}\|_Y + \|B\boldsymbol{v}\|_Y) \leq \|A\| + \|B\|.$$

Homogeneity is also immediate. Positive-definiteness follows from the fact that if $\|A\| = 0$ then $A\boldsymbol{x} = 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$. The property $\|A\boldsymbol{x}\|_Y \leq \|A\|_{X\to Y}\|\boldsymbol{x}\|_X$ follows from the definition. Finally, the multiplicative inequality follows from

$$\|AB\| = \sup_{\boldsymbol{v}:\|\boldsymbol{v}\|_X=1} \|AB\boldsymbol{v}\|_Z \leq \sup_{\boldsymbol{v}:\|\boldsymbol{v}\|_X=1} \|A\|_{Y\to Z}\|B\boldsymbol{v}\| = \|A\|_{Y\to Z}\|B\|_{X\to Y}$$

■

We have some simple examples of induced norms:

Example 34 (1-norm)**.** We claim

$$\|A\|_1 = \max_j \|\boldsymbol{a}_j\|_1$$

that is, the maximum 1-norm of the columns. To see this use the triangle inequality to find for $\|\boldsymbol{x}\|_1 = 1$

$$\|A\boldsymbol{x}\|_1 \leq \sum_{j=1}^n |x_j|\|\boldsymbol{a}_j\|_1 \leq \max_j \|\boldsymbol{a}_j\| \sum_{j=1}^n |x_j| = \max_j \|\boldsymbol{a}_j\|_1.$$

But the bound is also attained since if j is the column that maximises the norms then

$$\|A\boldsymbol{e}_j\|_1 = \|\boldsymbol{a}_j\|_1 = \max_j \|\boldsymbol{a}_j\|_1.$$

Note that

$$\|A\|_\infty = \max_k \|A[k,:]\|_1$$

that is, the maximum 1-norm of the rows.

An example that does not have a simple formula is

$$\|A\| := \|A\|_2$$

which requires the singular value decomposition.

Appendix E

ConditionNumbers

We have seen that floating point arithmetic induces errors in computations, and that we can typically bound the absolute errors to be proportional to $C\epsilon_{\mathrm{m}}$. We want a way to bound the effect of more complicated calculations like computing $A\boldsymbol{x}$ or $A^{-1}\boldsymbol{y}$ without having to deal with the exact nature of floating point arithmetic, as it will depend on the *data* A and $\boldsymbol{x}$. That is, we want to reduce floating point stability to a more fundamental property: *mathematical stability*: how does a mathematical operation like $A\boldsymbol{x}$ change in the precense of small perturbations (random noise or structured floating point errors)?

1. Backward error analysis: We introduce the concept of *backward error* analysis, which is a more practical way of understanding and bounding floating point errors.

2. Condition numbers: We introduce a *condition numbers*, which can capture the effect of perturbations in A for linear algebra operations. More precisely: matrix operations are mathematically *stable* when the condition number is small.

3. Bounding floating point errors for linear algebra: we see how simple operations like $A\boldsymbol{x}$ can be put into a backward error analysis framework, leading to bounds on the errors in terms of the condition number.

E.0.1 Backward error analysis

In the notes we have done forward error analysis, e.g., to understand $f(x) \approx f^{\mathrm{FP}}(x)$ we consider either the absolute

$$f^{\mathrm{FP}}(x) = f(x) + \delta_{\mathrm{a}}$$

or relative

$$f^{\mathrm{FP}}(x) = f(x)(1 + \delta_{\mathrm{r}})$$

errors of the *output*. More generally, for two vector spaces V and W (e.g. $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$) consider functions $\boldsymbol{f} = V \to W$. We write

$$\boldsymbol{f}^{\mathrm{FP}}(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x}) + \delta_{\mathrm{f}}$$

where we bound a norm of $\delta_{\mathrm{f}} \in W$ either *absolutely*:

$$\|\delta_{\mathrm{f}}\|_W \leq C\varepsilon$$

or *relative* to the true result:

$$\|\delta_{\mathrm{f}}\|_W \leq C\|\boldsymbol{f}(\boldsymbol{x})\|_W \varepsilon.$$

On the other hand, *backward error analysis* considers computations as errors in the *input*. That is, one writes the approximate function as the true function with a perturbed input: e.g. find $\tilde{\boldsymbol{x}} \in V$ such that

$$\boldsymbol{f}^{\mathrm{FP}}(\boldsymbol{x}) = \boldsymbol{f}(\tilde{\boldsymbol{x}}).$$

We study the *backward error*, the error in the input

$$\tilde{\boldsymbol{x}} = \boldsymbol{x} + \delta_{\mathrm{b}}$$

where $\delta_{\mathrm{b}} \in \mathbb{R}^n$ by bounding either the absolute error

$$\|\delta_{\mathrm{b}}\|_V \leq C\varepsilon$$

or relative error:

$$\delta_{\mathrm{b}}\|_V \leq C\|\boldsymbol{x}\|_V \varepsilon$$

We shall see that *some* algorithms (like `mul_rows`) lead naturally to backward error results.

E.0.2 Condition numbers

So now we get to a mathematical question independent of floating point: can we bound the *relative error* in approximating

$$A\boldsymbol{x} \approx (A + \delta A)\boldsymbol{x}$$

if we know a bound on the relative backward error $\|\delta A\|$? It turns out we can in turns of the *condition number* of the matrix:

Definition 52 (condition number)**.** For a square matrix A, the *condition number* (in p-norm) is

$$\kappa_p(A) := \|A\|_p \|A^{-1}\|_p$$

with the default being the 2-norm condition number, writable in terms of the singular values as:

$$\kappa(A) := \kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_1}{\sigma_n}.$$

Theorem 20 (relative forward error for matrix-vector)**.** *Assume we have the relative backward error bound $\|\delta A\| \leq \|A\|\varepsilon$. Then for*

$$(A + \delta A)\boldsymbol{x} = A\boldsymbol{x} + \delta_{\mathrm{f}}$$

the forward error has the relative error bound

$$\|\delta_{\mathrm{f}}\| \leq \|A\boldsymbol{x}\| \kappa(A) \varepsilon$$

Proof We can assume A is invertible (as otherwise $\kappa(A) = \infty$). Denote $\boldsymbol{y} = A\boldsymbol{x}$ and we have

$$\frac{\|\boldsymbol{x}\|}{\|A\boldsymbol{x}\|} = \frac{\|A^{-1}\boldsymbol{y}\|}{\|\boldsymbol{y}\|} \leq \|A^{-1}\|$$

Thus we have:

$$\frac{\|\delta_{\mathrm{f}}\|}{\|A\boldsymbol{x}\|} \leq \frac{\|\delta A\| \|\boldsymbol{x}\|}{\|A\boldsymbol{x}\|} \leq \underbrace{\|A\| \|A^{-1}\|}_{\kappa(A)} \varepsilon.$$

■

E.0.3 Bounding floating point errors for linear algebra

We now observe that errors in implementing matrix-vector multiplication using floating points can be captured by considering the multiplication to be exact on the wrong matrix: that is, `A*x` (implemented with floating point as `mul_rows`) is precisely $A + \delta A$ where δA has small norm, relative to A. That is, we have a bound on the *backward relative error*.

To discuss floating point errors we need to be precise which order the operations happened. We will use the definition `mul_rows(A,x)` (which is equivalent to `mul_cols(A,x)`). Note that each entry of the result is in fact a dot-product of the corresponding rows so we first consider the error in the dot product `dot(`$\boldsymbol{x}$`,`$\boldsymbol{y}$`)` as implemented in floating-point:

$$\mathrm{dot}(\boldsymbol{x}, \boldsymbol{y}) = \bigoplus_{k=1}^{n} (x_k \otimes y_k).$$

We first need a helper proposition:

Proposition 19 (products of errors)**.** *If $|\epsilon_i| \leq \epsilon$ and $n\epsilon < 1$, then*

$$\prod_{k=1}^{n} (1 + \epsilon_i) = 1 + \theta_n$$

for some constant θ_n satisfying

$$|\theta_n| \leq \underbrace{\frac{n\epsilon}{1 - n\epsilon}}_{E_{n,\epsilon}}.$$

Proof

See PS3 Q8(a).

■

Lemma 1 (dot product backward error) For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$,

$$\mathrm{dot}(\boldsymbol{x}, \boldsymbol{y}) = (\boldsymbol{x} + \delta\boldsymbol{x})^\top \boldsymbol{y}$$

where, assuming $n\epsilon_{\mathrm{m}} < 2$, the entries satisfy

$$|\delta x_k| \leq E_{n,\epsilon_{\mathrm{m}}/2} |x_k|.$$

Proof

This is related to PS4 Q3 but asks for the *backward error* instead of the *forward error*. Note

$$\mathrm{dot}(\boldsymbol{x}, \boldsymbol{y}) = \bigoplus_{j=1}^{n} (x_j \otimes y_j) = \bigoplus_{j=1}^{n} (x_j y_j)(1 + \delta_j) = x_1 y_1 (1 + \tilde{\theta}_n) + \sum_{j=2}^{n} x_j y_j (1 + \theta_{n-j+2})$$

where $|\tilde{\theta}_n|, |\theta_k| \leq E_{n,\epsilon_{\mathrm{m}}/2}$ (the subscript denotes the number of terms bounded by $\varepsilon_{\mathrm{m}}/2$). Thus we can define

$$\delta\boldsymbol{x} := \begin{bmatrix} x_1 \tilde{\theta}_n \\ x_2 \theta_n \\ \vdots \\ x_n \theta_2 \end{bmatrix}$$

where

$$|\delta x_k| \leq E_{n,\epsilon_\mathrm{m}/2}|x_k|.$$

■

We can use this to get a relative backward error bound on `mul_rows`:

Theorem 2 (matrix-vector backward error) For $A \in \mathbb{R}^{m\times n}$ and $\boldsymbol{x} \in \mathbb{R}^n$ (both with normal float entries) we have

$$\mathrm{mul_rows}(A, \boldsymbol{x}) = (A + \delta A)\boldsymbol{x}$$

where, assuming $n\epsilon_\mathrm{m} < 2$ and all operations are in the normalised range, the entries (denoting $\delta a_{kj} = \delta A[k, j] = \boldsymbol{e}_k^\top \delta A \boldsymbol{e}_j$) satisfy

$$|\delta a_{kj}| \leq E_{n,\epsilon_\mathrm{m}/2}|a_{kj}|.$$

Proof The bound on the entries of δA is implied by the previous lemma since each row is equivalent to a dot product. ■

Corollary 6 (Norms)**.**

$$\begin{aligned}
\|\delta A\|_1 &\leq E_{n,\epsilon_\mathrm{m}/2}\|A\|_1 \\
\|\delta A\|_2 &\leq \sqrt{\min(m,n)}E_{n,\epsilon_\mathrm{m}/2}\|A\|_2 \\
\|\delta A\|_\infty &\leq E_{n,\epsilon_\mathrm{m}/2}\|A\|_\infty
\end{aligned}$$

In particular,

$$\mathit{mul_rows}(A, \boldsymbol{x}) = A\boldsymbol{x} + \delta_\mathrm{f}$$

where

$$\|\delta_\mathrm{f}\| \leq \|A\boldsymbol{x}\|\kappa(A)E_{n,\epsilon_\mathrm{m}/2}$$

Proof

The 1-norm follow since

$$\|\delta A\|_1 = \max_j \sum_{k=1}^m |\delta a_{kj}| \leq E_{n,\epsilon_\mathrm{m}/2} \max_j \sum_{k=1}^m |a_{kj}| = E_{n,\epsilon_\mathrm{m}/2}\|A\|_1$$

and the proof for the ∞-norm is similar.

This leaves the 2-norm, which is a bit more challenging. We will prove the result by going through the Fröbenius norm and using

$$\|A\|_2 \leq \|A\|_F \leq \sqrt{r}\|A\|_2$$

where r is rank of A. So we deduce

$$\begin{aligned}
\|\delta A\|_2^2 \leq \|\delta A\|_F^2 &= \sum_{k=1}^m \sum_{j=1}^n |\delta a_{kj}|^2 \leq E_{n,\epsilon_\mathrm{m}/2}^2 \sum_{k=1}^m \sum_{j=1}^n |a_{kj}|^2 \\
&= E_{n,\epsilon_\mathrm{m}/2}^2 \|A\|_F^2 \leq E_{n,\epsilon_\mathrm{m}/2}^2 r\|A\|_2^2.
\end{aligned}$$

and the rank of A is bounded by $\min(m, n)$. The bound on the forward error then follows.

■

We can also bound the error of back-substitution in terms of the condition number. If one uses QR to solve $A\boldsymbol{x} = \boldsymbol{y}$ the condition number also gives a meaningful bound on the error. As we have already noted, there are some matrices where PLU decompositions introduce large errors, so in that case well-conditioning is not a guarantee of accuracy (but it still usually works).